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# Distributionally Robust Optimization: Endogenous Calibration of the Wasserstein Ambiguity Radius

Submitted by

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**Disclaimer**

The work is the responsibility of the author; in no way does it engage the responsibility of the University or of the supervising Professor.

**Reproducibility**

Replication code: https://github.com/jeyllani/mt-mfdro
Python package: https://pypi.org/project/mfdro

## Abstract

This thesis examines whether the timing of the Wasserstein ambiguity radius can be driven by observable information and have economic value distinct from its average level. We construct a misalignment signal from daily, weekly, and monthly return distributions, aggregated around a Wasserstein barycenter and compared using Sliced-Wasserstein distances. The signal modulates the radius of a $W_2$ ambiguity set in a linear-loss portfolio problem, evaluated through a point-in-time walk-forward design over 1995–2025. In the main universe, the net Sharpe ratio increases from 0.6892 to 0.7451, a gain of 0.0559. Neither the paired HAC test nor the paired bootstrap establishes a general improvement at the 5% level. Signal-ordering placebos reject the interchangeability of the observed temporal ordering, and the main specification remains significant after max-$T$ adjustment within both the four-cell and twelve-cell families. A modulation based on realized volatility nevertheless achieves a slightly higher point estimate, without a significant paired difference. The contribution is therefore primarily methodological: the level and timing of the ambiguity radius can be distinguished and tested, but the signal's incremental economic value has not been established.



**Keywords:** distributionally robust optimization; Wasserstein distance; portfolio allocation; ambiguity-radius timing; multi-frequency returns.

**JEL classification:** C61; D81; G11.



## Résumé

Ce mémoire étudie si la chronologie du rayon d'ambiguïté Wasserstein peut être pilotée par une information observable et posséder une valeur économique distincte de son niveau moyen. Nous construisons un signal de désalignement entre les distributions de rendements journalières, hebdomadaires et mensuelles, agrégées autour d'un barycentre de Wasserstein puis comparées au moyen de distances Sliced-Wasserstein. Ce signal module le rayon d'une boule $W_2$ dans un problème d'allocation à perte linéaire, évalué selon un protocole walk-forward point-in-time sur 1995–2025. Dans l'univers principal, le ratio de Sharpe net passe de 0,6892 à 0,7451, soit un gain de 0,0559. Les tests HAC et bootstrap ne concluent toutefois pas à une amélioration générale au seuil de 5 %. Les placebos chronologiques rejettent l'interchangeabilité du signal observé, et la spécification principale demeure significative après correction max-$T$ dans les familles de quatre et de douze cellules. Une modulation fondée sur la volatilité réalisée obtient néanmoins une performance ponctuelle légèrement supérieure, sans différence appariée significative. La contribution est ainsi principalement méthodologique : le niveau et la chronologie du rayon peuvent être distingués et testés, mais la valeur économique incrémentale du signal demeure non établie.

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# Executive Summary

Distributionally robust optimization protects a decision against uncertainty surrounding the data-generating distribution. In Wasserstein-based models, this uncertainty is represented by a ball centered on the empirical distribution. Its radius determines the range of alternative distributions under consideration: a small radius places greater confidence in the observed data, whereas a large radius leads to a more conservative allocation. This thesis examines whether, at a comparable average level of robustness, the timing of the radius can be driven by an observable distributional state and possess distinct economic value.

We construct a signal that measures misalignment among return distributions observed at multiple horizons. At each formation date, daily, weekly, and monthly returns for the same assets are extracted from a window strictly prior to the decision and rescaled to make the frequencies comparable. The resulting empirical measures are aggregated around a free-support Wasserstein barycenter, which represents their consensus, and their dispersion is then evaluated using Sliced-Wasserstein distances. The signal therefore increases when the distributional geometries associated with the different horizons diverge.

The framework separates a static anchor $\bar{\varepsilon}$ from a normalized modulation $m_\tau$, according to $\varepsilon_\tau = \bar{\varepsilon} m_\tau$. This decomposition allows static and dynamic portfolios to be compared while maintaining approximately comparable average robustness. Under the linear loss adopted here, the radius acts directly on the penalty associated with weight concentration. The problem is reformulated as a second-order cone program and solved at each formation date.

The analysis uses daily CRSP data from 1990 to 2025. The formation population is restricted point-in-time to U.S. common stocks active on the NYSE, AMEX, or NASDAQ; ADRs, REITs, listed funds, ETFs, and securities issued by non-U.S. issuers are excluded. Each security must have 60 consecutive calendar months of complete daily returns. The main universe comprises, each month, the 100 largest stocks above the $90^{\text{th}}$ NYSE percentile. The NYSE small–mid-cap universe comprises the 100 largest eligible stocks within the $P20$–$P50$ band. The walk-forward evaluation spans January 1995 to December 2025, incorporates delisting returns, and accounts for transaction costs.

The signal displays internal empirical validity. It reacts around several externally dated episodes of market stress, and its discriminatory power increases with the intensity of the misalignment introduced in a controlled experiment. Its dynamics also remain stable under the main numerical variants that preserve its economic definition. The regressions nevertheless reveal a substantial common component with volatility, particularly at the 36-month formation horizon. The signal's residual variation is therefore insufficient to establish structural independence or distinct predictive content.

In the main universe, the net Sharpe ratio increases from 0.6892 for the static portfolio to 0.7451 for the dynamic portfolio, a gain of 0.0559, compared with 0.0571 before costs. The dynamic portfolio combines a higher average return with lower volatility, tail risk, and maximum drawdown, while turnover rises only modestly. This point-estimate improvement nevertheless remains statistically imprecise: neither the HAC test applied to the mean return difference ($p = 0.287$) nor the paired bootstrap of the Sharpe-ratio difference ($p = 0.113$) establishes it at the 5% level.

Volatility-based rules further qualify this estimate. In the main universe, modulation based on 36-month realized volatility produces a slightly higher Sharpe ratio than $P^{(\rho)}$. In the NYSE small–mid-cap universe, $P^{(\rho)}$ has the highest point estimate. Because none of the paired differences is significant at the 5% level, the signal's incremental economic value has not been established.

The permutation tests address a distinct hypothesis: does performance depend on the observed temporal ordering of the signal? Simple monthly permutations and six- and twelve-month block permutations yield $p$-values of 0.001, 0.003, and 0.009, respectively. After controlling for multiplicity within the $\mathcal{F}_4$ family, only the main specification remains significant, with $p_{\text{max-}T_4} = 0.001$. It also remains significant within the extended $\mathcal{F}_{12}$ family, with $p = 0.034$. Large-cap stocks over 1995–2004 and high-dispersion states likewise remain significant, with adjusted $p$-values of 0.001 and 0.012. The evidence is therefore targeted rather than uniformly generalizable.

The geometry of the problem clarifies the economic scope of these findings. The static portfolio is already highly diversified and close to the region toward which the solution converges as the radius increases. An intermediate window nevertheless remains in which modulation produces a discernible gain. Under the prespecified identity form, $P^{(\rho)}$ improves on its static benchmark at the point estimate, without dominating equal weighting in absolute terms. The contrast becomes negative under the exponential and rank transformations, precluding generalization across functional forms.

The contribution of this thesis is primarily methodological. It introduces a multi-frequency distributional signal and a protocol that empirically separates the average level of a robustness parameter from the value of its timing. The point-in-time construction, signal-ordering placebos, regime attribution, and multiplicity control make this timing measurable and empirically testable. The results do not demonstrate the general superiority of an investment strategy; they show that the observed temporal ordering is not interchangeable with the permuted signal paths considered and is associated with a positive point estimate in some configurations, without establishing incremental value relative to volatility-based rules. Natural extensions include objective functions that directly incorporate covariance or tail risk, other asset classes, and learned modulation rules.

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# 1 Introduction

Financial markets bring together participants whose decision horizons differ substantially. Market makers adjust their positions over very short horizons, whereas institutional investors and pension funds operate over months or years. The heterogeneous market hypothesis of Müller et al. [1] places this coexistence at the core of price dynamics: agents operating at distinct horizons may interpret and incorporate information at different speeds. This perspective suggests that a change in the coherence among distributions observed at multiple frequencies may reveal a shift in the market's distributional state.

This issue is particularly important for asset allocation. Mean–variance optimization relies on unknown parameters that must be estimated, and its solutions are notoriously sensitive to estimation error, especially in expected returns [2, 3]. Distributionally robust optimization (DRO) addresses this fragility by replacing a single empirical distribution with a set of plausible probability laws. When this set is defined by a Wasserstein ball, the portfolio is optimized against the least favorable distribution contained in that ball while, for certain loss functions, retaining a tractable convex reformulation [4].

The extent of this protection nevertheless depends on the ambiguity radius. A radius that is too small leaves the portfolio exposed to estimation error; a large radius, by contrast, imposes substantial regularization and may drive the allocation toward highly diversified solutions. In many applications, the radius is calibrated from a theoretical bound, an operational rule, or cross-validation and then used as a reference level throughout the evaluation. Recent studies introduce time-dependent adjustments, but constructing a path directly from an observable distributional diagnostic remains relatively unexplored.

This thesis examines a different hypothesis: the degree of distributional instability may evolve over time, so the timing of the ambiguity radius may have value distinct from its reference level. The central question is whether a signal of misalignment across observation horizons can be constructed from the information available at each date and its timing used to modulate the ambiguity radius without introducing look-ahead bias. The associated economic question is whether this modulation improves a DRO portfolio relative to a static radius constructed around the same anchor.

To address these questions, we form empirical return distributions at daily, weekly, and monthly frequencies at each date. After rescaling them to make their geometries comparable, the resulting measures are aggregated around a multi-frequency barycenter. Their dispersion, evaluated using Sliced-Wasserstein distances, produces a scalar misalignment signal, $\widehat{\rho}_\tau^{\mathrm{hyb}}$. A high value indicates that the temporal representations of the same return process describe empirical distributions that are farther apart. The signal is neither an estimate of the distance to the unknown future distribution nor a direct measure of agent heterogeneity; it is used as an observable indicator of cross-horizon distributional instability.

The dynamic radius is then decomposed into a reference level $\bar{\varepsilon}$ and a point-in-time modulation $m_\tau$, constructed from $\sqrt{\widehat{\rho}_\tau^{\mathrm{hyb}}}$. This factorization separates two questions that are usually conflated: the choice of the reference level of robustness and the value of its timing. The dynamic portfolio can therefore be compared with a static portfolio that shares the same anchor, data, universe, and accounting conventions. The point-in-time expanding-window normalization of $m_\tau$ does not, however, impose exact ex post equality between their realized average radii. The comparison therefore evaluates the value of timing around a common anchor.

The contribution of this thesis is threefold. First, we construct an endogenous signal of cross-horizon misalignment and assess its empirical validity using externally dated stress episodes, a controlled placebo experiment, and sensitivity analyses. Second, we introduce a decomposition of the ambiguity radius into a level and a modulation, enabling an empirical assessment of the economic value associated with its timing around a common anchor. Third, we develop a point-in-time walk-forward protocol combining monthly reconstructed universes, delisting treatment, transaction costs, static benchmarks, signal-ordering placebos, and family-wise multiplicity control.

The results provide a nuanced answer. The signal reacts around several stress episodes, discriminates among the degrees of misalignment introduced in a controlled experiment, and contains a substantial volatility-related component without being fully explained by volatility.

In the main application, the net Sharpe ratio increases from 0.6892 for the static portfolio to 0.7451 for the dynamic portfolio, a gain of 0.0559. This improvement is not established at the 5% level by either the HAC test of the mean return difference or the paired bootstrap of the Sharpe-ratio gain. The observed timing nevertheless outperforms the rearrangements considered, and the main cell remains significant after the extended family-wise adjustment. This conclusion remains conditional: modulation based on 36-month realized volatility delivers a comparable point estimate, while some transformations of the signal reverse its sign. The results therefore establish neither general dominance over equal weighting nor dominance across all specifications examined.

The remainder of the thesis is organized as follows. Section 2 reviews the literature on portfolio robustness, Wasserstein DRO, and heterogeneous market horizons. Section 3 presents the data, the point-in-time construction of the investable universes, the treatment of delistings, and the formation of multi-frequency returns. Section 4.1 introduces the Wasserstein DRO framework, while Sections 4.2 and 4.5 develop the misalignment signal and the radius modulation, respectively. Section 5 describes the empirical protocol, and Section 6 presents the signal and portfolio results. Finally, Sections 7, 8, and 9 discuss their scope, limitations, and main implications.

# 2 Literature Review

This thesis connects two questions that are generally studied separately: protecting portfolios against estimation error and characterizing the state of the market across multiple horizons. Optimal transport provides the link between them: a measure of disagreement among multi-frequency distributions is constructed and then used to modulate the radius of a Wasserstein DRO problem.

## 2.1 Estimation Error and Portfolio Robustness

The mean–variance program of Markowitz [5] relies on estimated expected returns and a covariance matrix. Optimization can amplify their errors and produce unstable weights, particularly when expected returns are noisy [2, 3]. This fragility helps explain why an optimized strategy does not necessarily outperform a simple rule such as equal weighting out of sample [6].

Robustness addresses this problem by representing uncertainty explicitly. Parametric formulations protect the decision against a set of possible input values [7]. Distributionally robust optimization instead considers an entire family of plausible probability laws. Ambiguity sets defined through moment restrictions provide one tractable construction [8], but they do not describe the geometry of how probability mass moves between two distributions.

## 2.2 Wasserstein DRO and Radius Selection

A Wasserstein ball is centered on a nominal distribution, often the empirical measure, and contains the probability laws reachable within a transportation budget. The robust problem then minimizes the loss under the least favorable law in that ball. Mohajerin Esfahani and Kuhn [4] establish out-of-sample guarantees and convex reformulations for a broad class of problems based on $W_1$. The duality extends to more general transportation costs, including the $W_2$ framework adopted here [9, 10].[i]

[i] A Wasserstein distance measures the minimum effort required to transform one distribution into another by moving its probability mass. $W_1$ is based on the average distance traveled, whereas $W_2$ is the square root of the average squared distance. The quadratic cost assigns greater weight to large displacements: before taking the square root, a displacement twice as long contributes four times as much to the cost.

In portfolio problems, the radius therefore controls the degree of caution imposed on the weights; Blanchet et al. [11] study this mechanism directly in a Wasserstein distributionally robust mean–variance framework. Selecting the radius nevertheless remains difficult: concentration bounds give it a statistical interpretation but become conservative in high dimensions, whereas procedures such as cross-validation provide a more operational calibration.

The radius need not be constant. For example, Pun et al. [12] condition the ambiguity set on latent regimes and observable covariates. Their approach shows that protection can evolve over time, but its dynamics arise from a regime model rather than from a directly observed disagreement among multiple temporal representations of the distribution.

## 2.3 Market Horizons and Research Positioning

The economic motivation for the signal comes from the heterogeneous market hypothesis. Müller et al. [1] describe participants with different decision horizons; shocks may therefore be incorporated differently across frequencies. The HAR model of Corsi [13] translates this intuition into a combination of daily, weekly, and monthly components of realized volatility. These studies motivate a multi-horizon analysis, but they primarily summarize each frequency through a moment.

We instead compare the empirical distributions themselves. Optimal transport provides the required geometry [14]; the Wasserstein barycenter serves as a consensus among distributions [15]. In high dimensions, the Sliced-Wasserstein distance replaces multidimensional transport with an average of one-dimensional transport problems computed after projection [16]. It also has more favorable statistical properties than the full Wasserstein distance [17].

The closest studies fall along three complementary branches. Blanchet et al. [11] incorporate Wasserstein ambiguity into a mean–variance allocation problem without constructing the timing of the radius from a multi-frequency state. Pun et al. [12] allow protection to evolve, but this dynamic depends on latent regimes and covariates. Finally, Müller et al. [1] and Corsi [13] motivate the coexistence of multiple horizons while representing them primarily through moments such as volatility rather than through complete empirical distributions.

Within the scope of this review, we therefore did not identify a framework that combines these three dimensions. This thesis constructs daily, weekly, and monthly distributions, aggregates them around a Wasserstein barycenter, and uses their Sliced-Wasserstein dispersion to modulate a separately calibrated level of robustness. The contribution lies neither in a new DRO duality result nor in a new transportation distance, but in combining these tools to isolate and test the value of ambiguity-radius timing. The resulting signal is therefore neither a confidence radius nor an estimator of the distance to the future probability law.

# 3 Data and Construction of the Investment Universes

This section presents the data and conventions used to construct the point-in-time investment universes. The source panel consists of daily observations from January 1, 1990, to December 31, 2025. Before the universes are formed, a point-in-time classification layer restricts eligibility to U.S. common stocks and to regular, active market observations. The first five years provide the required minimum 60-month history, while the out-of-sample evaluation spans January 1995 to December 2025.

The CBOE VIX is used solely as an auxiliary series in the signal non-redundancy analysis. The section successively describes the data sources, security filters, construction of the two universes, aligned multi-frequency matrices, and treatment of delistings.

## 3.1 Data Sources

Market data come from the CRSP US Stock Database, accessed through Wharton Research Data Services [18]. Consistent with the CRSP CIZ nomenclature, the universe is restricted point-in-time to U.S. common stocks ($\texttt{SecurityType} = \texttt{EQTY}$, $\texttt{SecuritySubType} = \texttt{COM}$, $\texttt{ShareType} = \texttt{NS}$, $\texttt{USIncFlg} = \texttt{Y}$, and $\texttt{IssuerType} \in \{\texttt{ACOR}, \texttt{CORP}\}$). In the CIZ format, this filter is the operational equivalent of the historical CRSP share codes 10 and 11.

This filtering excludes ADRs, REITs, listed funds, ETFs, and common stocks of non-U.S. issuers from the formation population. Eligibility at a given date also requires a regular market observation $\texttt{ConditionalType} = \texttt{RW}$, an active trading status $\texttt{TradingStatusFlg} = \texttt{A}$, and a primary listing on the NYSE, AMEX, or NASDAQ. These criteria are evaluated row by row using only the information available at each date, without retrospective reclassification.

The compact source file nevertheless retains the complete daily history of every $PERMNO$ that satisfied the definition of a U.S. common stock at least once. This convention preserves subsequent changes in status or exchange and the observations required to process delistings. Rows that become ineligible are retained to track a security already held, but cannot enter a new formation population until they again satisfy the eligibility criteria.

The panel includes daily returns, prices, volumes, market capitalizations, exchange codes, and delisting information. The return variable used throughout the pipeline is `DlyRet`. In the adjusted panel used as input to the analyses, this variable already incorporates, on the exit date, the observed or imputed delisting return according to the convention described in Section 3.6.

## 3.2 Auxiliary Series

The signal non-redundancy test (Section 5.1.5) uses two volatility controls. Implied volatility is measured by the CBOE VIX index, whose official historical series has been available since 1990 [19].[ii] Realized volatility is constructed from the equal-weighted daily return of the universe. The definitions specific to the monthly and 36-month horizons are provided in Section 5.1.5.

## 3.3 Construction of the Investable Universe

At each month-end $M$, the classification and tradability criteria defined in Section 3.1 are evaluated on the final trading day. To belong to the formation population, a security must be observed on that date, must not have been delisted before formation, and must have a non-missing price and strictly positive market capitalization. The NYSE population used to compute the breakpoints is the subset of these eligible common stocks for which `PrimaryExch` = `N`. Security categories excluded upstream therefore enter neither the breakpoint calculations nor the market-capitalization rankings.

The 90$^{\text{th}}$ percentile of their month-end market capitalization defines the threshold

$$BP_{p90,M}.$$

This NYSE breakpoint is then applied to all eligible securities listed on the NYSE, AMEX, or NASDAQ:

$$ME_{i,M} > BP_{p90,M}.$$

Using the NYSE as the reference population follows size conventions intended to limit the influence of small-cap stocks [20]. Our construction nevertheless differs by requiring a

[ii] The series is retrieved from Yahoo Finance, ticker `^VIX`, through `yfinance`, and aggregated monthly by taking the mean.

continuous 60-month history, monthly point-in-time reconstruction, and final selection of the 100 largest eligible stocks.

Among the securities above this threshold, the 100 largest by market capitalization are retained. The composition of the universe may therefore change each month, while its cross-sectional dimension remains constant.

A security is eligible only if it has $W_{\text{hist}} = 60$ consecutive calendar months with complete daily returns on every trading day. This eligibility basis is maintained throughout the sensitivity analyses: only the depth of the estimation window varies, not the historical-availability rule.

The construction is strictly point-in-time. Universe membership, market capitalization, and historical returns are fixed at the close of month $M$. Returns in month $M+1$ are used exclusively to evaluate the portfolio formed at that date and never enter its construction.

The eligibility criteria are not reapplied retrospectively during the $M+1$ holding month. Once selected at date $M$, a security is tracked under the same $PERMNO$ until the end of its holding period or until a potential delisting. This distinction between formation eligibility and holding-period tracking prevents future information from entering portfolio selection.

## 3.4 NYSE Small–Mid-Cap Universe

For the cross-universe robustness analysis (Section 6.2.8), an NYSE small–mid-cap universe is constructed using the same point-in-time conventions and the same requirement of 60 complete months as the main universe. It is restricted to securities listed on the NYSE at the formation date.

At each month-end $M$, the $20^{\text{th}}$ and $50^{\text{th}}$ percentiles of the NYSE market-capitalization distribution define the thresholds $BP_{p20,M}$ and $BP_{p50,M}$. A security is eligible if

$$BP_{p20,M} < ME_{i,M} \leq BP_{p50,M}.$$

Among the securities that satisfy both this condition and the historical-data requirement, the 100 largest by market capitalization are retained.

Restricting the universe to the NYSE ensures that the population used to compute the thresholds is also the population from which constituents are selected. The resulting universe represents the upper portion of the $P20$–$P50$ band rather than the entire band. Its constant cardinality allows the two universes to be compared without observed differences arising mechanically from different numbers of assets.

## 3.5 Construction of Multi-Frequency Returns

At each formation date $M$, a matrix of daily `DlyRet` returns is constructed for the ordered list of 100 selected securities. It covers the relevant estimation window and ends at the close of month $M$. Weekly and monthly observations are obtained exclusively from this daily matrix.

For security $i$ and calendar interval $\mathcal{I}_{k,t}$ associated with frequency $k$, the aggregated return is defined by geometric compounding:

$$r_{i,t}^{(k)} = \prod_{s \in \mathcal{I}_{k,t}} \left(1 + r_{i,s}^{(d)}\right) - 1,$$

where $r_{i,s}^{(d)}$ denotes the daily CRSP return. Weekly intervals correspond to weeks ending on Friday, and monthly intervals to month-ends. Intervals do not overlap within a given frequency.

The same ordered list of $PERMNO$ identifiers is maintained in the daily, weekly, and monthly matrices. No frequency-specific imputation or deletion is permitted: a window is admissible only if the daily matrix is complete for all 100 securities on every trading day in the estimation period. Geometric compounding then preserves this completeness at the aggregated frequencies.

The three frequencies are therefore dependent temporal representations of the same daily process rather than independent information sources. Their scaling and conversion into multivariate empirical measures are defined in Section 4.2.1.

## 3.6 Treatment of Delistings

The treatment of delisted securities is essential in long-horizon performance studies. Omitting their terminal returns can introduce an upward bias because exits associated with poor performance are generally accompanied by large losses [21].

Delisting information is matched to the daily panel using the $PERMNO$ identifier. The exit date is then used to identify the effective event row.

Observations after that date are removed. When CRSP provides a delisting return `DelRet`, it is retained. When it is missing, the treatment depends on the category constructed from `DelReasonType`.

Reasons `BKPY` and `FING` are classified as bankruptcy or financial distress, whereas `DELQ`, `LP`, and `INSC` are classified as non-compliance. For these two categories, the missing return is drawn from a normal distribution with a mean of $-30\%$ for NYSE and AMEX securities and $-55\%$ for other securities. The standard deviation and truncation bounds are set, respectively, to 15% and $[-95\%, -20\%]$ for the first category, and to 10% and $[-60\%, -5\%]$ for the second. Reasons associated with a merger, a change of exchange, or an unspecified category receive a zero delisting return when it is missing.

The adjustment is applied only on the effective delisting date. The daily return used throughout the remainder of the pipeline is then

$$r_{i,t}^{\text{adj}} = (1 + r_{i,t})\,(1 + \text{DelRet}_{i,t}) - 1.$$

This value replaces `DlyRet` on the exit day; the delisting return is therefore not applied a second time in the portfolio engine.

The stochastic imputations were performed when the adjusted daily panel was constructed, and this realization was then frozen before the universes and point-in-time matrices were formed. All calculations reported in the thesis therefore rely on the same adjusted panel.

## 3.7 Final Sample

The source panel spans January 1, 1990, to December 31, 2025. The requirement of 60 complete months yields a first formation date in December 1994. The sample therefore contains 373 formation dates, from December 1994 to December 2025, while the realized evaluation covers 372 months, from January 1995 to December 2025. The decision formed in December 2025 for January 2026 is not included in the reported performance. Both universes therefore contain exactly 100 eligible common stocks at every formation date.

# 4 Methodology

This section presents the methodological framework used to construct a robust allocation when the future return distribution is unknown. At each formation date, the empirical distribution estimated from historical observations is only an approximation of this future distribution. The DRO framework adopted here accounts for this uncertainty by considering a set of plausible distributions defined by a Wasserstein ball around the nominal empirical measure.

The radius of this ball determines the intensity of robustness. A small radius keeps the allocation close to the empirical optimizer and leaves it more exposed to estimation noise, whereas a large radius imposes stronger regularization. The proposed construction nevertheless distinguishes two dimensions: the reference level $\bar{\varepsilon}$, which anchors the degree of robustness, and its temporal modulation, which determines when robustness should be increased or reduced.

This modulation is based on misalignment among empirical return distributions constructed at daily, weekly, and monthly frequencies. These three frequencies represent short, intermediate, and longer horizons. When their distributions move farther apart, distributional uncertainty is regarded as greater and the radius expands relative to its reference level. When they are more closely aligned, the radius contracts.

A comparison based solely on a few empirical moments would discard information contained in the joint return distribution. Estimating a continuous multivariate density would also be fragile in high dimensions and finite samples, particularly at the monthly frequency. Optimal transport makes it possible to compare the discrete empirical measures directly in the joint asset space, without imposing a parametric form or estimating a continuous density.

The construction then proceeds in three stages. First, the frequency-specific distributions are placed on comparable volatility scales. A Wasserstein barycenter subsequently defines their multivariate consensus, after which Sliced-Wasserstein distances measure their dispersion around that consensus. The resulting signal, denoted $\widehat{\rho}_{\tau}^{\mathrm{hyb}}$, ultimately modulates the radius around $\bar{\varepsilon}$. The remainder of this section presents, in turn, the DRO problem, the construction of the multi-frequency signal, and the calibration of the reference level.

## 4.1 Wasserstein DRO Framework

At each rebalancing date $\tau$, the investor selects an allocation before observing future returns. Their distribution remains unknown and is approximated only through an empirical measure constructed from the historical information available at that date. Optimization conducted exclusively under this nominal measure may therefore produce an allocation that is particularly sensitive to estimation noise.

The Wasserstein DRO framework replaces this single distribution with a set of plausible probability laws contained in a Wasserstein ball around the nominal measure. The portfolio is selected with respect to the least favorable distribution in this set. The radius of the ball therefore controls the intensity of robustness and provides the link to the misalignment signal constructed in the second part of the methodology.

The robust formulation is based on a second-order Wasserstein ball. General strong-duality results for Wasserstein DRO problems are established by Gao and Kleywegt [9] and Blanchet and Murthy [10]. We use the $p = 2$ case associated with the quadratic cost $c(r, r') = \|r - r'\|_2^2$.

This subsection successively defines the scenario space, the set of admissible portfolios, the loss function, and the nominal distribution. It then introduces the ambiguity ball and the conic reformulation used to solve the allocation problem.

### 4.1.1 Return Scenario Space

Consider a universe of $N$ risky assets.[iii] At each rebalancing date $\tau$, a market scenario is represented by a return vector

$$r = \left(r^{(1)}, \ldots, r^{(N)}\right)^{\top} \in \mathbb{R}^N.$$

[iii] Throughout the thesis, $N$ denotes the number of assets—that is, the dimension of the scenario space, denoted by $m$ in the concentration bound of Section 4.6 ($m = N$). The number of historical observations available at date $\tau$ is denoted by $T_\tau$. This convention differs from part of the DRO literature, in which $N$ denotes the sample size; the distinction should be kept in mind when convergence rates involve both the dimension and the number of observations.

The space $\mathbb{R}^N$ is therefore the return scenario space: each point corresponds to a simultaneous realization of the returns on the $N$ assets. The distance between two scenarios $r$ and $r'$ is measured using the Euclidean norm:

$$d(r, r') = \|r - r'\|_2.$$

The pair $(\mathbb{R}^N, \|\cdot\|_2)$ thus defines the metric space in which return distributions are compared. Its geometry is multivariate: two scenarios are close when their return vectors are close in the joint space of the $N$ assets.

A probability distribution on $\mathbb{R}^N$ describes a possible law for the future return vector. We consider the space

$$\mathcal{P}_2(\mathbb{R}^N) = \left\{ \mu \in \mathcal{P}(\mathbb{R}^N) : \int_{\mathbb{R}^N} \|r\|_2^2 \, d\mu(r) < \infty \right\},$$

that is, the set of distributions with a finite second moment. This condition ensures that the quadratic Wasserstein distances used below are well defined.

### 4.1.2 Admissible Portfolios and Loss Function

A portfolio allocation is represented by a weight vector

$$w = (w_1, \dots, w_N)^\top \in \mathbb{R}^N.$$

The set of admissible portfolios is denoted by $\mathcal{W}$. For example, in a fully invested long-only setting, one may define

$$\mathcal{W} = \left\{ w \in \mathbb{R}^N : \mathbf{1}^\top w = 1, \quad w_i \geq 0, \ i = 1, \dots, N \right\}.$$

Other constraints may be added depending on the empirical application, including individual weight limits, leverage constraints, sector bounds, or turnover constraints.

This section retains the general notation $\mathcal{W}$ to separate the DRO framework from the exact empirical specification of the portfolio constraints.

For return scenario $r$, the portfolio return is

$$R_p(w, r) = w^\top r.$$

The allocation decision therefore consists of choosing $w$ before observing the future realization of $r$. To express this problem as a minimization, we introduce a loss function associated with the portfolio and the realized scenario.

The problem is formulated in terms of a loss function

$$\ell(w, r),$$

which measures the adverse outcome associated with portfolio $w$ when return scenario $r$ occurs. Working with a loss function makes it possible to write the problem as a minimization: the larger $\ell(w, r)$, the less favorable the scenario is for the portfolio.

This thesis adopts the linear loss equal to the negative portfolio return:

$$\ell(w, r) = -w^\top r.$$

Minimizing its expectation is equivalent to maximizing expected return. This choice is deliberate and serves a methodological identification objective. The primary purpose of the analysis is not to compare multiple objective functions, but to isolate the transmission of a time-varying ambiguity radius into portfolio weights.

With this loss, Wasserstein robustness produces an explicit penalty on weight concentration, $\varepsilon_\tau \|w\|_2$, as shown in Equation (6). Holding the data, constraints, and average level of robustness fixed, changes in the allocation can therefore be linked directly to the timing of $\varepsilon_\tau$, without simultaneously introducing covariance dynamics, a risk-aversion parameter, or a preference for tail risk. The linear loss thus provides a controlled environment in which the modulation mechanism can be identified and interpreted transparently.

This choice does not imply that the linear loss is the most general or necessarily the best-performing economic objective. Mean–variance, quadratic, shortfall, or CVaR-type losses would provide additional channels for distributional information. Volatility, drawdown, and tail risk are evaluated here as out-of-sample outcomes but are not optimized directly.

### 4.1.3 Nominal Distribution and Empirical Optimization

At rebalancing date $\tau$, the investor does not know the true future return distribution. Only a set of historical observations collected over an estimation window is available. These observations define a nominal empirical distribution, denoted by

$$\widehat{\mu}_\tau^{\text{nom}}.$$

In the discrete case, this distribution is

$$\widehat{\mu}_\tau^{\text{nom}} = \frac{1}{T_\tau} \sum_{i=1}^{T_\tau} \delta_{r_{i,\tau}},$$

where $r_{i,\tau} \in \mathbb{R}^N$ is the $i$-th observed return vector in the window available at date $\tau$. Each historical observation is thus interpreted as an empirical market scenario assigned uniform mass $1/T_\tau$.

A conventional empirical approach selects the portfolio that minimizes average loss under this nominal distribution:

$$w_\tau^{(\text{emp})} \in \arg\min_{w \in \mathcal{W}} \mathbb{E}_{\widehat{\mu}_\tau^{\text{nom}}} \left[\ell(w, r)\right].$$

Because $\widehat{\mu}_\tau^{\text{nom}}$ is a discrete measure, this expectation reduces to an empirical average:

$$\mathbb{E}_{\widehat{\mu}_\tau^{\text{nom}}} \left[\ell(w, r)\right] = \frac{1}{T_\tau} \sum_{i=1}^{T_\tau} \ell(w, r_{i,\tau}).$$

This formulation selects the portfolio that minimizes average loss across the observed historical scenarios. It nevertheless assumes that the empirical measure provides a sufficiently reliable approximation to the future distribution. In finite samples, estimation noise, extreme observations, and regime changes may make the resulting allocation sensitive to the particular realization of the historical window. The DRO framework addresses this fragility by considering a neighborhood of plausible distributions around the nominal measure.

### 4.1.4 Wasserstein Distance and Ambiguity Ball

This neighborhood is defined using the Wasserstein distance, which is based on a transportation geometry: it measures the minimum cost required to move the mass of one distribution into another within the scenario space.

In our framework, scenarios are return vectors $r \in \mathbb{R}^N$. This geometry therefore accounts for both the probabilities assigned to scenarios and their locations in the joint asset space. In particular, it can reflect shifts in the mean, dispersion, or co-movements among assets without reducing the distributions to a small set of empirical moments.

Formally, for two distributions $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^N)$, the squared second-order Wasserstein distance is defined as follows [14]:

$$W_2^2(\mu, \nu) = \inf_{\pi \in \Pi(\mu,\nu)} \int_{\mathbb{R}^N \times \mathbb{R}^N} \|x - y\|_2^2 \, d\pi(x, y), \tag{1}$$

where $\Pi(\mu, \nu)$ denotes the set of admissible transport plans between $\mu$ and $\nu$. A plan $\pi \in \Pi(\mu, \nu)$ describes a reallocation of probability mass from $\mu$ to $\nu$, subject to preserving the masses specified by the two distributions. A detailed illustration of the discrete case, using two empirical measures of different sizes, is provided in Appendix A.1.

The Wasserstein distance is then

$$W_2(\mu, \nu) = \sqrt{W_2^2(\mu, \nu)}.$$

Accordingly, the constraint

$$W_2\left(\mathbb{Q}, \widehat{\mu}_\tau^{\text{nom}}\right) \leq \varepsilon_\tau$$

can equivalently be written as

$$W_2^2\left(\mathbb{Q}, \widehat{\mu}_\tau^{\text{nom}}\right) \leq \varepsilon_\tau^2.$$

Using the distance defined in Equation (1), the ambiguity set around the nominal distribution is

$$\mathcal{B}_{\varepsilon_\tau}\left(\widehat{\mu}_\tau^{\text{nom}}\right) = \left\{\mathbb{Q} \in \mathcal{P}_2(\mathbb{R}^N) : W_2\left(\mathbb{Q}, \widehat{\mu}_\tau^{\text{nom}}\right) \leq \varepsilon_\tau\right\}. \tag{2}$$

The ball $\mathcal{B}_{\varepsilon_\tau}(\widehat{\mu}_\tau^{\text{nom}})$ contains all distributions $\mathbb{Q}$ whose Wasserstein distance from the nominal measure is no greater than $\varepsilon_\tau$. The nominal measure is its center, while the radius determines the extent of the admissible distributional deformations.

When the nominal measure is empirical, historical observations serve as anchor points around which the future distribution may deform subject to a transportation-cost constraint. The nominal measure therefore summarizes the information available at date $\tau$, while the ambiguity ball acknowledges the uncertainty associated with its estimation. The DRO program defined next selects the allocation that minimizes its loss under the least favorable plausible distribution in this ball.

### 4.1.5 Primal DRO Program

Once the ambiguity set has been defined, the allocation problem can be formulated in a distributionally robust framework. At rebalancing date $\tau$, the investor no longer seeks only to minimize average loss under the nominal distribution $\widehat{\mu}_\tau^{\text{nom}}$. Instead, the investor seeks a portfolio whose performance remains robust across all plausible distributions in the Wasserstein ball $\mathcal{B}_{\varepsilon_\tau}(\widehat{\mu}_\tau^{\text{nom}})$ defined in Equation (2).

For a given portfolio $w$, its robust loss is defined as the largest expected loss across the admissible distributions:

$$\Phi_\tau(w;\varepsilon_\tau) = \sup_{\mathbb{Q}\in\mathcal{B}_{\varepsilon_\tau}(\widehat{\mu}_\tau^{\text{nom}})} \mathbb{E}_{\mathbb{Q}}\left[\ell(w,r)\right]. \tag{3}$$

This quantity provides a pessimistic assessment of the portfolio. It asks not only how the portfolio performs under the observed empirical distribution, but also how it would perform under the least favorable distribution among those that remain sufficiently close to that empirical distribution.

The primal DRO problem then selects the portfolio that minimizes this robust loss:

$$w_\tau^{(\text{DRO})} \in \arg\min_{w\in\mathcal{W}} \Phi_\tau(w;\varepsilon_\tau).$$

Substituting the definition of $\Phi_\tau(w;\varepsilon_\tau)$ yields the min–max form of the problem:

$$w_\tau^{(\text{DRO})} \in \arg\min_{w\in\mathcal{W}} \sup_{\mathbb{Q}\in\mathcal{B}_{\varepsilon_\tau}(\widehat{\mu}_\tau^{\text{nom}})} \mathbb{E}_{\mathbb{Q}}\left[\ell(w,r)\right]. \tag{4}$$

This formulation has two levels. The inner problem,

$$\sup_{\mathbb{Q}\in\mathcal{B}_{\varepsilon_\tau}(\widehat{\mu}_\tau^{\mathrm{nom}})} \mathbb{E}_{\mathbb{Q}}\left[\ell(w,r)\right],$$

seeks, for a given portfolio $w$, the least favorable distribution within the ambiguity ball. This adversarial distribution maximizes the portfolio's average loss while remaining at a Wasserstein distance no greater than $\varepsilon_\tau$ from the nominal distribution.

The outer problem,

$$\min_{w\in\mathcal{W}},$$

then selects the portfolio that minimizes this adverse loss. The robust portfolio is therefore not the one that is optimal only for the observed historical scenarios, but the one that limits loss under the plausible distributional worst case.

The interaction between the adversarial distribution in the inner problem and the allocation decision in the outer problem thus defines the min–max structure of the DRO program.

### 4.1.6 Role of the Ambiguity Radius

The radius $\varepsilon_\tau$ controls the size of the Wasserstein ambiguity ball. It therefore determines the range of alternative distributions included in the robust problem. More precisely, the radius sets the maximum perturbation allowed around the nominal distribution $\widehat{\mu}_\tau^{\mathrm{nom}}$.

When

$$\varepsilon_\tau = 0,$$

the ambiguity ball reduces to the nominal distribution:

$$\mathcal{B}_0\left(\widehat{\mu}_\tau^{\mathrm{nom}}\right) = \left\{\widehat{\mu}_\tau^{\mathrm{nom}}\right\}.$$

The DRO problem then coincides with the standard empirical problem:

$$\min_{w\in\mathcal{W}} \mathbb{E}_{\widehat{\mu}_\tau^{\mathrm{nom}}}\left[\ell(w,r)\right].$$

In the long-only linear specification adopted here, the radius governs a continuum between two poles. When $\varepsilon_\tau = 0$, the portfolio places full confidence in the estimated expected returns and remains exposed to their fragility. For $\varepsilon_\tau > 0$, the ball permits alternative distributions around the nominal measure: as the radius grows, the distributional adversary can move farther away and the allocation becomes more strongly regularized. In the limit as $\varepsilon_\tau \to \infty$, the robustness penalty dominates the return term and the portfolio converges to equal weighting, $\frac{1}{N}\mathbf{1}$. Figure 6.14 illustrates this continuum empirically.

Calibrating the radius therefore trades off the exploitation of empirical information against caution. A radius that is too small leaves the allocation sensitive to estimation noise, whereas an excessive radius pushes it toward an agnostic allocation at the cost of a potential loss in expected performance. This interpretation also clarifies the degeneracy of the theoretical bound in high dimensions examined in Section 4.6: a mechanically large radius is not neutral because it pushes the allocation toward equal weighting.

In a conventional approach, the radius may be fixed exogenously, through validation, or from a statistical rule. The proposed construction instead separates its reference level from its temporal modulation: the former sets the average intensity of robustness, while the latter evolves with the distributional uncertainty observed at each date. This decomposition is formalized in Section 4.5. The dual representation introduced next makes the primal program computationally tractable; applying it to the linear loss then yields the conic formulation actually used to construct the portfolios.

### 4.1.7 Dual Representation

The primal DRO program is conceptually clear but is not directly tractable in this form. For a given portfolio $w$, the inner problem is

$$\sup_{\mathbb{Q} \in \mathcal{B}_{\varepsilon_\tau}(\widehat{\mu}_\tau^{\mathrm{nom}})} \mathbb{E}_{\mathbb{Q}} \left[ \ell(w, r) \right].$$

This problem seeks the least favorable distribution among all distributions in the Wasserstein ball. The difficulty is that the optimization is not over a finite number of variables, but over a probability measure $\mathbb{Q}$. Even when the nominal distribution $\widehat{\mu}_\tau^{\mathrm{nom}}$ is empirical and therefore discrete, the adversarial distribution $\mathbb{Q}$ need not be restricted to the observed historical scenarios. It may move mass through the continuous scenario space $\mathbb{R}^N$, subject only to remaining within Wasserstein distance $\varepsilon_\tau$ of the nominal distribution.

The inner problem is therefore an optimization problem over a set of distributions rather than simply over a finite-dimensional parameter vector. This formulation is useful for the economic interpretation of DRO because it explicitly describes the search for an unfavorable plausible distribution. It is not, however, directly suited to numerical solution. The dual representation transforms this distributional problem into a more tractable form.

To understand this transformation, start from the ambiguity ball $\mathcal{B}_{\varepsilon_\tau}(\widehat{\mu}_\tau^{\mathrm{nom}})$ defined in Equation (2). Its constraint can be expressed in squared form as

$$W_2^2\left(\mathbb{Q}, \widehat{\mu}_\tau^{\mathrm{nom}}\right) \leq \varepsilon_\tau^2.$$

The inner problem then becomes

$$\sup_{\mathbb{Q} \in \mathcal{P}_2(\mathbb{R}^N)} \left\{ \mathbb{E}_{\mathbb{Q}}[\ell(w, r)] : W_2^2\left(\mathbb{Q}, \widehat{\mu}_\tau^{\mathrm{nom}}\right) \leq \varepsilon_\tau^2 \right\}.$$

This expression reveals a transportation-budget constraint: the adversarial distribution may increase the portfolio loss but cannot move arbitrarily far from the nominal distribution.

To dualize this constraint, introduce a Lagrange multiplier $\eta \geq 0$, which penalizes transportation cost relative to the nominal measure [9].

Formally, this yields an expression of the form

$$\inf_{\eta \geq 0} \left\{ \eta \varepsilon_\tau^2 + \sup_{\mathbb{Q} \in \mathcal{P}_2(\mathbb{R}^N)} \left[ \mathbb{E}_{\mathbb{Q}}[\ell(w, r)] - \eta W_2^2\left(\mathbb{Q}, \widehat{\mu}_\tau^{\mathrm{nom}}\right) \right] \right\}.$$

This step has a simple interpretation. The term $\mathbb{E}_{\mathbb{Q}}[\ell(w, r)]$ induces the adversarial distribution to increase the portfolio loss. The term

$$-\eta W_2^2\left(\mathbb{Q}, \widehat{\mu}_\tau^{\mathrm{nom}}\right)$$

penalizes distributions that move too far from the nominal distribution. The parameter $\eta$ therefore determines the implicit price of moving mass through the return space.

For the empirical measure $\widehat{\mu}_\tau^{\mathrm{nom}}$ defined in Section 4.1.3, the strong duality established by Gao and Kleywegt [9], applied to the $W_2$ distance and the linear loss adopted here, gives the following representation:

$$\sup_{\mathbb{Q}:W_2(\mathbb{Q},\widehat{\mu}_\tau^{\mathrm{nom}})\le\varepsilon_\tau} \mathbb{E}_{\mathbb{Q}}[\ell(w,r)] = \inf_{\eta\ge 0}\left\{\eta\varepsilon_\tau^2 + \frac{1}{T_\tau}\sum_{i=1}^{T_\tau}\sup_{r\in\mathbb{R}^N}\left[\ell(w,r) - \eta\|r - r_{i,\tau}\|_2^2\right]\right\}. \tag{5}$$

This formula is central. It replaces an optimization over all distributions $\mathbb{Q}$ with an optimization over a scalar $\eta$, combined with perturbation problems around each historical scenario $r_{i,\tau}$.

To interpret this expression, consider the inner term

$$\sup_{r\in\mathbb{R}^N}\left[\ell(w,r) - \eta\|r - r_{i,\tau}\|_2^2\right].$$

For each historical observation $r_{i,\tau}$, the adversary seeks a perturbed scenario $r$ that increases the portfolio loss. The farther this perturbed scenario moves from the historical observation, however, the larger the quadratic penalty

$$\eta\|r - r_{i,\tau}\|_2^2$$

becomes. The adversary therefore cannot freely select any extreme scenario; it trades off the increase in loss against the displacement cost in the return space.

The average over $i$ aggregates these local perturbations, while $\eta\varepsilon_\tau^2$ prices the overall transportation budget associated with the radius. The dual representation therefore replaces optimization over the probability law $\mathbb{Q}$ with scalar minimization in $\eta$ and a collection of local problems around the historical scenarios. Under the linear loss adopted here, these problems admit the closed-form solution derived below.

#### 4.1.8 Conic Formulation and Numerical Solution

The dual representation derived above transforms the DRO problem, initially formulated as an optimization over a set of distributions, into a more computationally tractable problem [4]. This step is essential in a rolling setting because the allocation problem must be solved repeatedly at each rebalancing date $\tau$.

Under the specification adopted in Section 4.1.2, the loss is linear:

$$\ell(w,r) = -w^\top r.$$

Substituting it into the dual representation gives the following robust loss:

$$\Phi_\tau(w;\varepsilon_\tau) = \inf_{\eta\geq 0}\left\{\eta\varepsilon_\tau^2 + \frac{1}{T_\tau}\sum_{i=1}^{T_\tau}\sup_{r\in\mathbb{R}^N}\left[-w^\top r - \eta\|r - r_{i,\tau}\|_2^2\right]\right\}.$$

The inner term corresponds to the worst possible perturbation of historical scenario $r_{i,\tau}$, penalized by its quadratic displacement cost. For $\eta > 0$,

$$\sup_{r\in\mathbb{R}^N}\left[-w^\top r - \eta\|r - r_{i,\tau}\|_2^2\right] = -w^\top r_{i,\tau} + \frac{\|w\|_2^2}{4\eta}.$$

Indeed, the optimal adversarial scenario is obtained by moving the historical observation in the direction unfavorable to the portfolio:

$$r_i^\star = r_{i,\tau} - \frac{w}{2\eta}.$$

Substituting this expression into the dual representation gives

$$\Phi_\tau(w;\varepsilon_\tau) = \inf_{\eta>0}\left\{\eta\varepsilon_\tau^2 - w^\top\widehat{\boldsymbol{\mu}}_{r,\tau} + \frac{\|w\|_2^2}{4\eta}\right\},$$

where

$$\widehat{\boldsymbol{\mu}}_{r,\tau} = \frac{1}{T_\tau}\sum_{i=1}^{T_\tau} r_{i,\tau}$$

denotes the vector of empirical mean returns.

The term that depends on $\eta$ is

$$h_w(\eta) = \eta\varepsilon_\tau^2 + \frac{\|w\|_2^2}{4\eta}, \qquad \eta > 0.$$

For $\varepsilon_\tau > 0$, its derivative is

$$h_w'(\eta) = \varepsilon_\tau^2 - \frac{\|w\|_2^2}{4\eta^2},$$

and its second derivative satisfies

$$h_w''(\eta) = \frac{\|w\|_2^2}{2\eta^3} > 0.$$

The function is therefore strictly convex. Its first-order condition gives

$$\eta^\star(w,\varepsilon_\tau) = \frac{\|w\|_2}{2\varepsilon_\tau}.$$

Substituting $\eta^\star$ separately into the two terms of $h_w(\eta)$ gives

$$\eta^\star \varepsilon_\tau^2 = \frac{\|w\|_2^2}{4\eta^\star} = \frac{\varepsilon_\tau \|w\|_2}{2}.$$

The two $\eta$-dependent components are therefore equal, and their sum is

$$\min_{\eta>0} h_w(\eta) = \varepsilon_\tau \|w\|_2.$$

When $\varepsilon_\tau = 0$, the robust value reduces continuously to the empirical loss. The robust value therefore has the closed form

$$\Phi_\tau(w; \varepsilon_\tau) = -w^\top \widehat{\boldsymbol{\mu}}_{r,\tau} + \varepsilon_\tau \|w\|_2. \tag{6}$$

This closed form is the starting point for the conic reformulation below.

The robust allocation problem becomes

$$\min_{w \in \mathcal{W}} \left\{ -w^\top \widehat{\boldsymbol{\mu}}_{r,\tau} + \varepsilon_\tau \|w\|_2 \right\}.$$

This expression highlights two components. The first term is the empirical objective. Explicit optimization over the adversarial distribution has been eliminated by duality: under the linear loss and quadratic transportation cost adopted here, its effect is entirely summarized by the penalty $\varepsilon_\tau \|w\|_2$. The larger the ambiguity radius $\varepsilon_\tau$, the more strongly the solution is induced to limit its sensitivity to adversarial perturbations of the nominal distribution.

The problem is convex because its objective is the sum of a linear term and a Euclidean norm. To obtain a formulation compatible with a conic solver, introduce an auxiliary variable $t \geq 0$ such that

$$\|w\|_2 \leq t.$$

The objective can then be written linearly in $(w, t)$:

$$-w^\top \widehat{\boldsymbol{\mu}}_{r,\tau} + \varepsilon_\tau t.$$

In the fully invested long-only case, the robust problem can therefore be reformulated as the following second-order cone program (SOCP):

$$
\begin{aligned}
\min_{w\in\mathbb{R}^N,\, t\in\mathbb{R}} \quad & -w^\top \widehat{\boldsymbol{\mu}}_{r,\tau} + \varepsilon_\tau t \\
\text{s.t.} \quad & \|w\|_2 \le t, \\
& \mathbf{1}^\top w = 1, \\
& w_i \ge 0, \qquad i = 1, \dots, N, \\
& t \ge 0.
\end{aligned} \tag{7}
$$

The first constraint is a second-order cone constraint. The remaining constraints define the admissible portfolio set: full investment and long-only positions. The variable $t$ has no independent economic interpretation; it represents the Wasserstein penalty in a numerically tractable conic form.

This SOCP, with variables $(w, t) \in \mathbb{R}^N \times \mathbb{R}$, is solved using MOSEK [22]. Its geometry and derivation from the primal DRO problem are presented in Appendix B, while the construction of its inputs and its numerical solution are described in the empirical protocol in Section 5.2.3.

The DRO program therefore establishes the mapping from the radius $\varepsilon_\tau$ to a robust allocation. The following sections construct its temporal modulation from cross-horizon misalignment, while Section 4.6 determines its reference level. The signal thus governs the relative timing of the radius without simultaneously setting its absolute scale.

## 4.2 Construction of the Distributional Misalignment Signal

### 4.2.1 Scaling and Construction of the Empirical Measures

The construction of the multi-frequency dispersion signal relies on comparing empirical distributions of returns observed at different horizons, whose construction is described in Section 3.5. An immediate difficulty arises, however: daily, weekly, and monthly returns are not naturally expressed on the same scale. A return computed over a longer horizon aggregates more price variation and therefore mechanically exhibits a larger magnitude.

This scale difference is problematic in our setting because Wasserstein distances are sensitive to the geometry of the distributions being compared. Without prior normalization, a substantial portion of the distance between the distributions could simply reflect the mechanical effect of the observation horizon rather than a genuine distributional misalignment across frequencies.

The issue is therefore not merely to normalize returns, but to adopt a correction consistent with the object to be measured. Several conventions are possible: a deterministic horizon-dependent factor, volatility estimated separately at each frequency, or a more general form $h_k^H$, where $H$ denotes a scaling exponent.

In this thesis, we adopt the deterministic normalization by $\sqrt{h_k}$. Because weekly and monthly returns are constructed through geometric compounding, a simple return aggregated over $h$ trading days is given by

$$R^{(h)} = \prod_{s=1}^{h}(1 + r_s) - 1.$$

To make the horizons comparable, each return is subsequently divided by $\sqrt{h_k}$. This transformation constitutes a deterministic scaling convention consistent with the usual temporal aggregation rule for volatility. It does not, however, assume that this relationship is an exact identity for realized returns or a structural law governing their dynamics.

Because it depends only on the horizon, this convention does not absorb the volatility regimes present within the estimation window. Residual differences across frequencies therefore remain in the distributions being compared and are precisely part of the misalignment that the signal is intended to measure.

Normalization by a rolling volatility estimate $\widehat{\sigma}_{k,\tau}$ would simultaneously absorb the horizon effect and part of the frequency-specific volatility regimes. It could therefore attenuate the cross-frequency information of interest. This possibility is nevertheless retained as a robustness specification, alongside the exponents

$$h_k^H, \qquad H \in \{0.4,\, 0.5,\, 0.6\}.$$

These variants are defined in Section 5.1.4, and their results are presented in Section 6.1.4. They assess the sensitivity of the conclusions to the scaling convention; they are not intended to estimate a structural exponent.

The normalization thus places the frequencies on a more comparable volatility scale without imposing equality of their distributions. The signal remains sensitive to residual differences in location, dispersion, shape, and tail thickness, particularly when a short-horizon frequency incorporates a shock more rapidly than longer horizons.

Let $h_k$ denote the approximate number of trading days associated with frequency $k$. In the main application, the selected frequencies are daily, weekly, and monthly, so that

$$\mathcal{K} = \{d, w, m\}, \qquad h_d = 1, \qquad h_w = 5, \qquad h_m = 21.$$

For each frequency $k \in \mathcal{K}$, returns are therefore placed on a comparable scale according to

$$\widetilde{r}_{k,i,\tau} = \frac{r_{k,i,\tau}}{\sqrt{h_k}}, \tag{8}$$

where $r_{k,i,\tau} \in \mathbb{R}^N$ denotes the vector of returns on the $N$ assets observed at the $i$th date of frequency $k$, within the rolling window ending at rebalancing date $\tau$.

At each formation date $\tau$, all three frequencies are constructed over the same ordered universe of $N$ assets. A given coordinate therefore identifies the same security in the daily, weekly, and monthly matrices. The corresponding empirical measures thus belong to the same joint space $\mathbb{R}^N$, a necessary condition for their geometric comparison.

This transformation multiplies every component of the return vector by a frequency-specific scalar. It therefore changes the scale of the observations, but neither rearranges the assets nor removes the cross-sectional information contained in each multivariate observation. The objective is not to make the daily, weekly, and monthly distributions identical, but to place them on a comparable scale before comparing their geometry.

Figure 4.1 illustrates this effect in a univariate setting using the empirical return distributions of Apple (AAPL). This illustration does not alter the multivariate nature of the computation, which remains conducted in the joint asset space $\mathbb{R}^N$.

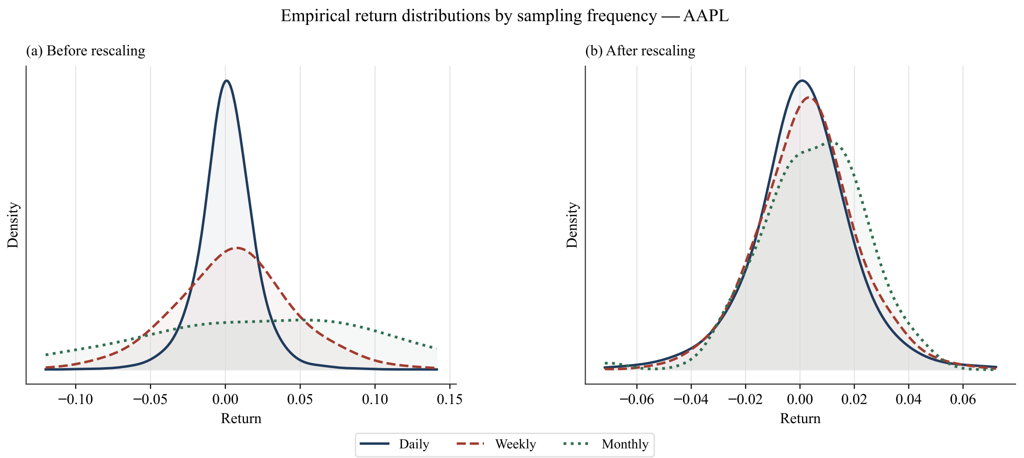


**Figure 4.1:** Effect of scaling multi-frequency returns. Panel (a) displays the empirical distributions before normalization: weekly and monthly returns mechanically exhibit greater dispersion than daily returns. Panel (b) displays the distributions after division by $\sqrt{h_k}$, which places the frequencies on a more comparable volatility scale without imposing equality of their distributions. The illustration is based on Apple (AAPL) returns observed over the 2008–2024 period.

At each rebalancing date $\tau$, the normalized observations at frequency $k$ are collected in the rolling matrix

$$\widetilde{X}_{k,\tau} = \begin{pmatrix} \widetilde{r}_{k,1,\tau}^{\top} \\ \widetilde{r}_{k,2,\tau}^{\top} \\ \vdots \\ \widetilde{r}_{k,T_{k,\tau},\tau}^{\top} \end{pmatrix} \in \mathbb{R}^{T_{k,\tau} \times N}.$$

Each row $\widetilde{r}_{k,i,\tau} \in \mathbb{R}^N$ is a multivariate observation containing the simultaneous normalized returns on the $N$ assets at a given date of frequency $k$. The quantity $T_{k,\tau}$ denotes the number of observations available at that frequency within the rolling window under consideration.

The matrix $\widetilde{X}_{k,\tau}$ is then interpreted as a discrete empirical measure on $\mathbb{R}^N$:

$$\widehat{\mu}_{k,\tau} = \frac{1}{T_{k,\tau}} \sum_{i=1}^{T_{k,\tau}} \delta_{\widetilde{r}_{k,i,\tau}}. \tag{9}$$

Figure 4.2 illustrates this conversion: each row of the rolling window becomes an atom of the empirical measure and receives uniform mass $1/T_{k,\tau}$.

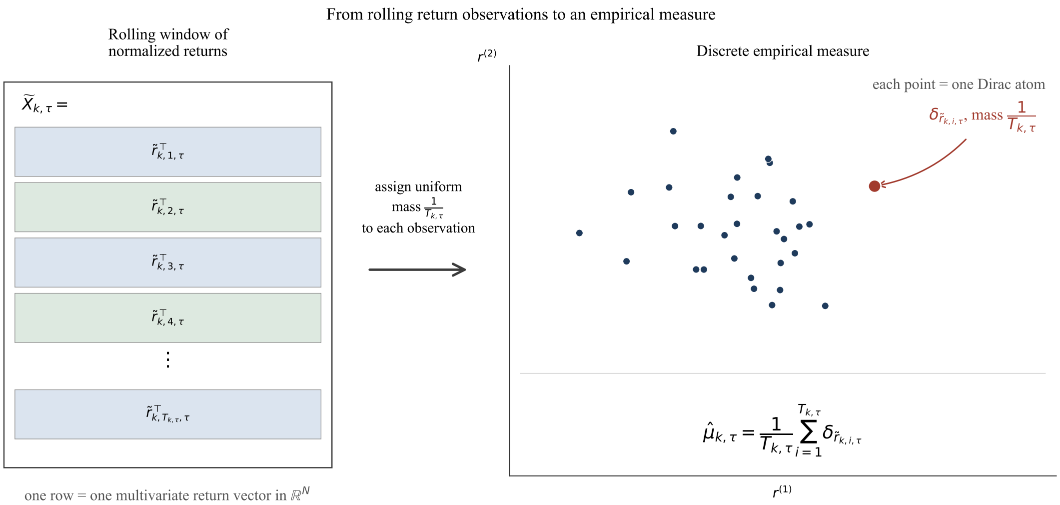


**Figure 4.2:** Construction of a discrete empirical measure from a rolling window of normalized returns. Each row of the matrix $\widetilde{X}_{k,\tau}$ corresponds to a multivariate observation and becomes a Dirac atom with uniform mass $1/T_{k,\tau}$ in the empirical measure $\widehat{\mu}_{k,\tau}$. The right-hand panel provides a schematic representation along two coordinates of the space $\mathbb{R}^N$.

Here, $\delta_{\widetilde{r}_{k,i,\tau}}$ denotes the Dirac measure at $\widetilde{r}_{k,i,\tau}$, that is, a unit mass concentrated on this multivariate observation. The uniform weight $1/T_{k,\tau}$ defines the empirical measure associated with the observations in the rolling window.

Thus, for each observation frequency, the window of normalized returns is converted into an empirical distribution on the joint asset space. These measures

$$\left(\widehat{\mu}_{k,\tau}\right)_{k\in\mathcal{K}}$$

are aggregated into a consensus distribution through the Wasserstein barycenter presented in Section 4.3; their dispersion around this consensus is subsequently measured using the Sliced-Wasserstein distances defined in Section 4.4.

## 4.3 Wasserstein Barycenter: Multi-Frequency Consensus

At each rebalancing date $\tau$, the construction described in Section 4.2.1 yields three normalized empirical measures

$$(\widehat{\mu}_{k,\tau})_{k\in\mathcal{K}}, \qquad \mathcal{K} = \{d, w, m\},$$

defined on the same joint space $\mathbb{R}^N$. Their supports need not have the same cardinality because the number of observations $T_{k,\tau}$ depends on the frequency.

The objective is to construct, from these three measures, a consensus distribution that represents their geometric center while preserving the multivariate information contained in the return vectors. This consensus will subsequently serve as a common reference for measuring the misalignment of each frequency.

### 4.3.1 Barycenter as a Fréchet Mean

The Wasserstein barycenter is defined as the distribution that minimizes the weighted sum of squared distances to the empirical measures [15]. Formally,

$$\widehat{\mu}_{\tau}^{*,W_2} \in \arg \min_{\nu\in\mathcal{P}_2(\mathbb{R}^N)} \sum_{k\in\mathcal{K}} \alpha_k W_2^2\left(\nu, \widehat{\mu}_{k,\tau}\right), \tag{10}$$

where $\mathcal{P}_2(\mathbb{R}^N)$ denotes the set of probability measures on $\mathbb{R}^N$ with a finite second moment, and

$$\alpha_k \geq 0, \qquad \sum_{k\in\mathcal{K}} \alpha_k = 1.$$

This definition generalizes the notion of a mean. In a Euclidean space, the mean of a set of points is the point that minimizes the sum of squared distances to those points. Here, the objects to be averaged are not points but empirical distributions. The Wasserstein barycenter can therefore be interpreted as a Fréchet mean in the space of probability measures [15, 23].

### 4.3.2 Weighting of the Barycentric Consensus

In the free-support barycenter construction, the three frequencies receive equal importance:

$$\alpha_d = \alpha_w = \alpha_m = \frac{1}{3}.$$

This choice constructs a consensus across three observation horizons rather than across their respective numbers of observations. In particular, it prevents the daily frequency from mechanically dominating the barycenter solely because its empirical support is denser.

Weights proportional to $T_{k,\tau}$ and $\log T_{k,\tau}$ concern only the aggregation of distances around this consensus. They are introduced when the signal is defined in Section 4.4.5 and are then evaluated under the sensitivity protocol defined in Section 5.1.4. The corresponding results are presented in Section 6.1.4.

### 4.3.3 Supports of Different Sizes and Optimal Transport

The empirical measures associated with the different frequencies need not have the same number of atoms. This creates no difficulty in the Wasserstein framework because their comparison relies on a transport plan rather than on a one-to-one correspondence between observations.

For a barycenter containing $M$ atoms and a frequency-specific measure containing $T_{k,\tau}$ atoms, the transport plan is an $M \times T_{k,\tau}$ matrix. The choice of $M$ therefore controls the numerical resolution of the barycenter support without requiring its cardinality to coincide with that of any particular frequency.

The mass constraints and the transport mechanism between supports of different sizes are presented in Appendix A.

### 4.3.4 Free-Support Representation of the Barycenter

In the implementation, the barycenter is represented by the discrete measure

$$\widehat{\mu}_\tau^{*,W_2} = \sum_{j=1}^{M} b_j \delta_{y_{j,\tau}}, \qquad y_{j,\tau} \in \mathbb{R}^N. \tag{11}$$

The points

$$y_{1,\tau}, \ldots, y_{M,\tau}$$

are the barycenter atoms. Each atom is a complete return vector in $\mathbb{R}^N$. The weights satisfy

$$b_j \geq 0, \qquad \sum_{j=1}^{M} b_j = 1.$$

The term "free support" means that the positions $y_{j,\tau}$ are neither imposed on a grid nor constrained to coincide with historical observations. They are optimized directly in the multivariate return space. The barycenter therefore does not simply select a subset of past observations: it constructs new representative atoms located at the center of the empirical distributions in the optimal-transport sense.

### 4.3.5 Iterative Computation of the Barycenter

The free-support barycenter is computed using an alternating procedure [24]. Starting from a current support $Y_\tau^{(q)}$, the algorithm first computes an optimal transport plan between the current barycenter and each of the three empirical measures. The positions of the barycenter atoms are then updated from the transported masses:

$$Y_\tau^{(q)} \longrightarrow \left(\pi^{(k,q)}\right)_{k\in\mathcal{K}} \longrightarrow Y_\tau^{(q+1)}.$$

The procedure is repeated until the convergence criterion is met or the maximum number of iterations is reached. The transport-plan constraints, the exact update rule, and a numerical example are presented in Appendix C.3.

### 4.3.6 Choice and Initialization of the Barycenter Support

In the main application, the masses of the $M$ atoms are uniform, $b_j = 1/M$, and only their positions are optimized. The reference specification sets $M = 50$, interpreted as a numerical resolution parameter for the consensus rather than as an economic parameter.

The initial support is obtained using a weighted version of the k-means++ initialization of Arthur and Vassilvitskii [25]. Each frequency receives the same total mass so that the daily frequency does not dominate the initialization solely because it contains more observations.

This step is purely numerical; its implementation is described in Appendix C.3.

Sensitivity to the support resolution is evaluated for $M \in \{36, 50, 72\}$, in accordance with the robustness protocol defined in Section 5.1.4; the results are presented in Section 6.1.4.

### 4.3.7 Construction of the Consensus in the Joint Asset Space

The barycenter is computed in the joint asset space. Each barycenter atom

$$y_{j,\tau} = \left(y_{j,\tau}^{(1)}, \ldots, y_{j,\tau}^{(N)}\right)^{\top}$$

is therefore a complete return vector for the $N$ assets.

This choice is essential for the portfolio application. Portfolio risk depends not only on the marginal distributions of the assets but also on their joint distribution. By constructing the barycenter directly in $\mathbb{R}^N$, the method preserves co-movements, cross-sectional dependencies, and market configurations in which several assets move simultaneously.

Conversely, constructing the barycenter after one-dimensional projection would amount to working with a collection of separate views of the data. These projections are useful for subsequently evaluating dispersion more efficiently, but they do not directly define a unique consensus distribution in the original asset space. In this thesis, the barycenter is therefore constructed in $\mathbb{R}^N$, whereas slicing is used only in the next step to measure dispersion around this consensus.

Figure 4.3 illustrates this construction: frequency-specific distributions are summarized by a common consensus in the joint asset space.

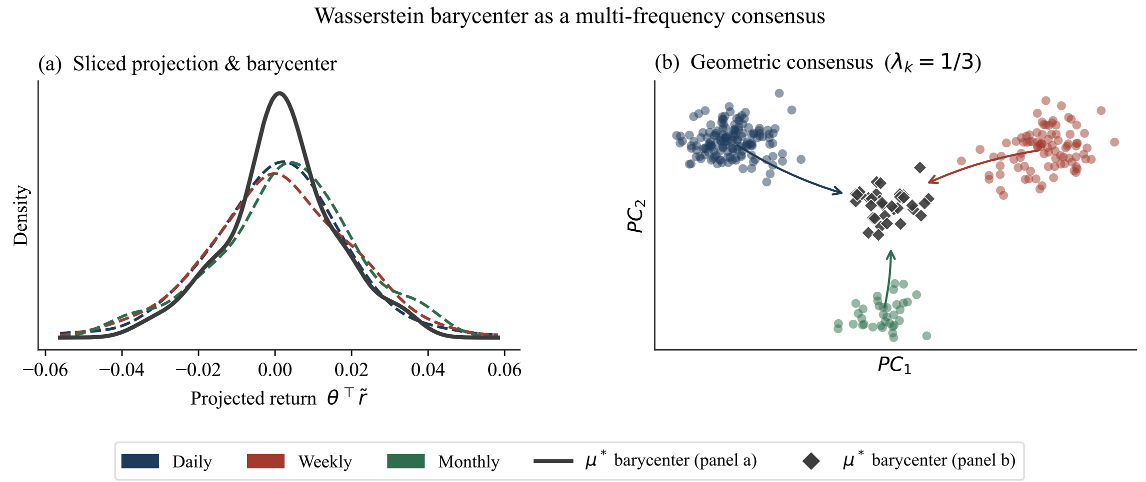


**Figure 4.3:** Illustration of the barycenter as a multivariate consensus distribution. Panel (a) represents a one-dimensional projection of the frequency-specific measures and of the barycenter previously constructed in $\mathbb{R}^N$. Panel (b) provides a schematic representation of the consensus in the plane spanned by the first two principal components. These projections are illustrative only: the barycenter used in the computations remains constructed in the joint space of the $N$ assets.

### 4.3.8 Nonconvexity of the Free-Support Problem

The free-support problem is not convex with respect to the positions of the atoms. The computed barycenter must therefore be interpreted as a local algorithmic approximation rather than as a guaranteed global solution. This limitation motivates both the weighted initialization described above and the sensitivity analysis with respect to the number of atoms presented in Section 6.1.4.

The barycenter thus provides the multivariate consensus around which the following section measures cross-horizon misalignment.

## 4.4 Sliced-Wasserstein Dispersion around the Barycenter

At each rebalancing date $\tau$, the preceding section constructed a multivariate consensus distribution

$$\widehat{\mu}_\tau^{*,W_2}$$

from the empirical measures associated with the different observation frequencies. The next step is to assess the extent to which each frequency-specific distribution departs from this consensus. For each $k \in \mathcal{K} = \{d, w, m\}$, the objective is therefore to quantify the discrepancy between $\widehat{\mu}_{k,\tau}$ and $\widehat{\mu}_\tau^{*,W_2}$.

A natural possibility would be to compute the $W_2$ distance directly between each empirical measure and the barycenter. Once the cost matrix has been constructed, the dimension of the transport program depends primarily on the cardinalities of the two supports. Constructing this matrix nevertheless retains a dependence on dimension $N$, since each cost is based on a Euclidean distance between two vectors in $\mathbb{R}^N$. In our application, this computation remains numerically feasible.

The principal motivation for slicing is therefore statistical. In high dimensions, convergence of an empirical measure under the $W_2$ distance deteriorates substantially with dimension. In the high-dimensional regime relevant to $W_2$, a canonical order of magnitude is $n^{-1/N}$, under the corresponding moment and regularity conditions [26, 27]. With $N$ close to 100 and finite rolling windows, empirical assessment of the multivariate distance may therefore be substantially affected by sampling noise.

Figure 4.4 illustrates this sparsification of empirical information as dimension increases while sample size remains fixed.

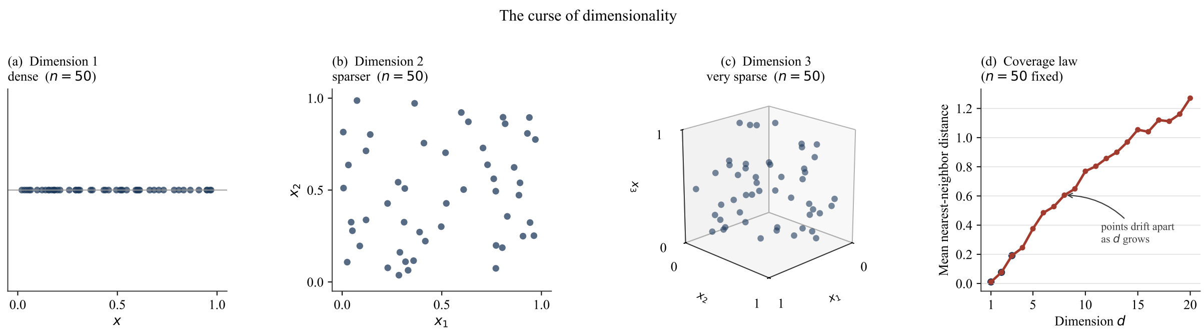


**Figure 4.4:** Illustration of the geometric sparsification of observations as dimension increases. The first three panels represent $n = 50$ points in the hypercube $[0, 1]^d$, for $d \in \{1, 2, 3\}$. The final panel reports their mean Euclidean distance to the nearest neighbor for $d = 1, \ldots, 20$. Its increase illustrates the deterioration in coverage at a fixed sample size; it is not a direct estimate of the $n^{-1/N}$ rate.

To establish the relevant orders of magnitude, consider the indicative factor $n^{-1/N}$ in the high-dimensional regime. With a three-year daily window containing approximately $n = 756$ observations, we obtain:

**Table 4.1:** Indicative factor $n^{-1/N}$ for $n = 756$

| Dimension $N$ | $n^{-1/N}$ | | Indicative error reduction |
|---|---|---|---|
| $N = 1$ | $756^{-1}$ | $\approx 1.3 \times 10^{-3}$ | Near-complete reduction |
| $N = 2$ | $756^{-1/2}$ | $\approx 0.036$ | Strong reduction |
| $N = 10$ | $756^{-1/10}$ | $\approx 0.51$ | Limited reduction |
| $N = 100$ | $756^{-1/100}$ | $\approx 0.94$ | Negligible reduction |

The rows $N = 1$ and $N = 2$ provide a numerical contrast only: they do not represent the exact theoretical convergence rates of $W_2$ in low dimensions. The asymptotic interpretation used here concerns the high-dimensional regime.

For $N = 100$, this factor remains close to one: increasing the sample size therefore reduces the error associated with multivariate geometry only very slowly. This calculation provides an illustrative order of magnitude only. It is not an error bound specific to our data, particularly because returns exhibit temporal dependence and their intrinsic dimension may be lower than the ambient dimension.

This difficulty is compounded by heterogeneous sample sizes across frequencies, since the monthly measure naturally has a smaller support. The use of Sliced-Wasserstein distances is intended to mitigate this dimensional sensitivity. The measures are projected onto one-dimensional directions, and the resulting distances are then aggregated. Under appropriate regularity conditions, projected distances benefit from convergence rates of order $n^{-1/2}$, independently of the ambient dimension [17]. This statistical improvement is accompanied by numerical error arising from the finite number of projections, which is examined separately below.

Slicing thus replaces a direct comparison in $\mathbb{R}^N$ with an average of one-dimensional views obtained along different projection directions. The agreement between the resulting signal and the discrete $W_2$ distance, computed without projection while holding the barycenter fixed, is evaluated in Section 6.1.4.

### 4.4.1 Dimensionality Reduction through Random Projections

Let

$$\theta_\ell \in S^{N-1}, \qquad \|\theta_\ell\|_2 = 1,$$

be a unit direction on the sphere in $\mathbb{R}^N$. For a normalized observation $\widetilde{r}_{k,i,\tau} \in \mathbb{R}^N$, its projection onto direction $\theta_\ell$ is the scalar

$$p_{k,i,\tau}^{(\ell)} = \langle \widetilde{r}_{k,i,\tau}, \theta_\ell \rangle \in \mathbb{R}.$$

Thus, each multivariate observation is transformed into a one-dimensional observation. In matrix form, if

$$\widetilde{X}_{k,\tau} \in \mathbb{R}^{T_{k,\tau} \times N},$$

then all projections at frequency $k$ are given by

$$p_{k,\tau}^{(\ell)} = \widetilde{X}_{k,\tau}\theta_\ell \in \mathbb{R}^{T_{k,\tau}}. \tag{12}$$

The barycenter is projected analogously. Writing

$$X_\tau^* = \begin{pmatrix} y_{1,\tau}^\top \\ y_{2,\tau}^\top \\ \vdots \\ y_{M,\tau}^\top \end{pmatrix} \in \mathbb{R}^{M \times N},$$

we obtain

$$p_{*,\tau}^{(\ell)} = X_\tau^* \theta_\ell \in \mathbb{R}^M. \tag{13}$$

Projection does not modify the masses of the measures. It changes only the positions of the atoms, which move from $\mathbb{R}^N$ to $\mathbb{R}$.

Figure 4.5 represents this step: once the measures have been projected, the Wasserstein distance can be read as a discrepancy between quantile functions.

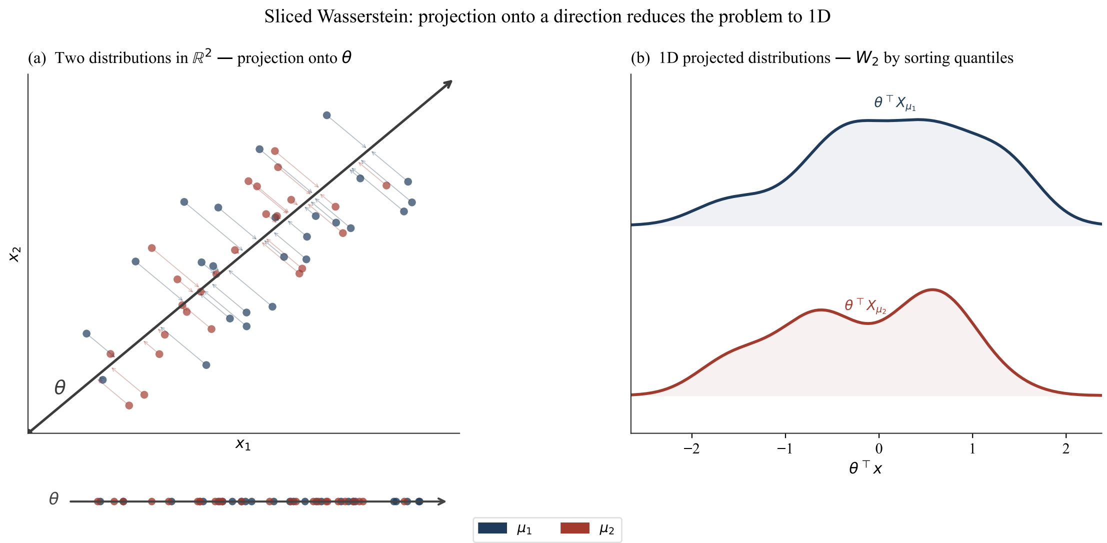


**Figure 4.5:** Slicing principle. The empirical measures and the barycenter are projected onto a unit direction $\theta_\ell$. Each multivariate observation becomes a scalar, allowing a transport problem in dimension $N$ to be replaced by a one-dimensional transport problem.

### 4.4.2 One-Dimensional Optimal Transport

For a given direction $\theta_\ell$, the projected measures are

$$\theta_\ell^\# \widehat{\mu}_{k,\tau} = \frac{1}{T_{k,\tau}} \sum_{i=1}^{T_{k,\tau}} \delta_{p_{k,i,\tau}^{(\ell)}},$$

and

$$\theta_\ell^\# \widehat{\mu}_\tau^{*,W_2} = \sum_{j=1}^{M} b_j \delta_{p_{*,j,\tau}^{(\ell)}}.$$

We then compute the squared Wasserstein distance between these two one-dimensional measures:

$$d_{k*,\tau}^{(\ell)} = W_2^2\left(\theta_\ell^\# \widehat{\mu}_{k,\tau}, \theta_\ell^\# \widehat{\mu}_\tau^{*,W_2}\right).$$

The advantage of this projection is that, in one dimension, optimal transport can be computed directly from the quantile functions. For two one-dimensional measures $\alpha$ and $\beta$,

$$W_2^2(\alpha, \beta) = \int_0^1 \left|F_\alpha^{-1}(u) - F_\beta^{-1}(u)\right|^2 du.$$

Thus, after projection, measuring the discrepancy between a frequency and the barycenter amounts to comparing the quantile functions of their projected distributions.

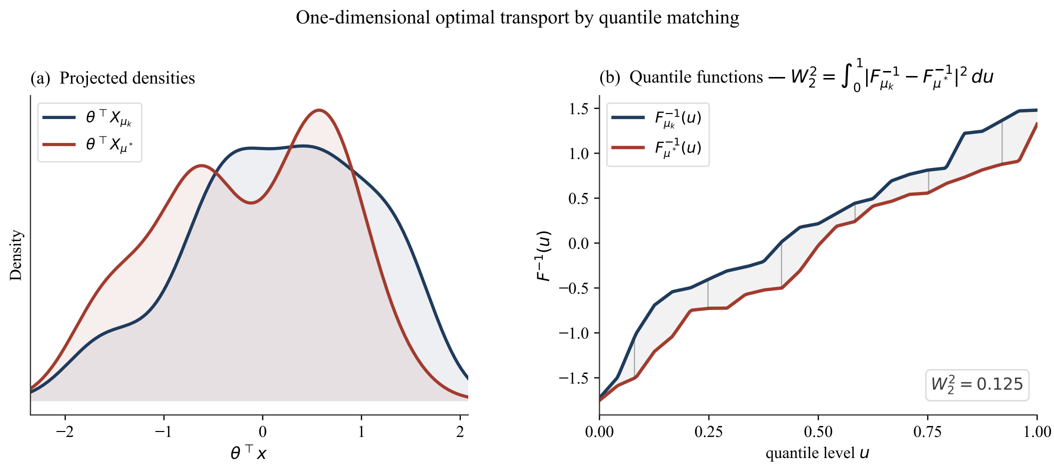


**Figure 4.6:** Slicing principle for two generic measures. Both measures are projected onto a unit direction $\theta_\ell$, so that each multivariate observation becomes a scalar. In the application, these two objects respectively correspond to a frequency-specific measure $\widehat{\mu}_{k,\tau}$ and the barycenter $\widehat{\mu}_\tau^{*,W_2}$.

### 4.4.3 Approximation Using Monte Carlo Projections

The squared Sliced-Wasserstein distance is defined as the average of the squared Wasserstein distances obtained after projection onto all unit directions in $\mathbb{R}^N$ [16]:

$$SW_2^2\left(\widehat{\mu}_{k,\tau}, \widehat{\mu}_\tau^{*,W_2}\right) = \int_{S^{N-1}} W_2^2\left(\theta^{\#}\widehat{\mu}_{k,\tau}, \theta^{\#}\widehat{\mu}_\tau^{*,W_2}\right) d\sigma(\theta),$$

where $\mathbb{S}^{N-1} = \{\theta \in \mathbb{R}^N : \|\theta\|_2 = 1\}$ denotes the set of unit directions.

In practice, this integral is approximated by Monte Carlo using $L$ independent directions $\theta_1, \ldots, \theta_L$:

$$\widehat{SW}^2_{2,L}\left(\widehat{\mu}_{k,\tau}, \widehat{\mu}_\tau^{*,W_2}\right) = \frac{1}{L}\sum_{\ell=1}^{L} d_{k*,\tau}^{(\ell)}. \tag{14}$$

The number of projections $L$ governs the trade-off between numerical accuracy and computational cost. As $L$ increases, the Monte Carlo approximation converges to the Sliced-Wasserstein distance.

### 4.4.4 Control of Monte Carlo Error

The numerical stability of the sliced approximation is evaluated over two 36-month windows: a stress window ending in September 2009 and a calm window ending in June 2016. For each window, the free-support barycenter is computed once and held fixed throughout the diagnostic.

For each number of projections

$$L \in \{10, 20, 50, 100, 200, 500, 1000\},$$

the signal is recomputed using $B_{\mathrm{MC}} = 30$ sets of random directions. Let

$$\widehat{\rho}^{(b)}_{\tau,L}, \qquad b = 1, \ldots, B_{\mathrm{MC}},$$

denote the value obtained in replication $b$. Its mean and dispersion across replications are

$$\overline{\rho}_{\tau,L} = \frac{1}{B_{\mathrm{MC}}} \sum_{b=1}^{B_{\mathrm{MC}}} \widehat{\rho}^{(b)}_{\tau,L}$$

and

$$s^{\mathrm{MC}}_{\tau,L} = \left[ \frac{1}{B_{\mathrm{MC}}} \sum_{b=1}^{B_{\mathrm{MC}}} \left( \widehat{\rho}^{(b)}_{\tau,L} - \overline{\rho}_{\tau,L} \right)^2 \right]^{1/2}.$$

Relative stability is measured by

$$CV^{\mathrm{MC}}_{\tau,L} = \frac{s^{\mathrm{MC}}_{\tau,L}}{\overline{\rho}_{\tau,L}}. \tag{15}$$

For a given replication, the same seed is retained across the different values of $L$. The directions are therefore nested: the initial directions used for a given value of $L$ are reused as $L$ increases. This construction based on common random numbers isolates the effect of the number of projections more cleanly.

This diagnostic measures only the numerical sensitivity of the signal to the draw of projection directions; it is neither a standard error associated with the economic time series nor a measure of its sampling uncertainty. The complete results, including those obtained for the retained specification $L = 200$, are presented in Appendix D, Figure D.1.

### 4.4.5 Hybrid Dispersion Signal

Starting from the Wasserstein barycenter $\widehat{\mu}^{*,W_2}_{\tau}$ defined in (10), the overall dispersion of the multi-frequency distributions around the consensus is defined as the weighted mean of the squared Sliced-Wasserstein distances:

$$\widehat{\rho}_{\tau}^{\text{hyb}} = \sum_{k \in \mathcal{K}} \lambda_{k,\tau} \widehat{SW}_{2,L}^{2} \left( \widehat{\mu}_{k,\tau}, \widehat{\mu}_{\tau}^{*,W_2} \right). \tag{16}$$

The barycenter weights $\alpha_k$ remain uniformly fixed at $1/3$, in accordance with the consensus construction defined in Section 4.3.2. The coefficients $\lambda_{k,\tau}$ enter at a distinct level: they weight the squared distance of each frequency from the barycenter in the final aggregation of the signal.

In the main specification,

$$\lambda_{k,\tau} = \frac{1}{3}, \qquad k \in \{d, w, m\}.$$

The sensitivity variants replace these uniform weights with

$$\lambda_{k,\tau}^{\text{obs}} = \frac{T_{k,\tau}}{\sum_{\ell \in \mathcal{K}} T_{\ell,\tau}}$$

or with

$$\lambda_{k,\tau}^{\log} = \frac{\log T_{k,\tau}}{\sum_{\ell \in \mathcal{K}} \log T_{\ell,\tau}}.$$

In these variants, only the relative weighting of distances in $\widehat{\rho}_{\tau}^{\text{hyb}}$ is modified; the free-support barycenter remains constructed using the uniform weights $\alpha_k = 1/3$.

This quantity measures the degree of fragmentation of the daily, weekly, and monthly distributions around their common barycenter. It is termed hybrid because the reference center is constructed as a Wasserstein barycenter in the original multivariate space $\mathbb{R}^N$, whereas dispersion around this center is evaluated using Sliced-Wasserstein distances.

It is important to emphasize that $\widehat{\rho}_{\tau}^{\text{hyb}}$ is not an exact Fréchet variance in the $W_2$ geometry: the center is the $W_2$ barycenter defined in Section 4.3, whereas dispersion is evaluated through projections. This qualification describes the hybrid nature of the construction; each component estimates a squared Sliced-Wasserstein distance according to (14).

## 4.5 Endogenous Ambiguity Radius

The preceding section defined the multi-frequency dispersion signal $\widehat{\rho}_{\tau}^{\text{hyb}}$ in (16). The objective is now to use its temporal profile to modulate the ambiguity radius of the DRO problem.

We posit an increasing relationship between the dispersion signal and the ambiguity radius. The selected specification uses the square root of the signal:

$$\varepsilon_\tau \ \propto \ \sqrt{\widehat{\rho}_\tau^{\mathrm{hyb}}}. \tag{17}$$

This choice ensures dimensional consistency between the two objects. Because $\widehat{\rho}_\tau^{\mathrm{hyb}}$ is a weighted average of squared Sliced-Wasserstein distances, it is expressed in squared return units. The radius $\varepsilon_\tau$, by contrast, is expressed in distance units. The square-root transformation therefore places the signal on the same dimensional scale as the radius.

This dimensional compatibility does not, however, constitute a structural identification. The signal measures disagreement among several historical representations of returns; it is neither an estimator of

$$W_2\big(\mathbb{P}_\tau^{\mathrm{fut}}, \widehat{\mu}_\tau^{\mathrm{nom}}\big)$$

nor a consequence of the duality of the robust problem. The barycenter and its dispersion are descriptive objects, whereas the ambiguity set is a decision-theoretic object centered on the nominal distribution.

We therefore use $\sqrt{\widehat{\rho}_\tau^{\mathrm{hyb}}}$ as a relative indicator of the temporal profile of ambiguity, rather than as a measure of its absolute level. The overall scale of the radius is determined separately by a reference level $\bar{\varepsilon}$, whose calibration is presented in Section 4.6. The empirical relevance of this construction is assessed in the walk-forward exercise rather than guaranteed by a concentration result.

Because the numerical level of the signal depends on the units used in the multi-frequency construction, we express the radius as the product of a level and a modulation:

$$\varepsilon_\tau = \bar{\varepsilon}\, m_\tau, \qquad m_\tau = \frac{\sqrt{\widehat{\rho}_\tau^{\mathrm{hyb}}}}{\overline{\sqrt{\widehat{\rho}^{\mathrm{hyb}}}}_\tau}. \tag{18}$$

The normalization

$$\overline{\sqrt{\widehat{\rho}^{\text{hyb}}}}_{\tau} = \frac{1}{J_\tau} \sum_{s \le \tau} \sqrt{\widehat{\rho}_s^{\text{hyb}}}, \qquad J_\tau = \big|\{\, s : s \le \tau \,\}\big|, \tag{19}$$

denotes the signal mean computed using only the information available at date $\tau$, that is, over an expanding window of dates $s \le \tau$. No observation subsequent to $\tau$ therefore enters the radius normalization, which rules out look-ahead bias.

This factorization separates two functions. The level $\bar{\varepsilon}$ determines the global robustness anchor, whereas the dimensionless modulation $m_\tau$ compares current dispersion with its own point-in-time expanding mean. This normalization keeps the modulation empirically centered around one without imposing that its full-sample mean be exactly equal to one.

When current dispersion is below its expanding mean, $m_\tau < 1$ and the radius narrows relative to $\bar{\varepsilon}$. When it is above that mean, $m_\tau > 1$ and the radius widens. This interpretation is relative to the available history: it does not define an absolute threshold of distributional coherence or fragmentation.

This separation delineates the methodological contribution of the thesis. The level $\bar{\varepsilon}$ remains a standard robustness parameter; the specific contribution lies in the endogenous modulation of this level using multi-frequency misalignment. By maintaining a common anchor for the static and dynamic portfolios, the framework isolates the value of the timing of robustness as it is transmitted nonlinearly to portfolio weights.

The complete methodological construction can therefore be summarized by the sequence

$$(\widehat{\mu}_{k,\tau})_{k \in \mathcal{K}} \longrightarrow \widehat{\mu}_\tau^{*,W_2} \longrightarrow \widehat{\rho}_\tau^{\text{hyb}} \longrightarrow m_\tau \longrightarrow \varepsilon_\tau = \bar{\varepsilon}\, m_\tau.$$

The barycenter provides the distributional consensus across horizons, $\widehat{\rho}_\tau^{\text{hyb}}$ measures dispersion around it, and $m_\tau$ extracts a relative modulation based exclusively on information available at date $\tau$. The reference level $\bar{\varepsilon}$ is calibrated separately in the following section.

## 4.6 Calibration of the Reference Level

The decomposition in (18) separates the radius into a reference level $\bar{\varepsilon}$ and an endogenous modulation. The former determines the common robustness anchor, whereas the latter redistributes its intensity over time.

Two calibration rules are considered, but they do not play symmetric roles in the empirical evaluation. The operational proxy inspired by a concentration bound provides the constant anchor for the primary contrast between $P^{(\text{stat})}$ and $P^{(\rho)}$. Cross-validation separately defines the auxiliary portfolio $P^{(\text{CV})}$, whose radius is selected sequentially from past performance and is not modulated by the signal.

The first calibration takes as its starting point the concentration result of Mohajerin Esfahani and Kuhn [4]. Their analysis explicitly concerns $W_1$. In the small-radius branch, under a light-tail assumption and outside the critical case $m = 2$, it yields, up to constants, a radius whose order depends on sample size $n$, nominal level $\beta$, and dimension $m$:

$$\varepsilon_n(\beta) \;\propto\; \left(\frac{\log(c_1/\beta)}{c_2\, n}\right)^{1/\max(m,2)}, \tag{20}$$

where $n$ is the number of observations, $m$ is the dimension of the scenario space, and $c_1, c_2$ are constants associated with the concentration assumptions. In our notation, $m = N = 100$ is the dimension of the return space and should not be confused with $M = 50$, the number of barycenter atoms. The exponent is therefore $1/100$. The sole factor associated with sample size becomes $n^{-1/100}$; for $n = 36$, it is approximately 0.965. This number is not a measure of the effective error, but it shows that the component of the bound that depends on $n$ decreases very slowly in dimension 100.

This slow rate illustrates the curse of dimensionality discussed in Section 4.4. Under standard conditions, the estimation error of a scalar quantity extracted from the distribution, such as a mean, may decrease at the parametric rate $n^{-1/2}$. By contrast, controlling the entire distribution in Wasserstein distance deteriorates toward $n^{-1/m}$ in the high-dimensional regime. Equation (20) is therefore reproduced to illustrate this dimensional deterioration; it is not applied as a $W_2$ guarantee.

A calibration that produced a very large radius would substantially reduce the discriminating power of the robustness level and push the portfolio toward equal weighting. This mechanism is detailed in Appendix B.4 and documented empirically in Section 6.2.7.

We therefore adopt the reduced rule

$$\bar{\varepsilon}^{\mathrm{ref}} = \sqrt{\frac{2\ln(1/\beta)}{n}}, \tag{21}$$

as a prespecified operational anchor inspired by the canonical form of exponential concentration bounds. The scale constant is normalized to one in the numerical return units used in the implementation. This rule is neither the inversion of the $W_1$ bound in dimension 100 nor a $W_2$ confidence radius. It defines the static benchmark whose sensitivity is subsequently evaluated empirically.

**Remark 4.6.1** (Status of the Parameter $\beta$)**.** In the reduced form (21), $\beta$ is a tuning parameter inherited from the structure of a concentration bound; it does not retain the exact interpretation of a coverage probability. The source result concerns $W_1$, whereas the portfolio implementation uses a $W_2$ ball. Moreover, the observations are dependent and heteroskedastic, the three frequencies are constructed from the same daily returns, and the light-tail assumptions are not verified. We therefore refer to a tuning parameter rather than a confidence level. Its variation over $\{0.05, 0.10, 0.20\}$ constitutes a sensitivity analysis, whose protocol is defined in Section 5.2.6 and whose results are presented in Section 6.2.9.2.

In the main specification, the window length is set to $n = W = 36$ and the tuning parameter to $\beta = 0.10$, which gives

$$\varepsilon_0 = \bar{\varepsilon}^{\mathrm{ref}} = \sqrt{\frac{2\ln(1/0.10)}{36}} \simeq 0.358.$$

The portfolio $P^{(\mathrm{stat})}$ retains this radius, whereas $P^{(\rho)}$ modulates its timing according to (18). The two portfolios thus share the same anchor.

The second calibration sequentially selects the radius that maximizes the Sharpe ratio of prior walk-forward performance, using a strictly sequential time-series cross-validation procedure. The first 36 holding months constitute an initialization period during which the radius is fixed at $\varepsilon_{\mathrm{init}}$, according to the convention specified in Section 5.2.1.

Beginning in the following month, at each rebalancing date, the level $\varepsilon_{\mathrm{CV},\tau}$ is selected from a prespecified grid of candidate values $\mathcal{E}$, detailed in Section 5.2.1, as the value that would have produced the highest Sharpe ratio over the history available before that date:

$$\varepsilon_{\mathrm{CV},\tau} = \arg\max_{\varepsilon \in \mathcal{E}} \ \mathrm{SR}\big(\{r_s(\varepsilon)\}_{s<\tau}\big) , \tag{22}$$

where $r_s(\varepsilon)$ denotes the realized walk-forward return at date $s$ of the portfolio associated with radius $\varepsilon$, and $\mathrm{SR}(\cdot)$ denotes the corresponding Sharpe ratio. The restriction $s < \tau$ ensures that only performance available before the rebalancing date enters the selection. The resulting radius is not modulated by the dispersion signal.

Independently of this cross-validation selection, sensitivity to the reference level is examined through an exogenous sweep over $\bar{\varepsilon}$. For each fixed level, the static portfolio and its signal-modulated counterpart are reconstructed using the same data, constraints, and accounting conventions. This exercise therefore selects no radius: it measures how the contrast between level and modulation evolves along the robustness continuum. The portfolios being compared are defined in Section 5.2.1, and the sweep protocol is presented in Section 5.2.6.

# 5 Empirical Protocol and Validation Strategy

This section presents the empirical protocol in two stages. The first evaluates the dispersion signal using dated stress episodes, a controlled placebo, robustness diagnostics, and a comparison with standard volatility measures. The second examines the value added by the endogenous radius when this signal is incorporated into the DRO allocation problem. This separation thus distinguishes validation of the signal itself from its portfolio application.

## 5.1 Signal Validation Protocol

### 5.1.1 Operational Signal Parameters

The hybrid dispersion signal defined in Section 4.4.5 is evaluated under the reference empirical configuration described below. At each formation date $\tau$, it is estimated for the 100 securities in the point-in-time universe using a rolling window of 36 calendar months ending at the close of $\tau$. This estimation window must be distinguished from the 60-month continuous-history filter used to determine security eligibility. The construction of the universe, the strict alignment of assets, and the aggregation of returns at daily, weekly, and monthly frequencies are described in Section 3.3 and Section 3.5.

The main specification uses deterministic normalization by $h_k^{1/2}$, with $(h_d, h_w, h_m) = (1, 5, 21)$, a free-support Wasserstein barycenter with $M = 50$ atoms initialized using weighted k-means++, and uniform frequency weights. Dispersion around the barycenter is estimated using $L = 200$ random projections and a grid of 200 quantiles per projection. The directions are generated from a deterministic seed specific to each universe and formation month. The same seed is retained across variants computed at a given date to reduce Monte Carlo noise in sensitivity comparisons.

The first available formation occurs in December 1994 and provides the signal used for the January 1995 allocation. The economic evaluation period therefore extends from January 1995 through December 2025. At each date, only observations available through the close of month $\tau$ enter the signal calculation; returns in the following month, $\tau + 1$, are reserved exclusively for portfolio evaluation.

The dynamics of the series obtained under this reference configuration are presented in

Section 6.1.1.

### 5.1.2 Response to Dated Stress Episodes

The first stage examines the signal dynamics around stress episodes dated independently of its construction. Five dates are selected ex ante: the collapse of the dot-com bubble in March 2000, the global financial crisis in September 2008, European sovereign-debt tensions in August 2011, the pandemic shock in March 2020, and monetary tightening in June 2022. These dates are neither selected nor adjusted according to the observed maxima of the signal.

For each date $\tau_c$, the diagnostic is conducted over a symmetric window $[\tau_c - H, \tau_c + H]$, with $H = 18$ months. The signal is expressed on the scale that enters the radius modulation and then standardized:

$$z_\tau = \frac{\sqrt{\widehat{\rho}_\tau^{\text{hyb}}} - \mu_{\sqrt{\rho}}}{\sigma_{\sqrt{\rho}}}, \tag{23}$$

where $\mu_{\sqrt{\rho}}$ and $\sigma_{\sqrt{\rho}}$ respectively denote the mean and standard deviation of $\sqrt{\widehat{\rho}_\tau^{\text{hyb}}}$ over the evaluation period. This full-sample standardization is used exclusively for descriptive purposes; it enters neither the point-in-time construction of the signal nor portfolio formation.

The event study assesses whether increases and reversals in the signal occur before, during, or after the selected stress dates. It is not a predictive test of crises and does not require the signal maximum to coincide with the month conventionally associated with each episode. Its purpose is to provide an external diagnostic of temporal concordance, which is complemented by the controlled misalignment test presented in the following section.

The results of this event study are presented in Section 6.1.2.

### 5.1.3 Discriminating Power: Controlled Placebo Test

This limitation motivates a controlled experiment in which a cross-frequency misalignment of known intensity is injected into the observed data.

Two distinct deterministic samples of $B_{\text{win}} = 300$ windows are drawn without replacement. The first is used for the reference joint shock; the second is shared by the detection-rate curves and the ROC curves.

For each window $b$, the signal is first recomputed from the observed daily, weekly, and monthly matrices without perturbation:

$$\left\{\widehat{\rho}_b^{(0)}\right\}_{b=1}^{B_{\text{win}}}.$$

Here, $H_0$ denotes the absence of an artificial perturbation; it does not imply that the historical data are free of any economic misalignment.

Within each of the two samples, an analysis-specific descriptive threshold is defined as the empirical 95% quantile of the corresponding $H_0$ distribution:

$$q_{0.95} = Q_{0.95}\left(\left\{\widehat{\rho}_b^{(0)}\right\}_{b=1}^{B_{\text{win}}}\right).$$

The proportion of unperturbed windows exceeding this threshold is mechanically close to 5% in the sample used for its calibration. This threshold therefore provides a common point of comparison for measuring the detection rate of the injected perturbations. It should not, however, be interpreted as an exact out-of-sample control of the Type I error rate, because the same set of windows is used to estimate both the reference distribution and its quantile.

An artificial perturbation designed to create a controlled misalignment across frequencies is then applied to the same formation windows. This setting defines the alternative regime $H_1$. The daily matrix is left unchanged and serves as the anchor, whereas the weekly and monthly matrices may be perturbed according to

$$\widetilde{\mathbf{r}}_{w,t}^{H_1} = \kappa\,\widetilde{\mathbf{r}}_{w,t}, \qquad \widetilde{\mathbf{r}}_{m,t}^{H_1} = \widetilde{\mathbf{r}}_{m,t} + \xi\,\widehat{\boldsymbol{\sigma}}_m,$$

where $\widehat{\boldsymbol{\sigma}}_m$ denotes the vector of empirical monthly standard deviations computed asset by asset within the window under consideration. The parameter $\kappa \geq 1$ amplifies the scale of the weekly matrix, whereas $\xi \geq 0$ shifts each monthly series by a multiple of its empirical standard deviation. The first transformation affects both the mean level and the dispersion of weekly returns; it must therefore not be interpreted as a dilation centered around their mean.

The pair $(\kappa, \xi) = (2, 2)$ defines the reference joint shock. Sensitivity to perturbation intensity is then examined separately along each dimension, while holding the other frequency unchanged, over the grids

$$\kappa \in \{1,\ 1.25,\ 1.5,\ 2,\ 3\}, \qquad \xi \in \{0,\ 0.5,\ 1,\ 2,\ 3\}.$$

The values $\kappa = 1$ and $\xi = 0$ respectively correspond to no weekly and no monthly perturbation.

For each window $b$ and each intensity, the signal is fully recomputed after perturbation, yielding

$$\left\{\widehat{\rho}_b^{(1)}\right\}_{b=1}^{B_{\text{win}}},$$

to be compared with the unperturbed distribution

$$\left\{\widehat{\rho}_b^{(0)}\right\}_{b=1}^{B_{\text{win}}}.$$

Within a given window, regimes $H_0$ and $H_1$ use exactly the same assets, the same column order, and the same random projection directions. This construction based on common random numbers neutralizes the numerical variability arising from the sliced approximation and attributes the difference between the two signal values exclusively to the injected perturbation.

The discriminating power of the signal is evaluated as a binary classification problem: using only the value of $\widehat{\rho}$, the objective is to distinguish the observed windows without artificial perturbation ($H_0$) from the same windows after injection of a misalignment ($H_1$). Two complementary measures are retained: an empirical detection rate evaluated at threshold $q_{0.95}$, followed by the area under the ROC curve, which does not depend on the choice of any particular threshold.

For a given perturbation intensity, the empirical detection rate is defined by

$$\widehat{\pi}_{\text{det}} = \frac{1}{B_{\text{win}}} \sum_{b=1}^{B_{\text{win}}} \mathbf{1}\left\{\widehat{\rho}_b^{(1)} > q_{0.95}\right\}.$$

It measures the proportion of artificially perturbed windows whose signal exceeds the 95th percentile of the unperturbed distribution. A high value indicates that the injected perturbation generally produces a signal above the values observed under $H_0$.

This quantity is an empirical measure of sensitivity under the placebo protocol. It is not, however, equated with theoretical power equal to $1-\beta$: the threshold and detection rate are estimated from a finite set of historical windows, and $H_0$ denotes the absence of an artificial perturbation rather than a fully specified probabilistic model. Moreover, its interpretation depends on the threshold $q_{0.95}$. The analysis is therefore complemented by the ROC curve and its area, which evaluate the separation between $H_0$ and $H_1$ over all possible thresholds.

The ROC curve plots, over all thresholds $q$, the true-positive rate against the false-positive rate. It thus describes the trade-off between detection and false alarms independently of any particular threshold.

The area under this curve, or AUC, summarizes this trade-off in a single number. It has the following probabilistic interpretation:

$$\mathrm{AUC} = \mathbb{P}\left(\widehat{\rho}^{(1)} > \widehat{\rho}^{(0)}\right),$$

that is, the probability that a perturbed window receives a higher signal than an unperturbed window. An AUC of 0.5 corresponds to random classification, whereas an AUC close to 1 indicates near-perfect separation between the two regimes.

The AUC is evaluated separately along the two intensity dimensions introduced above. For each value of $\kappa$, the weekly frequency is amplified while the monthly frequency remains unperturbed ($\xi = 0$). Conversely, for each value of $\xi$, the monthly frequency is shifted while the weekly frequency remains unperturbed ($\kappa = 1$). Each intensity therefore defines its own distribution $\{\widehat{\rho}_b^{(1)}\}_{b=1}^{B_{\mathrm{win}}}$, ROC curve, and AUC.

The uncertainty associated with each AUC is evaluated using $B_{\mathrm{boot}} = 500$ replications of a paired bootstrap over formation windows [28]. In each replication, the same set of $B_{\mathrm{win}}$ indices is sampled with replacement and then applied jointly to $\{\widehat{\rho}_b^{(0)}\}$ and $\{\widehat{\rho}_b^{(1)}\}$. The AUC is recomputed on this paired sample, and the 95 % bootstrap interval is given by the empirical 2.5 % and 97.5 % quantiles of the bootstrap distribution. This resampling preserves the pairing between $H_0$ and $H_1$ within each window. It does not, however, explicitly reproduce the temporal dependence induced by the overlap among the 36-month formation windows; the resulting intervals are therefore interpreted as diagnostics of AUC stability rather than as exact temporal coverage intervals.

The results of the controlled placebo are presented in Section 6.1.3.

### 5.1.4 Robustness to Construction Choices

After evaluating, in the preceding controlled setting, the ability of the signal to respond to cross-frequency misalignment, the next stage examines its sensitivity to the numerical and geometric choices underlying its construction. Seven dimensions are considered. Correlations with the reference specification are computed over the full sample and over three subperiods: 1995–2004, 2004–2015, and 2015–2025. The boundary years 2004 and 2015 are intentionally included in both adjacent windows; these comparisons are descriptive diagnostics rather than independent tests.

The first dimension concerns barycenter cardinality. The signal is recomputed for $M \in \{36, 50, 72\}$ atoms, and the resulting series are compared both globally and by subperiod. Low sensitivity to $M$ indicates that the signal does not reflect an artifact arising from the resolution of the barycenter support.

The second dimension concerns frequency weights. The reference uniform weighting is compared with weights proportional to the number of observations and log-proportional weights to verify that the signal dynamics do not depend on the relative weight assigned to each horizon.

The third dimension concerns return normalization. Deterministic normalization by $\sqrt{h_k}$ is compared with normalization by realized volatility. This comparison is not strictly a robustness exercise but rather a specification comparison: the two normalizations measure different objects, and realized-volatility normalization is expected to attenuate part of the regime misalignment preserved by $\sqrt{h_k}$.

The fourth dimension extends the preceding comparison within the deterministic family. Rather than contrasting rescaling by $\sqrt{h_k}$ with realized-volatility normalization, the scaling exponent itself is varied by imposing $h_k^H$ for $H \in \{0.4, 0.5, 0.6\}$. The exponent is imposed exogenously and is never estimated within the window: estimating it would reintroduce precisely the drawback attributed to realized-volatility normalization, namely a data-dependent scale factor that would absorb part of the misalignment that the signal is intended to measure. This lever therefore tests whether the timing of the signal depends on the scaling assumption underlying the reference specification $H = 1/2$.

The fifth dimension concerns the metric used to evaluate dispersion around the barycenter. In the reference specification, this dispersion is approximated using projected Wasserstein distances. It is compared with a variant in which the discrete Wasserstein distance is computed without projection between each empirical frequency-specific distribution and the same free-support barycenter, using the POT library [29]. The term “exact” qualifies the evaluation of the distance conditional on the selected barycenter; it means neither that the estimated barycenter is free of error nor that this variant represents a population truth.

Agreement between the two series is examined globally, by subperiod, and around the prespecified stress episodes. This comparison assesses the extent to which the timing of the signal depends on the projection approximation. Any divergences are documented without being attributed ex ante to either estimator: they may arise from either the sliced approximation or the sensitivity of the multivariate distance to high dimension and finite sample sizes.

The sixth dimension concerns barycenter geometry. The reference specification uses a free-support barycenter in $\mathbb{R}^N$, which exploits the joint structure of returns across assets. It is compared with a one-dimensional quantile barycenter constructed frequency by frequency. This comparison assesses whether the timing of the signal depends specifically on the selected multivariate geometry or whether the temporal content of the misalignment remains present under a simpler barycentric construction.

The seventh dimension concerns Monte Carlo convergence of the sliced approximation described in Section 4.4.4. The signal is recomputed, with 30 replications, for each number of projections on the grid

$$\mathcal{L} = \{10, 20, 50, 100, 200, 500, 1000\},$$

over two contrasting windows: a stress window ending in September 2009 and a calm window ending in June 2016. The barycenter support is computed only once per window, and the same seeds are used for each value of $L$. Stability is measured using the coefficient of variation of $\widehat{\rho}$, which assesses the numerical precision of the reference choice $L = 200$.

The results of these sensitivity analyses are presented in Section 6.1.4, whereas convergence as a function of the number of projections is documented in Appendix D.

#### 5.1.5 Relation to Volatility Measures

The final diagnostic examines the extent to which the transformed signal

$$S_\tau = \sqrt{\widehat{\rho}_\tau^{\,\mathrm{hyb}}},$$

which enters the construction of the dynamic radius, is linearly associated with standard volatility measures. The objective is neither to establish a causal relationship nor to conduct a spanning test, but to determine what fraction of the observed variation in $S_\tau$ remains unexplained by implied and realized volatility.

This monthly signal is compared with volatility controls defined at two horizons, thereby avoiding a comparison based exclusively on monthly measures. At the first horizon, $\mathrm{IV}_\tau^{(m)}$ is the monthly average of the daily VIX (Section 3.2), and $\mathrm{RV}_\tau^{(m)}$ is annualized volatility computed from the equal-weighted daily return of the active universe. These variables describe contemporaneous volatility at the monthly frequency.

At the second horizon, $\mathrm{IV}_\tau^{(36)}$ is the 36-month moving average of the monthly VIX series. This variable is a historical smoothing of the VIX, not a 36-month-horizon implied forecast. $\mathrm{RV}_\tau^{(36)}$ is computed at each formation date as the annualized volatility of the equal-weighted daily return on the 100 securities belonging exactly to the formation universe, over the same 36-month retrospective window used to construct the signal. This measure thus remains aligned with the universe, the window, and the information available at the formation date.

For each horizon $h \in \{m, 36\}$, three specifications are estimated: implied volatility alone, realized volatility alone, and both variables jointly. The joint specification is

$$S_\tau = \alpha + \beta_{\mathrm{IV}} \mathrm{IV}_\tau^{(h)} + \beta_{\mathrm{RV}} \mathrm{RV}_\tau^{(h)} + u_\tau. \tag{24}$$

The univariate specifications assess the explanatory content of each measure in isolation without overinterpreting the coefficients of the joint regression when the two regressors are strongly correlated.

Persistence is documented using first-order autocorrelation and an augmented Dickey–Fuller test [30] applied to the five series under consideration: $S_\tau = \sqrt{\widehat{\rho}_\tau^{\text{hyb}}}$, monthly implied volatility, monthly realized volatility, the 36-month moving average of the VIX, and realized volatility over the 36-month formation window. To limit the risk of spurious regression in the level estimates [31], all specifications are also estimated in first differences:

$$\Delta S_\tau = \alpha + \beta_{\text{IV}} \Delta \text{IV}_\tau^{(h)} + \beta_{\text{RV}} \Delta \text{RV}_\tau^{(h)} + u_\tau. \tag{25}$$

The level estimates describe common persistent components, whereas first differences provide a more conservative diagnostic of short-run co-movements.

The coefficients are estimated by ordinary least squares. Inference relies on Newey–West HAC standard errors with 36 lags [32] to account for the autocorrelation and overlap induced by the signal-construction window. The coefficient of determination is reported for the univariate and joint specifications, at both horizons and under both transformations.

This diagnostic is interpreted as an analysis of contemporaneous linear non-redundancy. An $R^2$ below one indicates that part of the signal variation is not reproduced by the selected volatility controls; on its own, however, this result is insufficient to establish that this component constitutes predictive information or an independent economic factor.

The results of this comparison are presented in Section 6.1.5.

## 5.2 Portfolio Evaluation Protocol

This section specifies how the cross-horizon dispersion signal constructed in Section 4.4.5, and subsequently transformed into an ambiguity radius in Section 4.5, is translated into allocation decisions and evaluated out of sample. Its own validation protocol is defined in Section 5.1. The objective is to isolate a precise question: does the endogenous modulation of the ambiguity radius $\varepsilon_\tau$ by the cross-horizon disagreement signal create economic value beyond a properly calibrated constant radius, and does that value arise from the robustness level or from its timing?

To answer this question, we first define the family of portfolios being compared and the calibration of their ambiguity radii (Section 5.2.1). We then specify weight construction, holding mechanics, and the treatment of delistings (Section 5.2.3), followed by the walk-forward protocol and transaction costs (Section 5.2.4). The value of timing is evaluated using a three-level permutation protocol (Section 5.2.5), before its robustness to the principal construction choices is examined (Section 5.2.6). The corresponding performance metrics and inference procedures are defined in Section 5.3.

### 5.2.1 Portfolios Compared

We consider an ordered family of portfolios, applied separately to each of the point-in-time universes defined in Sections 3.3 and 3.4. Within a given universe, the portfolios use the same estimation window, holding periods, and accounting framework; they differ only in the rule that determines the ambiguity radius. Table 5.1 summarizes the notation.

Two portfolios bound the continuum described in Section 4.1.6. The equal-weighted portfolio $P^{(\mathrm{EW})}$, defined by $w_i = 1/N$, represents the diversified limit obtained when $\varepsilon$ becomes sufficiently large. At the other extreme, $P^{(\mathrm{emp})}$ represents the empirical limit $\varepsilon \downarrow 0$, numerically approximated by the floor $\varepsilon_{\min} = 10^{-4}$. These two portfolios position the DRO allocations but do not constitute the central comparison. Their empirical positioning is examined in Section 6.2.6.

The reference static portfolio $P^{(\mathrm{stat})}$ uses a constant interior radius $\varepsilon_0$. Its dynamic counterpart $P^{(\rho)}$ applies, at each formation date $\tau$,

$$\varepsilon_\tau^{(\rho)} = \Pi_{[\varepsilon_{\min},\varepsilon_{\max}]}\left(\varepsilon_0\, m_\tau\right), \qquad m_\tau = \frac{\sqrt{\rho_\tau}}{\left\langle\sqrt{\rho}\right\rangle_\tau},$$

where, to simplify notation, $\rho_\tau$ denotes the hybrid signal $\widehat{\rho}_\tau^{\,\mathrm{hyb}}$ defined in Section 4.4.5. The operator $\Pi$ denotes projection onto the safety region, and $\left\langle\sqrt{\rho}\right\rangle_\tau$ is the expanding mean computed using information available through date $\tau$. This rule is the empirical implementation of the decomposition in (18). The radius determined at the close of $\tau$ governs the portfolio held during month $\tau+1$. Its empirical path is presented in Section 6.2.1.

The comparison

$$\left(P^{(\mathrm{stat})}, P^{(\rho)}\right)$$

constitutes the primary contrast. The expanding normalization keeps the modulation around an approximately unit value: the two portfolios therefore share the same anchor, while the second redistributes robustness over time. The observed difference measures the joint effect of this timing and its nonlinear transmission to portfolio weights, rather than an abstract causal effect of the signal independently of the allocation problem. This contrast is evaluated in Section 6.2.3.

As an exploratory diagnostic, we also construct two modulations based exclusively on volatility. For $j \in \{\mathrm{IV}, \mathrm{RV}_{36}\}$, they are defined by

$$m_\tau^{(j)} = \frac{x_\tau^{(j)}}{\left\langle x^{(j)}\right\rangle_\tau}, \qquad \varepsilon_\tau^{(j)} = \Pi_{[\varepsilon_{\min},\varepsilon_{\max}]}\left(\varepsilon_0 m_\tau^{(j)}\right),$$

with

$$x_\tau^{(\mathrm{IV})} = \mathrm{IV}_\tau^{(m)}, \qquad x_\tau^{(\mathrm{RV}_{36})} = \mathrm{RV}_\tau^{(36)}.$$

Both measures are defined in Section 5.1.5. The first benchmark uses the VIX at its natural short-term horizon, aggregated over the formation month; the second uses the realized volatility of the exact 100 securities in the formation universe over the same point-in-time 36-month window as the signal. This choice is not an exhaustive grid of horizons: it contrasts a forward-looking market control with a backward-looking control directly matched to the construction of $\rho$, without multiplying intermediate horizons ex post.

The portfolios $P^{(\mathrm{IV})}$ and $P^{(\mathrm{RV}_{36})}$ retain the same anchor, the same safety bounds, the same universes, the same allocation framework, and the same accounting conventions as $P^{(\rho)}$. They are used solely to examine whether the economic effect associated with $\rho$ differs from volatility-based timing rules. Their results are presented in Section 6.2.9.3.

We retain $P^{(\mathrm{CV})}$ separately. Its radius $\varepsilon_{\mathrm{CV},\tau}$ is selected sequentially from the past performance of the candidate strategies. This branch is an alternative calibration diagnostic: it is neither modulated by $\rho$ nor used to identify the primary timing effect. Its behavior is examined in Section 6.2.7.

**Table 5.1:** Family of portfolios compared. The central contrast compares the constant interior radius $P^{(\mathrm{stat})}$ with its point-in-time modulation $P^{(\rho)}$.

| Notation | Radius | Role |
|---|---|---|
| $P^{(\mathrm{EW})}$ | $\varepsilon \to \infty$ | equal-weight, diversified limit |
| $P^{(\mathrm{emp})}$ | $\varepsilon_{\min} = 10^{-4}$ | numerical proxy for the empirical limit |
| $P^{(\mathrm{stat})}$ | $\varepsilon_0$ | static DRO reference |
| $P^{(\rho)}$ | $\Pi_{[10^{-4},\,2]}(\varepsilon_0 m_\tau)$ | central point-in-time modulation |
| $P^{(\mathrm{IV})}$ | $\Pi_{[10^{-4},\,2]}(\varepsilon_0 m_\tau^{(\mathrm{IV})})$ | exploratory implied-volatility benchmark |
| $P^{(\mathrm{RV}_{36})}$ | $\Pi_{[10^{-4},\,2]}(\varepsilon_0 m_\tau^{(\mathrm{RV}_{36})})$ | exploratory matched realized-volatility benchmark |
| $P^{(\mathrm{CV})}$ | $\varepsilon_{\mathrm{CV},\tau}$ | auxiliary sequential calibration |

### 5.2.2 Calibration of the Reference Level

The principles underlying both calibrations are defined in Section 4.6. We specify their implementation in the empirical evaluation here.

For the main specification, the reduced form of Equation (21) is evaluated using a window of $n = W = 36$ months and tuning parameter $\beta = 0.10$. It provides the interior reference point

$$\varepsilon_0 = \sqrt{\frac{2\ln(1/\beta)}{W}} \simeq 0.358.$$

Sensitivity to the choice of $\beta$ is examined in Section 5.2.6.

For $P^{(\mathrm{CV})}$, the sequential rule in Equation (22) is applied to a prespecified logarithmic grid of 16 radii between 0.05 and 10. The first 36 holding months constitute an initialization period during which the radius is fixed at the numerical median of the grid, $\varepsilon_{\text{init}} = 0.718$. Located between the two central grid points, this value is reserved for initialization and does not constitute a seventeenth candidate in the subsequent selection.

Beginning in the following month, $\varepsilon_{\text{CV},\tau}$ maximizes the gross Sharpe ratio computed over the expanding set of realized walk-forward returns strictly preceding formation date $\tau$. If several candidates are exactly tied, the smallest radius is retained under a prespecified tie-breaking rule. The selected portfolio is then evaluated, like the other strategies, using both gross and net returns.

Finally, the radius of $P^{(\rho)}$ is projected onto the safety region

$$\varepsilon_\tau^{(\rho)} \in [10^{-4}, 2].$$

This projection applies to the central modulation and its robustness variants, but not to the $P^{(\text{CV})}$ diagnostic, whose grid is deliberately extended to 10.

### 5.2.3 Weight Construction and Reinvestment

At each formation date $\tau$, the conic program (7) is solved with MOSEK using the point-in-time matrix of $W = 36$ monthly returns for the $N = 100$ eligible assets. It produces a long-only, fully invested target-weight vector $w_\tau = (w_{1,\tau}, \ldots, w_{N,\tau})^\top$,

$$w_{i,\tau} \geq 0, \qquad \sum_{i=1}^{N} w_{i,\tau} = 1.$$

Failure to obtain an admissible solution constitutes a computational failure: no substitute portfolio, including an equal-weighted portfolio, replaces the optimizer. The transmission of the radius to weight concentration is examined in Section 6.2.2.

In the main specification, weights are formed at the close of month $\tau$ and held during month $\tau + 1$. For each asset, the holding-period growth factor is constructed from adjusted daily returns:

$$H_{i,\tau+1} = \prod_{d \in \mathcal{D}_{i,\tau+1}} \left(1 + \text{DlyRet}_{i,d}\right), \qquad R_{i,\tau+1}^{\text{hold}} = H_{i,\tau+1} - 1.$$

For a security that remains listed, the product covers all trading days in the month. If a delisting occurs, it stops on exit date $d_i^*$. The DlyRet return already incorporates the delisting return on that date; the latter is therefore applied exactly once and is never added separately.

This treatment is consistent with the construction of adjusted returns described in Section 3.6.

The portfolio's gross wealth factor during the holding month is

$$G_{p,\tau+1} = \sum_{i=1}^{N} w_{i,\tau} H_{i,\tau+1}, \qquad R^{\text{gross}}_{p,\tau+1} = G_{p,\tau+1} - 1.$$

At the end of the period, the weights of securities that remain listed have drifted according to

$$w^-_{i,\tau+1} = \frac{w_{i,\tau} H_{i,\tau+1}}{G_{p,\tau+1}}.$$

When a security is delisted during the holding period, its terminal value is not redistributed among the other assets. It is transferred to a zero-return cash account until the next rebalancing:

$$w^-_{\text{cash},\tau+1} = \frac{\sum_{i \in \mathcal{L}_{\tau+1}} w_{i,\tau} H_{i,\tau+1}}{G_{p,\tau+1}},$$

where $\mathcal{L}_{\tau+1}$ denotes the set of securities delisted during the holding period, whereas $\mathcal{S}_{\tau+1}$ denotes the set of securities still listed at its end.

The surviving risky weights and cash then satisfy

$$\sum_{i \in \mathcal{S}_{\tau+1}} w^-_{i,\tau+1} + w^-_{\text{cash},\tau+1} = 1.$$

At the next rebalancing, cash and surviving positions finance the new target-weight vector. A security that simply leaves the universe without being delisted is liquidated as part of this rebalancing; its weight is not redistributed according to a predetermined proportional rule.

In the main monthly specification, a missing daily return on a position actually held, outside a delisting event, is admitted only if it belongs to a prespecified finite list. The relevant observation is replaced by a zero return; no other missing value is admitted. The treatment of quarterly and annual holding calendars is specified in Section 5.2.6.

### 5.2.4 Walk-Forward Convention and Transaction Costs

The evaluation follows a strict walk-forward convention. At the close of each formation month $\tau$, the universe, estimation matrix, signal $\rho_\tau$, its expanding normalization, and the ambiguity radius are constructed exclusively from the information then available. The target weights are subsequently applied to the daily returns in month $\tau+1$, which never enter their own determination.

The decision record comprises 373 formation dates. The 372 realized holding periods cover January 1995 through December 2025; the final decision, formed in December 2025 for a holding period in January 2026, is retained as an out-of-sample forecast but excluded from performance measurement. This separation also applies to the normalization of $m_\tau$ and the sequential calibration of $\varepsilon_{\mathrm{CV},\tau}$.

At each rebalancing, the target weights $w_{i,\tau}$ are compared with the risky weights actually held immediately before trading, $w_{i,\tau}^-$, after the previous month's positions have drifted. Let $\mathcal{A}_\tau^-$ denote the set of risky positions held before trading and $\mathcal{A}_\tau$ the set of assets in the new target portfolio, with the weight of an asset absent from either vector defined as zero. Two-way turnover is

$$\mathrm{TO}_\tau = \sum_{i \in \mathcal{A}_\tau^- \cup \mathcal{A}_\tau} \left| w_{i,\tau} - w_{i,\tau}^- \right| .$$

The cash account resulting from delistings does not enter this sum directly. Its reinvestment is nevertheless recorded through the purchases required to reach the new risky target weights. At the first formation, the portfolio starts entirely in cash; its risky turnover is therefore one.

Let $\kappa_\tau$ denote the per-side unit cost expressed as a proportion of portfolio value. Transaction costs are

$$\mathrm{TC}_\tau = \kappa_\tau \, \mathrm{TO}_\tau .$$

The net return associated with the subsequent holding period is then

$$R^{\text{net}}_{p,\tau+1} = R^{\text{gross}}_{p,\tau+1} - \text{TC}_\tau.$$

Costs reduce the net return but do not modify either the normalization of the target weights or their relative drift during the holding period.

**Table 5.2:** Time-varying transaction-cost schedule. The per-side unit cost declines over time and is applied to the two-way turnover at each rebalancing. The decline reflects the structural reduction in execution costs for large-capitalization stocks over the sample period.

| Period | Per-side cost $\kappa_\tau$ (bps) | Market regime |
|---|---|---|
| 1995–1999 | 50 | High-friction environment |
| 2000–2009 | 30 | Post-decimalization transition |
| 2010–2025 | 20 | Modern low-friction regime |

We adopt a stylized time-varying schedule motivated by the historical decline in execution frictions documented by Frazzini et al. [33]. The per-side cost is set to 50 basis points for 1995–1999, 30 for 2000–2009, and 20 for 2010–2025 (Table 5.2). The same schedule is applied to the static and dynamic portfolios, both capitalization universes, and all robustness specifications to preserve the comparability of the contrasts.

For the NYSE small–mid-cap universe, this common schedule is a controlled comparison convention rather than a segment-specific estimate of execution costs, which are likely higher than those of large-capitalization stocks.

Net-of-cost results provide the primary basis for the economic evaluation and inference. Gross returns are retained as an audit that distinguishes the allocation effect from transaction frictions.

### 5.2.5 Timing Validity: Placebo Tests

The comparison between $P^{(\text{stat})}$ and $P^{(\rho)}$ measures the value associated with the observed timing of the modulation. A positive gain could nevertheless result merely from varying the radius, from signal persistence, or from the ex post selection of a favorable segment. We therefore organize permutation inference into three complementary levels.

In each replication, all else held constant, only the temporal ordering of the modulation vector $m_\tau$ is rearranged. The universes, estimation matrices, holding-period returns, delistings, transaction costs, and accounting rules remain unchanged; in particular, neither asset identities nor their realized returns are permuted. The dynamic portfolio is re-solved in full and reconstructed over all 372 months from the reordered modulation.

We use $B_{\text{perm}} = 1000$ replications. Net results constitute the primary basis for inference, while gross results are retained as an audit. The $p$-values are computed using the correction defined in Section 5.3.2.

**Level 1 — Timing placebos.**

The first level evaluates the dynamic large-capitalization portfolio over the full sample. Three distinct null distributions are constructed:

- the simple monthly permutation freely rearranges the 372 values of $m_\tau$, preserving their marginal distribution but destroying their temporal order;
- the six-month block permutation rearranges contiguous blocks while preserving the order of observations within each block;
- the twelve-month block permutation applies the same construction to annual blocks.

For each null, the placebo statistic is the annualized Sharpe ratio, denoted SR, of the reconstructed dynamic portfolio; it is compared with the SR of the portfolio based on the observed temporal ordering. Block permutations preserve part of the local dependence in the signal and therefore provide a more demanding control than the simple monthly permutation. This level produces three $p$-values, presented in Section 6.2.4.

**Level 2 — Restricted four-cell family.**

The second level evaluates whether the timing gain remains significant after accounting for selection across universes and subsamples. The statistic for cell $j$ is

$$\Delta\mathrm{SR}_j = \mathrm{SR}\left(P_j^{(\rho)}\right) - \mathrm{SR}\left(P_j^{(\mathrm{stat})}\right).$$

The restricted family is

$$\mathcal{F}_4 = \{L_{\mathrm{all}}, L_{\mathrm{exDC}}, S_{\mathrm{all}}, S_{\mathrm{exDC}}\},$$

where $L$ denotes the large-capitalization universe, $S$ the NYSE small–mid-cap universe, and exDC the sample excluding January 1999 through December 2001.

The full sample remains the main specification. As a sensitivity diagnostic, we also report results after excluding January 1999 through December 2001 to assess whether the gain is concentrated around the dot-com episode. This split is treated as exploratory rather than as an independent main specification.

In each replication, the same simple monthly permutation is applied simultaneously to the modulations of both universes. The four statistics are therefore produced jointly from the same temporal rearrangement, which preserves their cross-sectional dependence and their dependence across subsamples. Their maximum forms the max-$T_4$ reference distribution under the procedure defined in Section 5.3.2. For each of the four cells, we report the marginal $p$-value and the max-$T_4$-adjusted $p$-value. Cell $S_{\text{exDC}}$ is designated as the reporting target before execution of the joint permutations to illustrate the stability of the contrast in the secondary universe after the dot-com period is excluded. This designation organizes family-wise reporting but does not confer confirmatory status: the full sample remains the main specification, and the dot-com exclusion remains an exploratory sensitivity analysis.

**Level 3 — Extended twelve-cell family.**

The third level retains exactly the same monthly replications and extends the preceding family to twelve cells. It adds the six combinations of the two universes and the nonoverlapping subperiods 1995–2004, 2005–2014, and 2015–2025, together with two large-capitalization cells corresponding to dispersion regimes respectively above and below the median of $\sqrt{\rho}$ computed over the full sample. The resulting family is denoted $\mathcal{F}_{12}$.

The maximum of the twelve statistics in each replication defines the max-$T_{12}$ distribution. The corresponding adjusted $p$-values are computed for all twelve cells. Cell $S_{\text{exDC}}$ is retained as an illustrative target to allow direct comparison of the adjustments obtained under $\mathcal{F}_4$ and $\mathcal{F}_{12}$. This final level is not a new independent experiment: $\mathcal{F}_4 \subset \mathcal{F}_{12}$, and both tests rely on the same joint draws. Expanding the family therefore provides a more conservative multiplicity control. The results for Levels 2 and 3 are presented in Section 6.2.8.

### 5.2.6 Robustness of the Conclusions

The value attributed to radius timing should not depend on an isolated point of the robustness continuum, a particular construction choice, or a single historical period. It must also be compared with timing rules based on standard risk variables. We therefore organize the robustness analysis around four complementary diagnostics.

For the useful range of the radius, the construction choices, and temporal stability, the main outcome remains

$$\Delta\mathrm{SR}^{\mathrm{net}} = \mathrm{SR}\big(P^{(\rho)}\big) - \mathrm{SR}\big(P^{(\mathrm{stat})}\big)\,.$$

The volatility diagnostic instead compares $\mathrm{SR}(P^{(\rho)})$ with $\mathrm{SR}(P^{(\mathrm{IV})})$ and $\mathrm{SR}(P^{(\mathrm{RV}_{36})})$. In all cases, net results constitute the primary basis, while gross returns are retained as an audit.

**Useful range of the reference radius.**

The first diagnostic examines whether the value of timing persists when the reference point $\varepsilon_0$ is moved along the robustness continuum. We consider a prespecified logarithmic grid of ten radii between 0.10 and 2, to which the reference point $\varepsilon_0 \simeq 0.358$ is added.

For each baseline radius $\bar{\varepsilon}$, the static portfolio retains that radius, whereas the dynamic portfolio uses

$$\varepsilon_\tau^{(\rho)} = \Pi_{[10^{-4},2]}\left(\bar{\varepsilon} m_\tau\right).$$

Both paths are solved in full and evaluated under the same accounting framework. We report both the net timing gain and the mean concentration of the static portfolio, measured by $\|w\|_2$. This sweep is a diagnostic of the region in which modulation produces economically discernible effects; it is not used to select a new reference radius ex post.

**Robustness to construction choices.**

The second diagnostic varies one lever at a time while holding all other components identical to the main specification. Four dimensions are examined:

$$W_\rho \in \{36, 48, 60\} \text{ months},$$
$$\text{modulation form} \in \{\text{identity}, \exp(z_\tau), \text{expanding rank}\},$$
$$\text{rebalancing frequency} \in \{\text{monthly}, \text{quarterly}, \text{annual}\},$$
$$\beta \in \{0.05, 0.10, 0.20\}.$$

Setting $x_\tau = \sqrt{\rho_\tau}$, the three functional forms are based on

$$m_\tau = \frac{g_\tau}{\langle g \rangle_\tau},$$

where all quantities are computed on an expanding basis using information available through $\tau$. The reference form uses $g_\tau = x_\tau$. The exponential form uses

$$g_\tau = \exp(z_\tau), \qquad z_\tau = \frac{x_\tau - \langle x \rangle_\tau}{s_\tau(x)},$$

with a neutral initialization $z_\tau = 0$ during the first twelve observations. The third form defines $g_\tau$ as the percentile rank of $x_\tau$ among observations available through $\tau$. Signals computed with $W_\rho = 48$ and 60 months are constructed under the sensitivity protocol defined in Section 5.1.4. They are validated inputs to the allocation stage and are not re-estimated within it.

The quarterly and annual variants modify only the transaction dates. Between two scheduled rebalancings, risky positions continue to drift with realized returns, delisting proceeds remain in cash, and no asset is introduced solely because it enters a subsequent monthly universe. The new target vector is computed only at the next rebalancing date.

Quarterly and annual holding calendars may expose additional security–date pairs with a missing daily return outside a delisting event. They are subject to the same convention: only observations recorded in a prespecified finite list are replaced by zero, and no other missing value is admitted. An observation reused across several configurations is counted only once in this list.

**Volatility-based timing benchmarks.**

The third diagnostic compares $P^{(\rho)}$ with the two volatility modulations defined in Section 5.2.1. All three strategies are solved over the full sample, separately in the main universe and in the NYSE small–mid-cap universe. They share exactly the same formation dates, estimation matrices, holding-period returns, transaction costs, and delisting-treatment rules.

This diagnostic is neither a new placebo nor an extension of families $\mathcal{F}_4$ and $\mathcal{F}_{12}$. It addresses a distinct question: does the performance of $P^{(\rho)}$ differ from that of economically comparable rules whose timing depends exclusively on volatility? The paired comparisons are described in Section 5.3.2.

**Temporal stability.**

The fourth diagnostic examines the stability of the gain over three nonoverlapping subperiods:

$$1995\text{–}2004, \qquad 2005\text{–}2014, \qquad 2015\text{–}2025.$$

For each subperiod, $\Delta\mathrm{SR}^{\mathrm{net}}$ is computed separately for the large-capitalization universe and the NYSE small–mid-cap universe. The portfolios are not re-estimated specifically for each subperiod: the corresponding returns are extracted from the full walk-forward paths under continuous accounting. These six cells are descriptive and also enter the max-$T_{12}$ family defined above.

Across all four diagnostics, the point-in-time universes, estimation matrices, delisting treatment, position drift, cost schedule, and turnover definition remain identical to those of the main specification. Optimizer failure always constitutes a hard failure and is never replaced by an equal-weighted allocation. The objective is not to select the most favorable configuration but to determine the limits within which the conclusion regarding the value of timing remains stable. The corresponding results are presented in Section 6.2.9.

## 5.3 Portfolio Metrics and Statistical Inference

The empirical evaluation proceeds in two stages. We first describe the economic profile of the portfolios using performance, risk, and reallocation metrics. We then evaluate the central contrast using paired procedures, supplemented by descriptive conditional attribution and permutation tests with multiplicity control.

### 5.3.1 Descriptive Metrics

Let $\{R_{p,t}\}_{t=1}^{T}$ denote the series of monthly returns for portfolio $p$, with empirical mean

$$\bar{R}_p = \frac{1}{T}\sum_{t=1}^{T} R_{p,t},$$

and empirical standard deviation

$$\widehat{\sigma}_p = \left[\frac{1}{T-1}\sum_{t=1}^{T}\left(R_{p,t} - \bar{R}_p\right)^2\right]^{1/2}.$$

The annualized mean return, annualized volatility, and annualized Sharpe ratio are, respectively,

$$\widehat{\mu}_p^{\text{ann}} = 12\bar{R}_p, \qquad \widehat{\sigma}_p^{\text{ann}} = \sqrt{12}\,\widehat{\sigma}_p, \qquad \text{SR}_p = \sqrt{12}\,\frac{\bar{R}_p}{\widehat{\sigma}_p}.$$

Consistent with the empirical implementation, the risk-free rate is set to zero.

Writing cumulative wealth as

$$V_{p,0} = 1, \qquad V_{p,t} = \prod_{s=1}^{t}(1 + R_{p,s}),$$

maximum drawdown is

$$\text{MDD}_p = \max_{1\le t\le T}\left(1 - \frac{V_{p,t}}{\max_{0\le s\le t} V_{p,s}}\right).$$

We also report the date of the deepest drawdown trough and the maximum duration, expressed in consecutive months, for which wealth remains below its previous high.

Tail risk is measured using empirical Conditional Value-at-Risk at 95 %:

$$\text{CVaR}_{0.95,p} = -\frac{\sum_{t=1}^{T} R_{p,t}\,\mathbf{1}\{R_{p,t} \le q_{0.05,p}\}}{\sum_{t=1}^{T}\mathbf{1}\{R_{p,t} \le q_{0.05,p}\}},$$

where $q_{0.05,p}$ denotes the empirical 5 % quantile. This quantity is expressed at the monthly frequency and is not annualized.

Finally, recurring reallocation intensity is summarized by mean two-way turnover:

$$\overline{\mathrm{TO}}_p = \frac{1}{T-1}\sum_{t=2}^{T} \mathrm{TO}_{p,t},$$

where $\mathrm{TO}_{p,t}$ is defined in Section 5.2.4. The first rebalancing, which corresponds to the initial deployment of the portfolio from cash and for which $\mathrm{TO}_{p,1} = 1$, is retained in the accounting path and realized costs but excluded from the descriptive average of recurring turnover.

All these metrics are computed from both gross and net-of-cost returns. Unless otherwise stated, net results constitute the primary basis for evaluation; gross results are retained as an audit of the effect of transaction frictions.

### 5.3.2 Statistical Inference and Conditional Attribution

Unless otherwise stated, the central inference concerns the main large-capitalization universe and the full sample. It compares the portfolios defined in Section 5.2.1,

$$\left(P^{(\mathrm{stat})}, P^{(\rho)}\right).$$

The expanding normalization is intended to keep the modulation around one. This contrast therefore primarily evaluates the temporal redistribution of robustness around the same anchor; the effects of projecting the radius and of its nonlinear transmission to the weights form part of the measured effect.

**Mean return difference.**

For each basis $b \in \{\text{gross}, \text{net}\}$, we define the paired monthly difference

$$d_t^{(b)} = R_{p,t}^{(\rho,b)} - R_{p,t}^{(\mathrm{stat},b)}, \qquad \bar{d}^{(b)} = \frac{1}{T}\sum_{t=1}^{T} d_t^{(b)}.$$

The null hypothesis is

$$H_0 : \mathbb{E}\left[d_t^{(b)}\right] = 0.$$

It is evaluated using a paired mean test with Newey–West HAC standard errors [32], which are robust to heteroskedasticity and autocorrelation in the monthly differences. We use 12 monthly lags and compute a two-sided $p$-value from the normal approximation.

**Sharpe-ratio difference.**

For each basis $b \in \{\text{gross}, \text{net}\}$, the difference in annualized Sharpe ratios is

$$\Delta\mathrm{SR}^{(b)} = \mathrm{SR}^{(b)}\big(P^{(\rho)}\big) - \mathrm{SR}^{(b)}\big(P^{(\text{stat})}\big)\,.$$

Uncertainty surrounding $\Delta\mathrm{SR}^{(b)}$ is evaluated using a paired stationary bootstrap [34]. In each of the 10 000 replications, the monthly returns of both portfolios are jointly resampled in blocks of random length, equal to ten months on average, and the Sharpe-ratio difference is then recomputed. This procedure preserves the pairing of the strategies and part of the temporal dependence in returns. We report a 95 % percentile confidence interval and a two-sided finite-sample-corrected $p$-value.

The corresponding results are presented in Section 6.2.3.

**Exploratory comparisons with volatility benchmarks.**

For $j \in \{\mathrm{IV}, \mathrm{RV}_{36}\}$, we supplement the primary contrast with the paired difference

$$\Delta\mathrm{SR}^{\text{net}}_{\rho,j} = \mathrm{SR}_{\text{net}}\big(P^{(\rho)}\big) - \mathrm{SR}_{\text{net}}\big(P^{(j)}\big)\,.$$

This statistic is computed separately in the main universe and the NYSE small–mid-cap universe, producing four comparisons. For each, uncertainty is evaluated using the same paired stationary-bootstrap principle: the monthly returns of both strategies are jointly resampled in each of 10 000 replications, with a mean block length of ten months. We report a 95 % percentile interval and a two-sided finite-sample-corrected $p$-value. The reported inference concerns net returns.

These comparisons address a hypothesis different from that of the permutation tests: they assess whether the performance of $P^{(\rho)}$ differs from that of an observed volatility rule, rather than whether its temporal ordering is interchangeable with chance. They are exploratory, are not adjusted for multiplicity, and belong to neither $\mathcal{F}_4$ nor $\mathcal{F}_{12}$. They therefore do not constitute an equivalence test between strategies. The results are presented in Section 6.2.9.3.

**Descriptive conditional attribution.**

This inference is supplemented by a descriptive conditional attribution intended to locate the periods during which the difference between $P^{(\rho)}$ and $P^{(\mathrm{stat})}$ is most pronounced.

The first partition is based on the value of $\sqrt{\rho_\tau}$ observed at the portfolio formation date. Denoting its full-period median by

$$q_\rho = \mathrm{Med}\left(\{\sqrt{\rho_\tau}\}_{\tau=1}^{T}\right),$$

we distinguish

$$\mathcal{H} = \left\{t : \sqrt{\rho_{\tau(t)}} \geq q_\rho\right\}, \qquad \mathcal{L} = \left\{t : \sqrt{\rho_{\tau(t)}} < q_\rho\right\},$$

where $\tau(t)$ denotes the formation date associated with return month $t$. Each regime contains 186 monthly observations.

These same regimes constitute the two dispersion cells in the $\mathcal{F}_{12}$ family defined in Section 5.2.5.

For each regime $A \in \{\mathcal{H}, \mathcal{L}\}$, we compute the Sharpe ratios of the static and dynamic portfolios and their difference

$$\Delta\mathrm{SR}_A = \mathrm{SR}_A\big(P^{(\rho)}\big) - \mathrm{SR}_A\big(P^{(\mathrm{stat})}\big).$$

To locate the source of the mean return difference, we also compute the annualized contribution of each regime:

$$C_A = \frac{12}{T} \sum_{t \in A} \left(R_{p,t}^{(\rho)} - R_{p,t}^{(\mathrm{stat})}\right).$$

This quantity is the annualized mean difference observed within the regime, weighted by its frequency in the sample. By construction, $C_{\mathcal{H}} + C_{\mathcal{L}}$ equals the annualized mean difference over the full sample. This additive decomposition concerns mean returns, not Sharpe-ratio differences.

The second partition examines windows of $\pm 12$ months around the five stress episodes dated in Section 5.1.2. Each window thus contains 25 monthly returns, including the central month. Within each window, we report the static and dynamic Sharpe ratios,

$\Delta$SR, and the annualized mean return difference.

These decompositions do not constitute additional hypothesis tests. In particular, because threshold $q_\rho$ is computed over the full sample, the dispersion regimes are descriptive categories rather than a point-in-time investment rule. Their purpose is to clarify the temporal origin of the gain without assigning it a causal interpretation. The corresponding results are presented in Section 6.2.5.

**Permutation $p$-values.**

The permutations described in Section 5.2.5 generate the null distributions against which the observed statistics are compared. For an observed statistic $T^{\text{obs}}$ and its placebo values $\{T^{(b)}\}_{b=1}^{B_{\text{perm}}}$, we use a one-sided upper-tail test, because the alternative hypothesis is that informative timing produces higher performance or a larger performance gain than the permuted signal orderings.

The statistic depends on the level of inference being considered. For the three Level 1 timing placebos,

$$T^{\text{obs}} = \text{SR}^{\text{obs}}\big(P^{(\rho)}\big), \qquad T^{(b)} = \text{SR}^{(b)}\big(P^{(\rho)}\big).$$

For the Level 2 and Level 3 families, the statistic for cell $j$ is

$$T_j^{\text{obs}} = \Delta\text{SR}_j^{\text{obs}}, \qquad T_j^{(b)} = \Delta\text{SR}_j^{(b)},$$

with

$$\Delta\text{SR}_j = \text{SR}_j\big(P^{(\rho)}\big) - \text{SR}_j\big(P^{(\text{stat})}\big).$$

For each statistic, the number of exceedances is

$$r = \sum_{b=1}^{B_{\text{perm}}} \mathbf{1}\big\{T^{(b)} \geq T^{\text{obs}}\big\}.$$

The empirical $p$-value is computed using the finite-sample correction [28, 35]:

$$p_{\text{perm}} = \frac{r+1}{B_{\text{perm}}+1}.$$

The $+1$ term includes the observed statistic in the randomization experiment and prevents a strictly zero $p$-value.

We use $B_{\text{perm}} = 1000$ replications, which sets the minimum resolution at

$$\frac{1}{B_{\text{perm}} + 1} = \frac{1}{1001} \simeq 0.001.$$

The results of the three timing placebos are presented in Section 6.2.4.

**Family-wise multiplicity control using max-$T$.**

Marginal $p$-values evaluate each cell separately. When the timing gain is examined simultaneously across several universes and subsamples, selecting the most favorable cell nevertheless increases the probability of obtaining a large statistic by chance. We control this multiplicity using a max-$T$ procedure [36]. For $q \in \{4, 12\}$, let $\mathcal{F}_q$ denote the family of cells defined in Section 5.2.5. In each replication $b$, the placebo statistics are computed jointly for all cells in the family, and their maximum is retained:

$$M_q^{(b)} = \max_{j \in \mathcal{F}_q} \Delta\text{SR}_j^{(b)}.$$

The adjusted $p$-value associated with an observed cell $j \in \mathcal{F}_q$ is

$$p_{\max\text{-}T_q,j} = \frac{1 + \sum_{b=1}^{B_{\text{perm}}} \mathbf{1}\left\{M_q^{(b)} \geq \Delta\text{SR}_j^{\text{obs}}\right\}}{B_{\text{perm}} + 1}.$$

In each replication, the same monthly rearrangement is applied to both universes, after which the dynamic portfolios are reconstructed in full. The statistics thus rely on a common draw, preserving dependence across universes, subsamples, and nested cells.

The same draws feed the families $\mathcal{F}_4 \subset \mathcal{F}_{12}$. Consequently, $M_{12}^{(b)} \geq M_4^{(b)}$, and the adjustment associated with the extended family is at least as conservative for a given cell. The retained statistic is $\Delta$SR, not a standardized $t$-statistic. Marginal and family-wise results are presented in Section 6.2.8.

# 6 Results

## 6.1 Validation of the Dispersion Signal

This subsection documents the validity of the $\widehat{\rho}_{\tau}^{\mathrm{hyb}}$ signal before it is incorporated into the allocation problem. Consistent with the protocol in Section 5.1, the analysis proceeds through four diagnostics: the signal's response to independently dated stress episodes, its discriminating power in a controlled placebo setting, its robustness to construction choices, and, finally, its relationship with conventional volatility measures.

### 6.1.1 Overview

Under the reference configuration defined in Section 5.1.1, Figure 6.1 presents the signal over the full period on the $\sqrt{\widehat{\rho}_{\tau}^{\mathrm{hyb}}}$ scale that enters the ambiguity radius, together with the expanding mean used for its normalization. The signal remains low and stable during calm periods and rises markedly during the main episodes of market stress—particularly the 2008 financial crisis and the 2020 pandemic shock—providing initial evidence of its responsiveness to market regimes. The following sections formalize this interpretation.

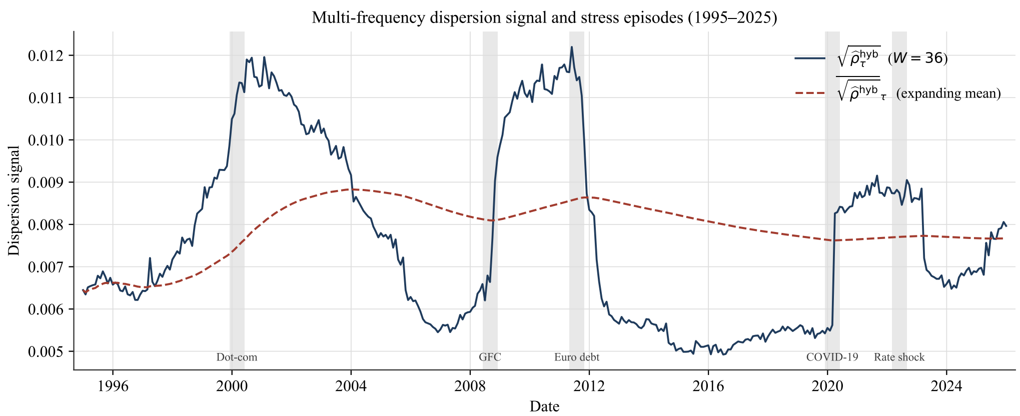


**Figure 6.1:** Multi-frequency dispersion signal $\sqrt{\widehat{\rho}_{\tau}^{\mathrm{hyb}}}$ ($W = 36$-month window) over 1995–2025. The dotted curve represents the expanding mean $\overline{\sqrt{\widehat{\rho}^{\mathrm{hyb}}}}_{\tau}$ used to normalize the radius; the shaded bands mark externally dated stress episodes.

### 6.1.2 Response to Dated Stress Episodes

Consistent with the protocol in Section 5.1.2, Figure 6.2 examines the behavior of the signal around five stress episodes whose dates are exogenous to our construction: the bursting of the dot-com bubble (March 2000), the global financial crisis (September 2008), the European sovereign debt crisis (August 2011), the COVID-19 shock (March 2020), and the 2022 monetary tightening episode (June 2022). For each peak $\tau_c$, we report the standardized signal $z_\tau$ (Equation 23) over a symmetric ±18-month window. Each panel has its own vertical scale: the object of interest is the local dynamics around each episode, rather than a comparison of amplitudes across crises.

Three temporal patterns emerge. First, the global financial crisis and the pandemic shock are accompanied by a rapid transition in the signal around the external date. In both cases, $z_\tau$ moves from a negative level before the peak to a positive level thereafter. The increase begins around the external date and continues during the following months, particularly in the case of the GFC. The shock date therefore corresponds more closely to a regime change in the signal than to its maximum.

Second, the dot-com episode exhibits a more gradual pattern. The signal rises during the months preceding March 2000 and remains elevated thereafter. This profile is consistent with a gradual accumulation of inter-frequency misalignment during the formation and correction of the bubble. Because the event study is descriptive, however, this lead does not by itself constitute a test of predictive ability.

Third, the European sovereign debt crisis and the monetary tightening episode illustrate the possible gap between the external dating of an event and the signal's own dynamics. The signal is already positive before August 2011 and then declines markedly after that date. Around the June 2022 interest-rate shock, it remains relatively stable for several months before declining later. The external peak therefore does not necessarily coincide with a local maximum of $\widehat{\rho}_\tau^{\text{hyb}}$.

Overall, this analysis documents a temporal association between the signal and several independently dated stress episodes, with heterogeneous patterns across types of shocks. It supports the signal's sensitivity to strained market conditions without, at this stage, establishing a causal relationship or predictive ability. The placebo test presented next more directly evaluates its ability to distinguish inter-frequency misalignment from estimation noise.

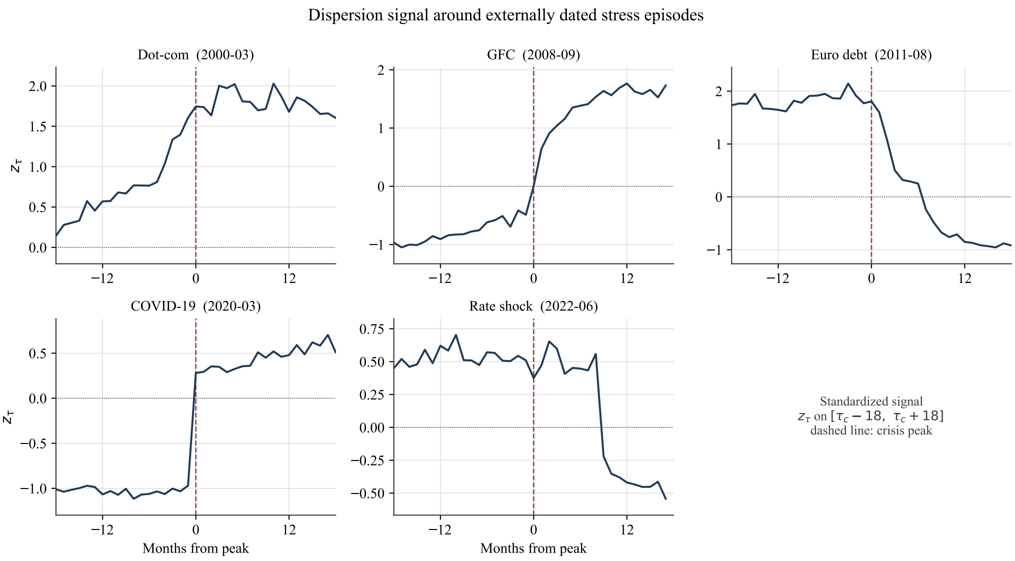


**Figure 6.2:** Response of the signal to externally dated stress episodes. Standardized signal $z_\tau = (\sqrt{\widehat{\rho}_\tau^{\text{hyb}}} - \mu)/\sigma$ over a $\pm 18$-month window centered on the peak $\tau_c$ of each episode (vertical dashed line). Each panel has its own vertical scale. Reference specification ($W = 36$, $M = 50$, uniform weights, $L = 200$).

### 6.1.3 Discriminating Power: Controlled Placebo Test

The preceding event study provides a descriptive diagnostic based on historical episodes. We now use the controlled placebo defined in Section 5.1.3 to examine whether the signal distinguishes unperturbed observed windows ($H_0$) from windows in which inter-frequency misalignment of known intensity is artificially injected ($H_1$).

Consistent with the protocol, the reference joint shock and the dose–response paths rely on two distinct deterministic draws of $B_{\text{win}} = 300$ windows, sampled without replacement. Within each comparison, $H_0$ and $H_1$ share the same formation window and the same random directions of the estimator. The comparison is therefore paired: only the injected perturbation distinguishes the two regimes.

Under the reference joint shock $(\kappa, \xi) = (2, 2)$, separation is complete in the controlled experiment: the AUC reaches 1.000, and all 300 realizations under $H_1$ exceed the threshold corresponding to the 95 % quantile of $H_0$. The mean of $\widehat{\rho}^{\text{hyb}}$ rises from $6.49 \times 10^{-5}$ under $H_0$ to $5.21 \times 10^{-4}$ under $H_1$, a ratio of 8.0, with a Cohen's effect size of $d = 2.42$.

This separation characterizes the reference joint shock. To assess the signal's sensitivity to less pronounced perturbations, Figure 6.3 reports the empirical detection rate along each of the two axes while holding the other at its unperturbed value.

Along the weekly-volatility axis, this rate rises from 5.0 % in the absence of perturbation to 18.3 %, 26.0 %, 59.3 %, and 100 % as $\kappa$ successively increases to 1.25, 1.5, 2, and 3. Along the monthly-drift axis, it reaches 21.0 %, 57.0 %, and then 100 % for $\xi = 0.5$, 1, and 2, respectively. The response is therefore monotonic along both axes but faster for the drift shift: the indicative 80 % threshold is reached at $\xi = 2$, compared with $\kappa = 3$ for the volatility shock. This progression shows that separation does not rely solely on the reference joint shock.

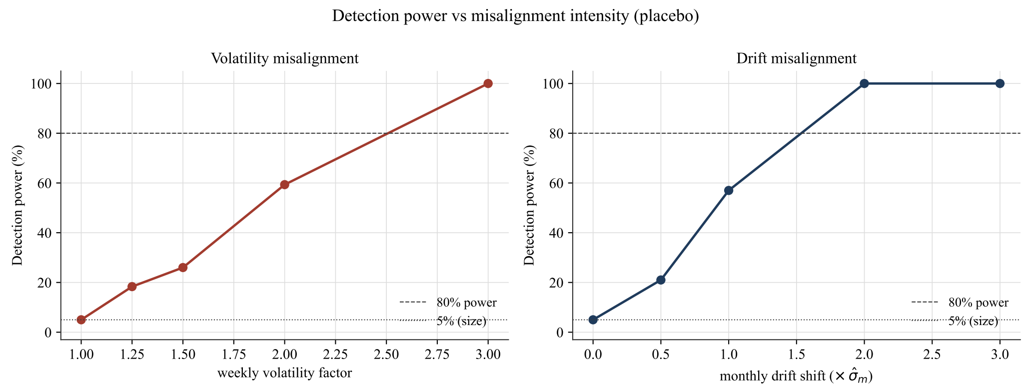


**Figure 6.3:** Empirical placebo detection rate as a function of misalignment intensity. Fraction of $H_1$ draws exceeding the 95 % quantile of $H_0$, along the volatility axis (left) and the drift axis (right), with the other axis held at its null value. Dotted lines: empirical reference level of 5 % and indicative threshold of 80 %.

Figure 6.4 complements this diagnostic by evaluating separation over all possible thresholds. Along the weekly-volatility axis, the AUC rises from 0.613 [0.610, 0.620] for $\kappa = 1.25$ to 0.733 [0.722, 0.745] for $\kappa = 1.5$, then to 0.919 [0.902, 0.933] for $\kappa = 2$, before reaching 1.000 for $\kappa = 3$. Along the monthly-drift axis, it increases from 0.696 [0.685, 0.708] for $\xi = 0.5$ to 0.890 [0.873, 0.906] for $\xi = 1$, and reaches 1.000 from $\xi = 2$ onward.

The AUCs therefore increase monotonically with perturbation intensity, and their diagnostic bootstrap intervals exclude 0.5 from the first nonzero intensity onward. As with the detection rate, separation increases more rapidly along the monthly-drift axis than along the weekly-volatility axis. The signal therefore distinguishes increasing degrees of injected misalignment without the result relying exclusively on the joint specification $(\kappa, \xi) = (2, 2)$.

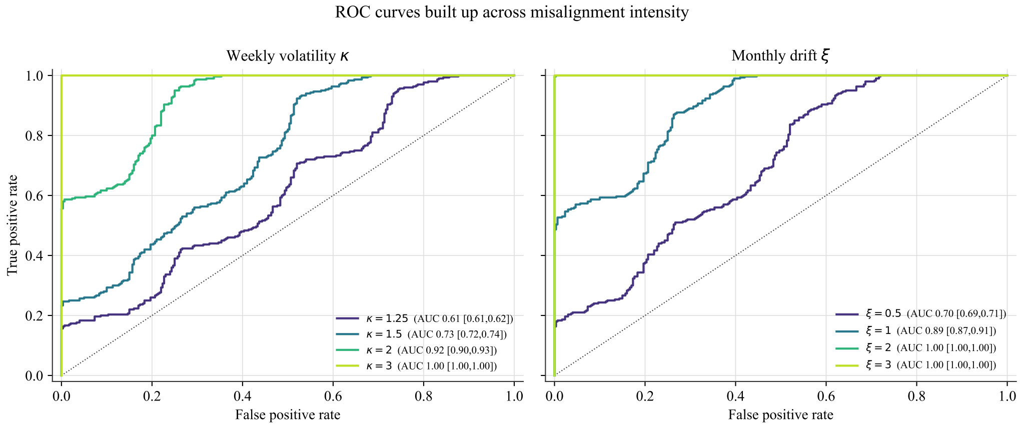


**Figure 6.4:** Signal discrimination by the intensity of the injected misalignment. ROC curves of the signal for increasingly large shocks along the weekly-volatility axis $\kappa$ (left) and the monthly-drift axis $\xi$ (right), with the other axis held at its null value. The unperturbed case ($\kappa = 1$ or $\xi = 0$) corresponds to no discrimination (AUC $= 0.5$) and is not plotted. Each curve is annotated with its AUC and its 95 % paired-bootstrap interval, interpreted as a stability diagnostic ($B_{\text{boot}} = 500$ resamples with replacement of the $B_{\text{win}} = 300$ pairs of $H_0/H_1$ windows). Dotted diagonal: random classification.

### 6.1.4 Robustness to Construction Choices

Consistent with the protocol defined in Section 5.1.4, we examine whether the dynamics of $\widehat{\rho}_\tau^{\text{hyb}}$ remain stable as the main numerical, metric, and geometric construction choices vary. Six dimensions are studied in this section: barycenter cardinality, frequency weighting, inter-frequency scaling, the scaling exponent, the transport distance, and barycenter geometry. The seventh dimension announced in the protocol, concerning the Monte Carlo precision associated with the number of projections, is documented separately in Appendix D.

Table 6.1 reports, for each variant, its correlation with the reference specification over the full sample and the three subperiods defined in the protocol. These correlations measure the linear temporal concordance of the series rather than equality of their levels. The event studies presented subsequently complement this diagnostic by showing where any differences are concentrated around stress episodes.

**Table 6.1:** Robustness of the signal dynamics to construction choices: correlation of each variant with the reference specification ($\widehat{\rho}^{\text{hyb}}$, 1995–2025).

| Lever | Variant | Correlation with the reference | | | |
|---|---|---|---|---|---|
| | | Overall | 1995–2004 | 2004–2015 | 2015–2025 |
| Cardinality $M$ | $M = 36$ | 0.999 | 0.999 | 0.999 | 1.000 |
| | $M = 72$ | 0.999 | 1.000 | 0.999 | 0.999 |
| Frequency weights $\lambda_k$ | $\lambda_k \propto T_k$ | 0.988 | 0.994 | 0.997 | 0.990 |
| | $\lambda_k \propto \log T_k$ | 0.999 | 0.999 | 1.000 | 0.999 |
| Scaling | realized vol. | 0.203 | −0.561 | 0.053 | 0.618 |
| Scaling exponent $H$ | $h_k^{0.4}$ | 0.991 | 0.993 | 0.996 | 0.990 |
| | $h_k^{0.6}$ | 0.998 | 0.996 | 0.999 | 0.998 |
| Distance | discrete $W_2$ (no slicing) | 0.958 | 0.998 | 0.983 | 0.916 |
| Barycenter | 1-D quantile | 0.883 | 0.934 | 0.984 | 0.963 |

Table 6.1 distinguishes three degrees of sensitivity. Barycenter cardinality and frequency weights produce very close agreement with the reference, with full-sample correlations ranging from 0.988 to 0.999. Varying the exponent within the deterministic family also largely preserves the dynamics (0.991 for $H = 0.4$ and 0.998 for $H = 0.6$).

The discrete $W_2$ distance evaluated without projection (0.958) and the one-dimensional quantile barycenter (0.883) retain substantial, though less close, temporal concordance. By contrast, realized-volatility normalization has a full-sample correlation of only 0.203, confirming that it is not a mere numerical adjustment: it changes the object being measured by absorbing part of the scale heterogeneity across frequencies. The following figures locate these differences around stress episodes.

#### 6.1.4.1 Barycenter Cardinality

Figure 6.5 compares the reference signal with the variants obtained using $M = 36$ and $M = 72$ atoms.

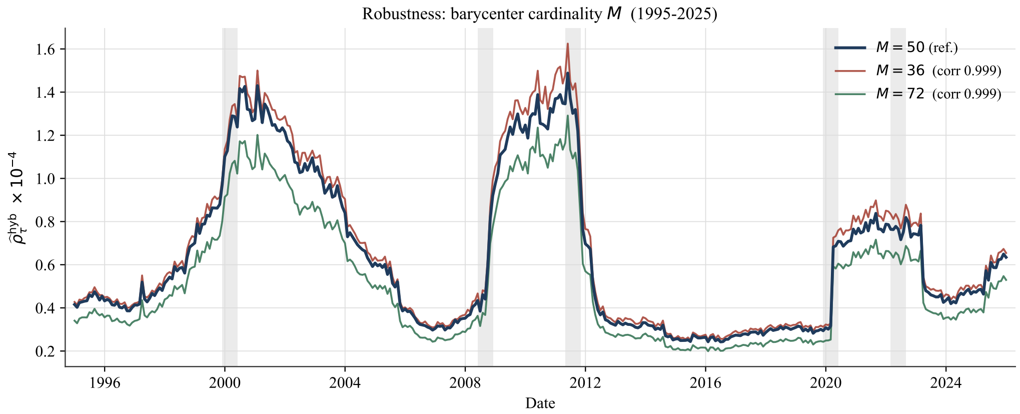


**Figure 6.5:** Robustness to barycenter cardinality $M$. Raw $\widehat{\rho}_\tau^{\mathrm{hyb}}$ signal ($\times 10^{-4}$ scale) for $M \in \{36, 50, 72\}$ over 1995–2025. The level varies with $M$, while the series remain strongly correlated with the reference (corr = 0.999). Shaded bands: stress episodes.

Cardinality primarily affects the level of the signal. A finer barycenter ($M = 72$) produces lower values, whereas a coarser support ($M = 36$) raises them. The correlation of 0.999 with the reference specification nonetheless shows that the temporal profiles remain very close. Over the range considered, the signal dynamics therefore appear relatively insensitive to barycentric support resolution.

#### 6.1.4.2 Frequency Weights

Figure 6.6 compares the reference uniform weighting with weights proportional and log-proportional to the number of observations.

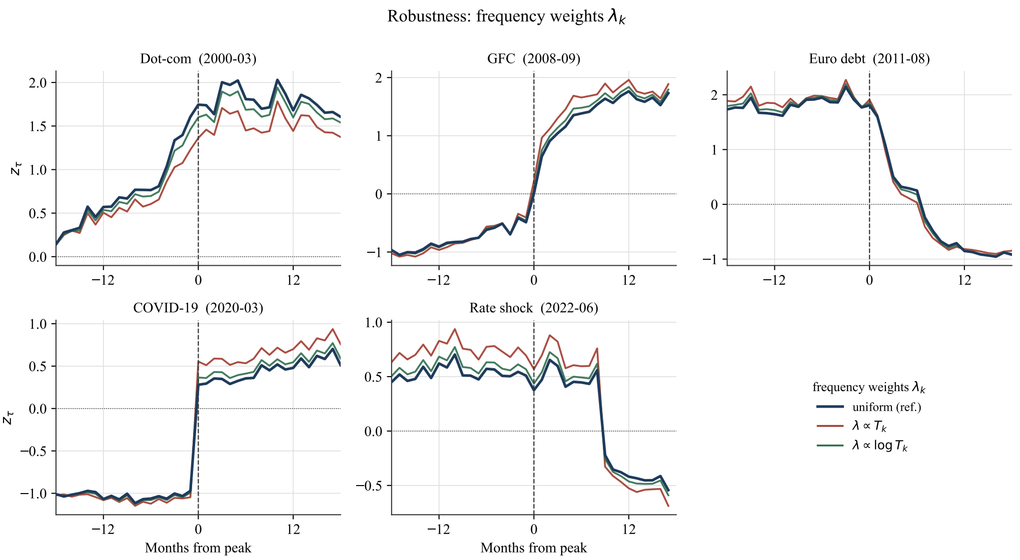


**Figure 6.6:** Robustness to frequency weights $\lambda_k$. Standardized signal $z_\tau$ around each stress episode (±18 months), under uniform weighting (reference), observation-proportional weighting $\lambda \propto T_k$, and log-proportional weighting $\lambda \propto \log T_k$. Dashed line: external crisis peak.

Both alternative weighting schemes remain strongly correlated with the uniform specification. Log-proportional weighting is almost indistinguishable from the reference over the full sample (corr = 0.999), whereas weighting proportional to the number of observations exhibits slightly lower, but still high, concordance (corr = 0.988). The differences under the latter are most visible around the dot-com episode.

This difference is consistent with the structure of the weights: $\lambda_k \propto T_k$ mechanically assigns a dominant role to the daily frequency, which has the largest number of observations. The figure does not, however, permit the observed differences to be attributed to any particular economic mechanism. Uniform weights remain the reference specification because of their neutrality across horizons, not because the alternative weights are empirically invalidated.

#### 6.1.4.3 Inter-Frequency Scaling

Figure 6.7 compares deterministic rescaling by $\sqrt{h_k}$ with normalization by the realized volatility of each frequency. Unlike the preceding diagnostics, this comparison changes the normalization family rather than a single numerical parameter.

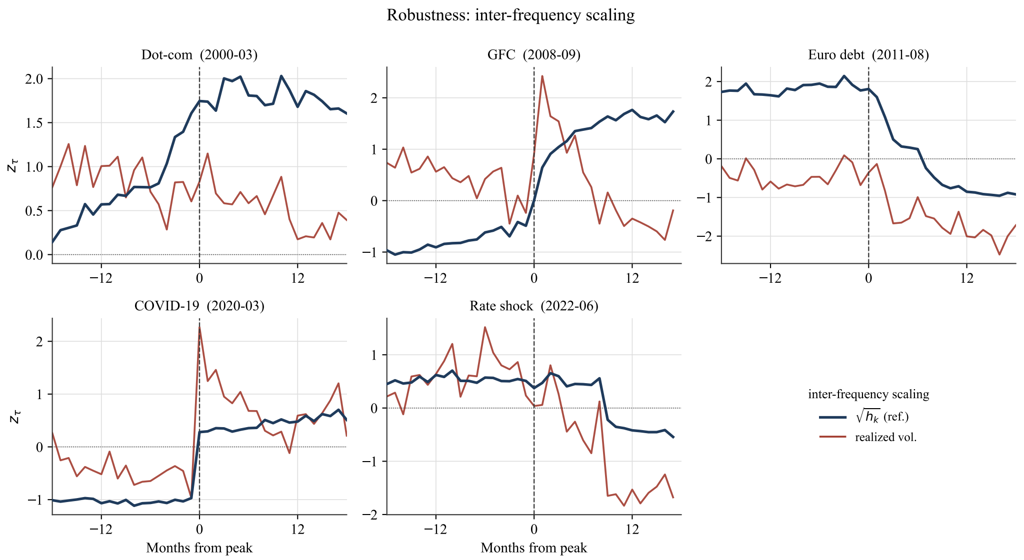


**Figure 6.7:** Sensitivity to inter-frequency scaling. Standardized signal $z_\tau$ around each stress episode ($\pm 18$ months), under temporal rescaling by $\sqrt{h_k}$ (reference) and normalization by the realized volatility of each frequency. Unlike the other levers, the two series diverge sharply and even have opposite signs in some instances: normalizing by volatility may attenuate part of the scale misalignment that the signal is intended to measure. Dashed line: external crisis peak.

The low full-sample correlation with the reference (corr = 0.203) confirms that the two normalizations produce markedly different dynamics. Around the global financial crisis, the realized-volatility-normalized variant exhibits a brief peak immediately after the external date and then declines rapidly, whereas the reference signal rises more gradually and remains elevated. Around the sovereign debt crisis, the two series even move in opposite directions. Volatility normalization therefore changes not only the amplitude of the signal but also some of its temporal profiles.

This divergence follows directly from the construction of the two normalizations. During periods of stress, realized volatility rises simultaneously across frequencies. Normalizing each frequency by its own volatility therefore tends to absorb part of the inter-horizon scale difference, even though this difference is one of the phenomena the signal is designed to measure. This variant should consequently be interpreted not as a simple alternative numerical adjustment, but as a specification that measures a different object. Rescaling by $\sqrt{h_k}$, grounded in the temporal aggregation of returns, is therefore integral to the economic interpretation of $\widehat{\rho}_\tau^{\text{hyb}}$.

#### 6.1.4.4 Scaling Exponent

This sensitivity to the choice of normalization must be distinguished from a change in the exponent within the deterministic family. When $H$ moves from its reference value of 0.5 to 0.4 or 0.6, the full-sample correlations remain 0.991 and 0.998, respectively (Table 6.1).

Figure 6.8 shows that this variation primarily affects the level of the signal, whereas Figure 6.9 confirms that the local profiles around stress episodes remain broadly concordant. This stability follows from the deterministic nature of the $h_k^H$ factor, which is fixed by horizon and independent of the observations in the window. The results therefore indicate that the signal dynamics are relatively insensitive to moderate variations in $H$ around 1/2, without implying exact invariance to the exponent.

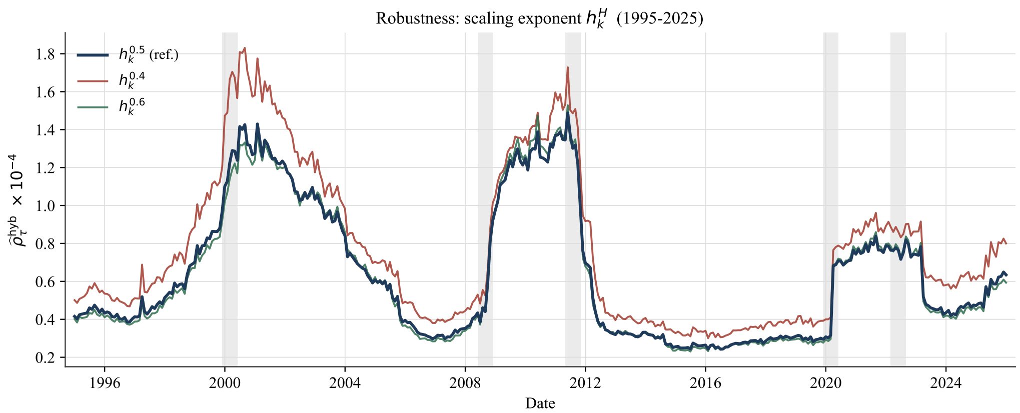


**Figure 6.8:** Sensitivity to the scaling exponent over the full series. Raw $\widehat{\rho}_\tau^{\text{hyb}}$ signal over 1995–2025 under $h_k^H$, $H \in \{0.4, 0.5, 0.6\}$. The temporal profiles remain highly concordant, with differences primarily concentrated in their levels (correlations of 0.991 and 0.998 with the $H = 1/2$ reference). Shaded bands: stress episodes.

Figure 6.9 complements this full-sample view by isolating the windows surrounding the five stress episodes. It makes it possible to verify whether the strong overall concordance conceals any local differences during periods of stress.

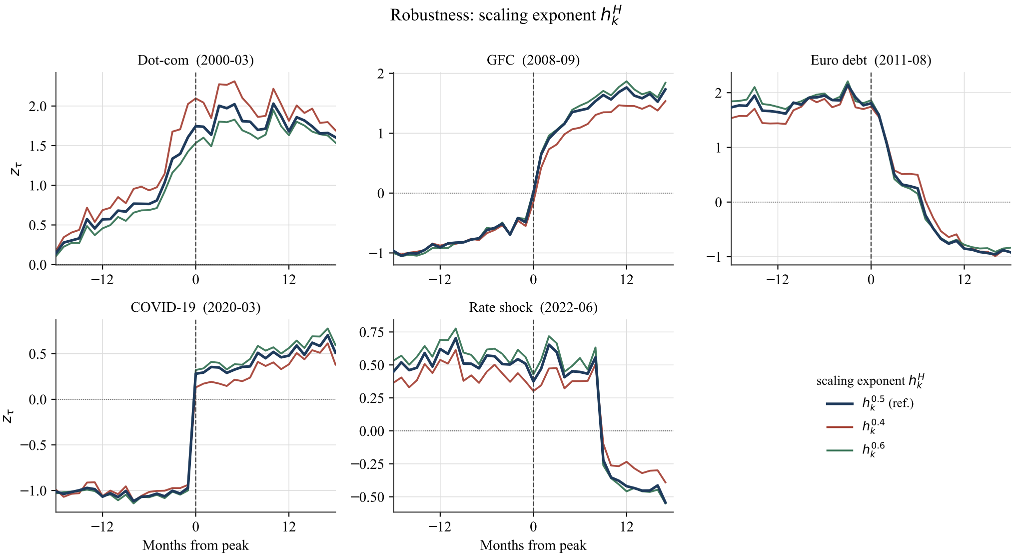


**Figure 6.9:** Sensitivity to the scaling exponent around crises. Standardized signal $z_\tau$ ($\pm18$ months) under $h_k^H$, $H \in \{0.4, 0.5, 0.6\}$. The profiles around the five episodes remain similar despite some local differences in amplitude. Dashed line: external crisis peak.

#### 6.1.4.5 Transport Distance

Figure 6.10 compares the reference Sliced-Wasserstein distance with the discrete $W_2$ distance, evaluated without projection around the same free-support barycenter. This variant removes the projection approximation when evaluating distances, without thereby providing a ground truth independent of the barycenter construction.

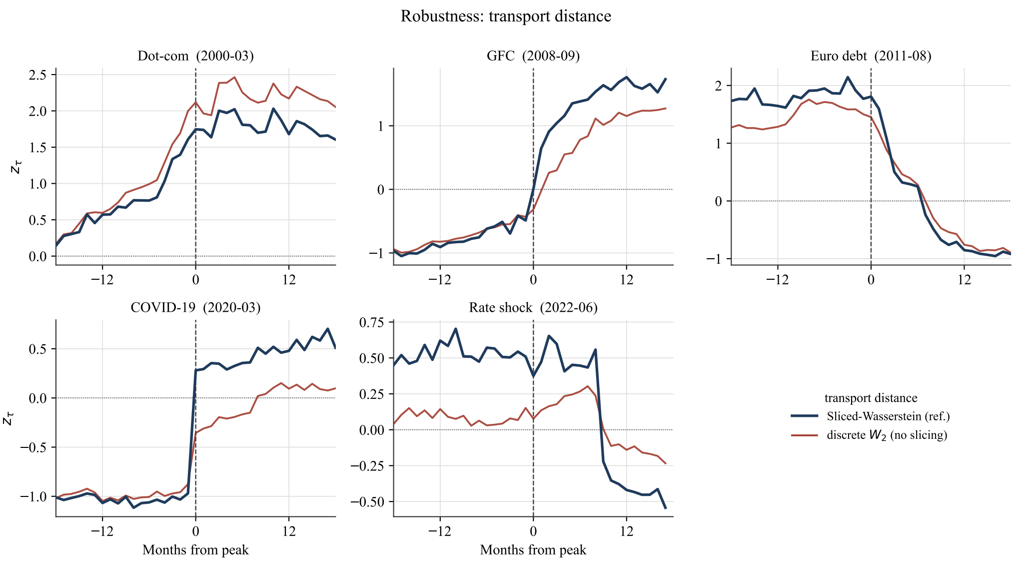


**Figure 6.10:** Robustness to the transport distance. Standardized signal $z_\tau$ around each stress episode ($\pm18$ months), under the Sliced-Wasserstein distance (reference) and the discrete $W_2$ distance, evaluated without projection around the same barycenter. The main temporal profiles remain similar despite some local differences in amplitude. Dashed line: external crisis peak.

The full-sample correlation between the two series reaches 0.958, indicating strong concordance without numerical identity. It remains high over the first two subperiods (0.998 and 0.983) and then declines to 0.916 over 2015–2025. Around stress episodes, the main temporal transitions remain similar, but differences in amplitude emerge, particularly after the pandemic shock and during the monetary tightening episode.

This comparison shows that the main temporal profiles do not depend exclusively on the projection approximation. It does not, however, determine which of the two metrics is the more accurate estimator: the discrete $W_2$ distance and the Sliced-Wasserstein distance have different statistical properties in high dimensions. The result should therefore be interpreted as a diagnostic of concordance, rather than as an unequivocal validation of one by the other.

#### 6.1.4.6 Barycenter Geometry

Figure 6.11 compares the free-support barycenter in $\mathbb{R}^N$ with a one-dimensional quantile construction performed separately within each projection.

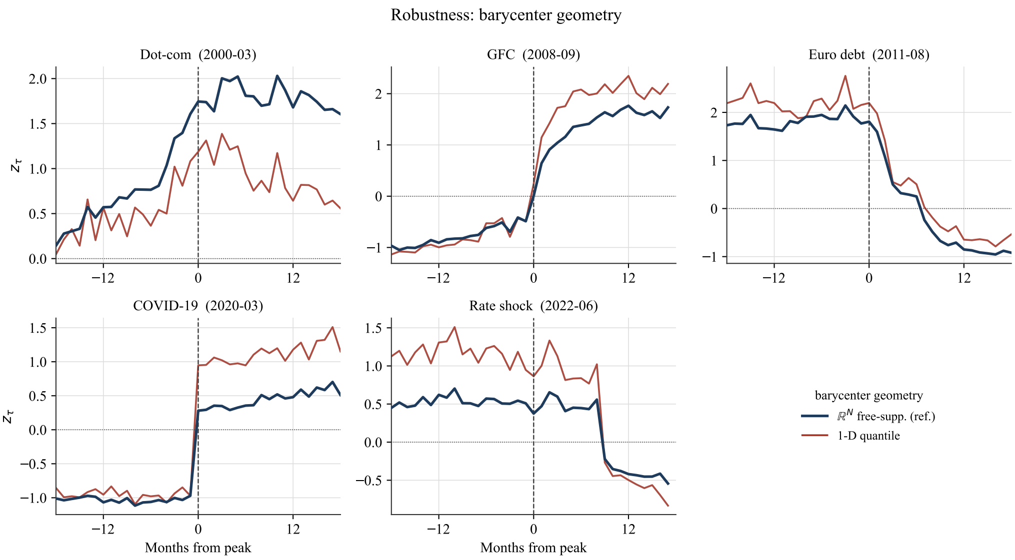


**Figure 6.11:** Robustness to barycenter geometry. Standardized signal $z_\tau$ around each stress episode ($\pm$18 months), under the free-support barycenter in $\mathbb{R}^N$ (reference) and the purely one-dimensional quantile barycenter. The main temporal profiles remain recognizable, but the differences are more pronounced than for the purely numerical variants. The one-dimensional construction replaces the common multivariate support with a separate quantile barycenter for each projection. Dashed line: external crisis peak.

The full-sample correlation with the reference reaches 0.883. It equals 0.934, 0.984, and 0.963, respectively, over the three subperiods. The one-dimensional construction therefore retains a substantial part of the signal's temporal profile without producing a series numerically identical to that obtained from the multivariate barycenter.

The observed differences are more pronounced than for the purely numerical variants. They arise from the difference in construction: the reference specification uses a common multivariate support, whereas the one-dimensional variant constructs a separate quantile barycenter within each projection. The result therefore indicates substantial temporal concordance, but not invariance to barycenter geometry.

### 6.1.5 Relationship with Volatility Measures

Consistent with the protocol in Section 5.1.5, we evaluate the extent to which $\sqrt{\widehat{\rho}_\tau^{\mathrm{hyb}}}$ shares its empirical variation with two conventional volatility measures. The first is market-implied volatility, denoted $\mathrm{IV}_\tau$ and measured by the CBOE VIX (see Note ii). The second is realized volatility, denoted $\mathrm{RV}_\tau$, constructed from daily returns.

The analysis is conducted at the monthly frequency and over a 36-month horizon corresponding to the reference formation window. At each horizon, the relationships are examined in levels and first differences, using univariate and joint specifications. The objective is to quantify the component shared by the series, without interpreting the regressions as a causal test or as evidence of structural independence.

Figure 6.12 first provides a descriptive comparison of the three standardized monthly series. The signal and the two volatility measures frequently rise during the main episodes of stress, revealing substantial co-movement. Their paths do not, however, overlap: the amplitude, persistence, and speed at which increases dissipate differ across episodes. This graphical evidence motivates a formal assessment of their relationship but does not by itself establish that the signal contains a distinct component.

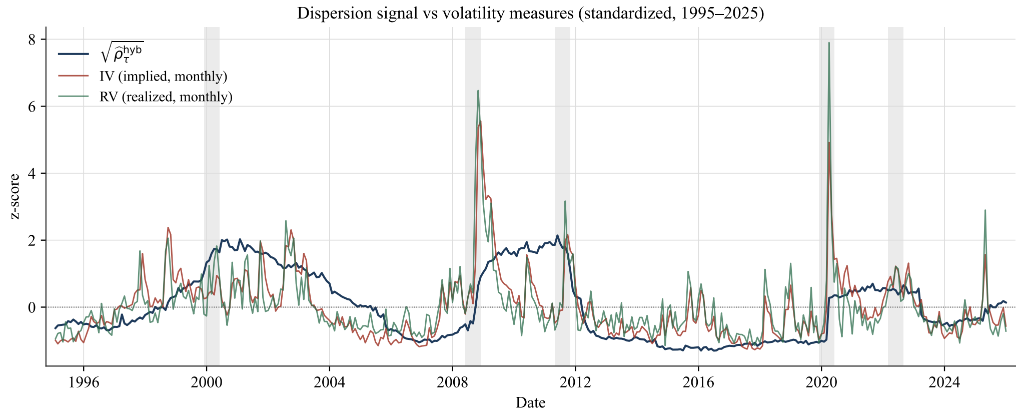


**Figure 6.12:** Dispersion signal and monthly volatility measures, standardized series (1995–2025). Signal $\sqrt{\widehat{\rho}_\tau^{\text{hyb}}}$, implied volatility $\text{IV}_\tau$, and realized volatility $\text{RV}_\tau$, each centered and standardized. Shaded bands: externally dated stress episodes.

Table 6.2 documents first-order autocorrelation and the augmented Dickey–Fuller test for the signal and the four volatility measures. These diagnostics help assess the persistence of the series before the regressions in levels and first differences are interpreted.

**Table 6.2:** First-order autocorrelation and augmented Dickey–Fuller test for $\sqrt{\widehat{\rho}_\tau^{\text{hyb}}}$, implied volatility, and realized volatility at the monthly and 36-month horizons over 1995–2025. The null hypothesis of the ADF test is the presence of a unit root.

| Series | AR(1) | ADF stat. | $p$-value | Unit root |
|---|---|---|---|---|
| $\sqrt{\widehat{\rho}^{\text{hyb}}}$ | 0.990 | −2.72 | 0.070 | not rejected |
| IV (monthly) | 0.843 | −4.96 | 0.000 | rejected |
| RV (monthly) | 0.637 | −9.06 | 0.000 | rejected |
| IV (36m) | 0.998 | −4.31 | 0.000 | rejected |
| RV (36m) | 0.992 | −3.05 | 0.030 | rejected |

The signal has a first-order autocorrelation of 0.990, and the null hypothesis of a unit root is not rejected at the 5 % level ($p = 0.070$). Monthly implied and realized volatility are less persistent, with autocorrelations of 0.843 and 0.637, respectively, and the ADF tests reject a unit root in both cases. The 36-month transformations are highly persistent (0.998 for IV and 0.992 for RV), although the ADF tests also reject a unit root at conventional significance levels. Results in levels should therefore be interpreted cautiously alongside the first-difference estimates.

Table 6.3 reports the coefficients of determination from the univariate and joint specifications at the monthly and 36-month horizons, in both levels and first differences.

**Table 6.3:** Regressions of $\sqrt{\widehat{\rho}_{\tau}^{\,\mathrm{hyb}}}$ on implied and realized volatility at the monthly and 36-month horizons, in levels and first differences, over 1995–2025. The first panel reports the coefficients of determination from the univariate and joint specifications; the second reports the coefficients from the joint specification in levels. Inference is based on Newey–West HAC standard errors with 36 lags. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

| | Monthly horizon | | 36-month horizon | |
|---|---|---|---|---|
| | Levels | Differences | Levels | Differences |
| $R^2$*: signal explained by...* | | | | |
| implied vol. (IV) alone | 0.289 | 0.101 | 0.728 | 0.232 |
| realized vol. (RV) alone | 0.133 | 0.074 | 0.699 | 0.569 |
| IV and RV jointly | 0.346 | 0.102 | 0.747 | 0.573 |
| *Joint coefficients (levels, HAC)* | | | | |
| IV | $+0.0003^{***}$ | | $+0.0002^{*}$ | |
| RV | $-0.0107^{***}$ | | $+0.0121$ | |
| corr(IV, RV) | 0.885 | | 0.915 | |
| $n$ | 372 | 371 | 372 | 371 |

At the monthly horizon, implied volatility alone accounts for 28.9 % of the variation in the signal in levels, compared with 13.3 % for realized volatility. Their joint specification raises $R^2$ to 34.6 %. In first differences, explanatory power reaches 10.1 % for IV, 7.4 % for RV, and 10.2 % for their combination. At this frequency, adding RV therefore contributes virtually no explanatory power beyond that provided by IV alone.

The relationship is stronger when the horizons are matched to the 36-month formation window. In levels, the $R^2$ values reach 72.8 % for IV, 69.9 % for RV, and 74.7 % in the joint specification. In first differences, RV explains 56.9 % of the variation in the signal, compared with 23.2 % for IV; their combination reaches 57.3 %. The incremental information contributed by IV beyond RV therefore remains limited in the latter specification.

The two volatility measures are themselves strongly correlated (0.885 at the monthly horizon and 0.915 at 36 months). This collinearity makes their joint coefficients less directly interpretable than the comparison of their $R^2$ values. In particular, the negative coefficient on RV in the joint monthly regression should not be interpreted as an autonomous negative economic relationship: RV is positively associated with the signal when entered alone.

These results reveal a substantial common component between the signal and volatility, particularly at the formation horizon. They do not, however, establish complete empirical identity: the residual share reaches 25.3 % in levels and 42.7 % in first differences at 36 months, compared with 65.4 % and 89.8 %, respectively, at the monthly horizon. This unexplained share does not, by itself, constitute evidence of economic independence or non-redundancy; it indicates more cautiously that the two volatility measures do not fully reproduce the signal's variation in these linear specifications.

## 6.2 Portfolio Results

This subsection evaluates the portfolios defined in Section 5.2 using the metrics and inferential procedures presented in Section 5.3. The analysis follows the transmission of the signal through to realized performance: we first examine the modulation of the ambiguity radius and its effect on the weights, and then compare the static and dynamic portfolios. We subsequently assess the validity of the signal's timing, locate the periods with which the gain is associated, and position the strategies within the robustness continuum. Finally, we study cross-validation calibration, generalization to the NYSE small–mid-cap universe, and the robustness of the results.

### 6.2.1 The Radius at Work

Consistent with the calibration defined in Section 5.2.2, the static portfolio retains the interior radius $\varepsilon_0 = 0.3577$, whereas the dynamic portfolio applies the point-in-time modulation constructed from the dispersion signal. Figure 6.13 presents the radii associated with the 372 holding months between 1995 and 2025.

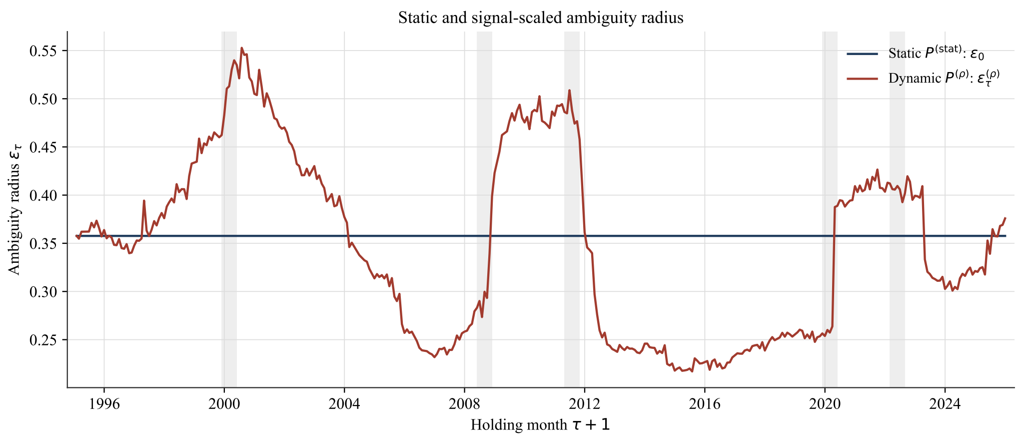


**Figure 6.13:** Paths of the static and dynamic ambiguity radii over 1995–2025. Portfolio $P^{(\text{stat})}$ retains $\varepsilon_0 = 0.3577$, whereas $P^{(\rho)}$ uses $\varepsilon_0\sqrt{\rho_\tau}/\langle\sqrt{\rho}\rangle_\tau$. The radius is computed at the close of the formation month and associated with the following holding month. The shaded bands represent independently dated stress episodes.

The dynamic radius ranges from 0.217 to 0.553, with a median of 0.348. Its highest values are concentrated notably around the bursting of the technology bubble, whereas its level contracts during periods of low dispersion. The path therefore translates the temporal variation in the signal documented in Section 6.1 into the decision space.

Its mean, equal to 0.348, remains close to the static level without being strictly identical to it. Expanding normalization therefore keeps the two radii on a comparable scale but does not impose retrospective equality of their full-sample means. Moreover, the dynamic radius remains strictly between the numerical floor $\varepsilon_{\min} = 10^{-4}$ and the safety cap $\varepsilon_{\max} = 2$. Projection is therefore never active in the central comparison.

This first step establishes the transmission of the signal to the ambiguity radius. It is not, however, sufficient to demonstrate that this modulation materially affects the allocation. The next subsection directly examines the relationship between the radius and weight concentration.

### 6.2.2 From the Radius to the Weights

The radius has economic consequences only through its transmission to the weights. As explained in Section 5.2.3, a larger radius strengthens the penalty on concentration and moves the allocation toward the diversified limit. We document this mechanism by solving the DRO problem at every formation date over a prespecified grid of constant radii. Concentration is measured by

$$\|w_\tau(\varepsilon)\|_2 = \left(\sum_{i=1}^{N} w_{i,\tau}(\varepsilon)^2\right)^{1/2},$$

whose minimum under the budget constraint is $1/\sqrt{N} = 0.10$ for $N = 100$.

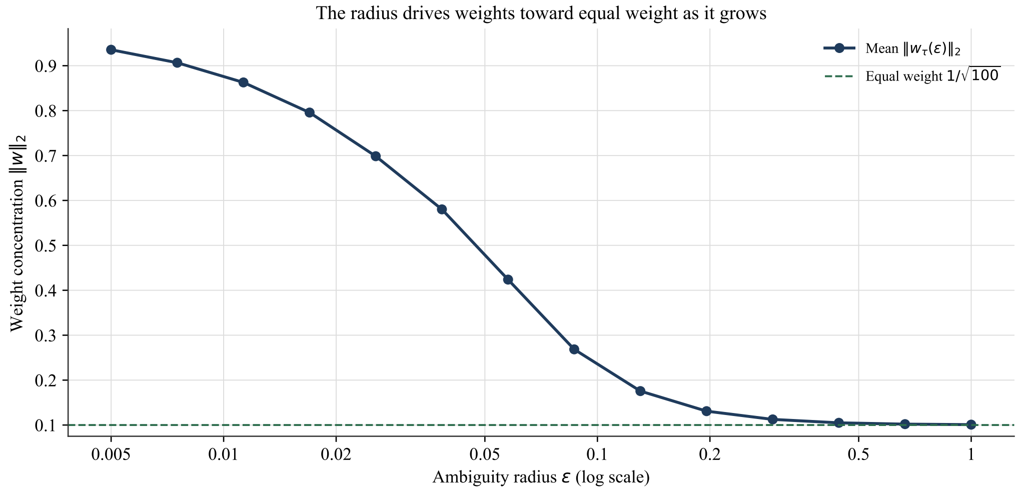


**Figure 6.14:** Mean weight concentration $\|w_\tau(\varepsilon)\|_2$ as a function of a constant radius $\varepsilon$, across all formation dates. The horizontal axis is logarithmic. The dashed line represents the concentration of the equally weighted portfolio, $1/\sqrt{100} = 0.10$. The numerical floor $\varepsilon_{\min} = 10^{-4}$, at which mean concentration equals 0.999, is indicated separately because it lies outside the displayed vertical scale.

The relationship is clearly decreasing. Near the numerical floor, the empirical optimizer is almost fully concentrated: mean concentration reaches 0.999, while its median and ninth decile both equal 1. It remains high at $\varepsilon = 0.005$, where it equals 0.936, before declining to 0.268 around $\varepsilon = 0.087$ and to 0.131 around $\varepsilon = 0.196$. At larger radii, the curve gradually flattens near the equally weighted limit; it reaches 0.1009 at $\varepsilon = 1$.

At the operational radius $\varepsilon_0 = 0.3577$, the static portfolio's mean concentration equals 0.1080. The portfolio is therefore already in a highly diversified region, but before the curve becomes completely flat. This geometric proximity does not, however, mean that its weight vector is exactly equal-weighted: $\|w\|_2$ summarizes the dispersion of the weights, not their composition. Cross-sectional differences remain and still allow radius modulation to materially alter the allocation.

The choice of $\varepsilon_0$ precedes the evaluation of the continuum and follows the operational rule defined in Section 5.2.2. Its proximity to the equal-weight plateau is therefore an empirical outcome of this calibration, rather than a point selected ex post to maximize the timing gain.

The figure thus establishes the intended mechanical channel: robustness rapidly reduces concentration, after which its marginal effect weakens as the portfolio approaches the equally weighted limit. The following comparison determines whether the residual adjustments generated by the timing of $\rho_\tau$ translate into realized performance.

### 6.2.3 Static versus Dynamic: Central Comparison

The central comparison contrasts $P^{(\text{stat})}$, which retains radius $\varepsilon_0$, with $P^{(\rho)}$, which applies the point-in-time modulation defined in Section 5.2.1. The two portfolios use the same universe, estimation matrices, and accounting framework. Only the path of the radius, and hence that of the target weights, differs. Net-of-cost results constitute the primary basis for evaluation; gross results are retained as an audit.

**Table 6.4:** Performance, risk, and reallocation of portfolios $P^{(\text{stat})}$ and $P^{(\rho)}$ over 1995–2025, net of transaction costs. The HAC test concerns the mean difference in paired monthly returns. The interval and $p$-value for $\Delta$SR are obtained using a paired stationary bootstrap.

| | $P^{(\text{stat})}$ | $P^{(\rho)}$ |
|---|---|---|
| Return $\mu$ (%) | 11.20 | 11.66 |
| Volatility $\sigma$ (%) | 16.26 | 15.64 |
| Sharpe | 0.689 | 0.745 |
| Max drawdown | 0.617 | 0.528 |
| date | 2009-02 | 2003-02 |
| duration (months) | 169 | 155 |
| $\text{CVaR}_{95}$ | 0.106 | 0.098 |
| Turnover (two-way) | 0.1116 | 0.1152 |
| $\Delta$SR | +0.0559 | |
| Paired HAC mean test | $t = 1.07$, $p = 0.287$ | |
| Stationary bootstrap $\Delta$SR | $[-0.0053, +0.1256]$, $p = 0.113$ | |

The dynamic portfolio has an annualized mean return of 11.66 %, compared with 11.20 % for the static portfolio. Its annualized volatility is also lower (15.64 % versus 16.26 %), so that its Sharpe ratio reaches 0.745, compared with 0.689. The observed gain is therefore

$$\Delta \text{SR}^{\text{net}} = +0.0559.$$

The other risk measures move in the same direction. Maximum drawdown declines from 0.617 to 0.528. The longest continuous period spent below the previous peak falls from 169 to 155 months. The maximum-drawdown trough occurs in February 2009 for the static portfolio, compared with February 2003 for the dynamic portfolio. Monthly 95 % CVaR declines from 0.106 to 0.098. In point estimates, $P^{(\rho)}$ therefore combines a higher mean return with lower volatility, tail-risk, and maximum-drawdown measures.

Recurring two-way turnover increases from 0.1116 to 0.1152. Transaction costs nevertheless reduce the contrast only marginally: the Sharpe-ratio gain is +0.0571 before costs and +0.0559 after costs. Their effect on the difference between the two strategies therefore remains limited.

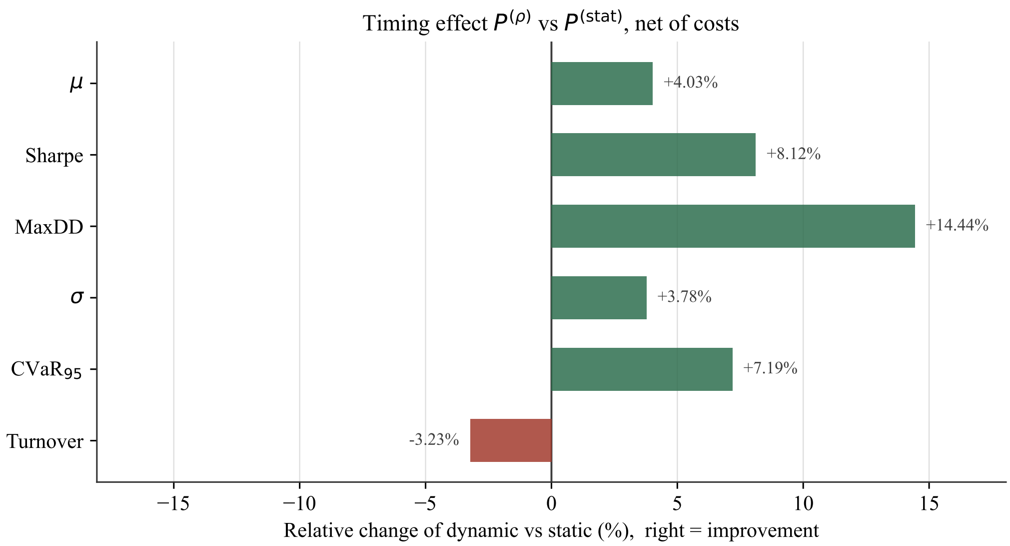


**Figure 6.15:** Relative change in the net metrics of $P^{(\rho)}$ compared with $P^{(\text{stat})}$. Changes are oriented so that a positive value always represents an improvement: the sign is therefore reversed for volatility, maximum drawdown, CVaR, and turnover.

Figure 6.15 confirms that the point-estimate improvement is economically visible. The Sharpe ratio rises by 8.12 % in relative terms and the mean return by 4.03 %. The relative reductions in volatility, maximum drawdown, and CVaR reach 3.78 %, 14.44 %, and 7.19 %, respectively. In return, turnover rises by 3.23 %.

Statistical uncertainty nevertheless calls for caution in interpreting these differences. The paired mean-difference test with HAC standard errors yields $t = 1.07$ and $p = 0.287$. It therefore does not reject equality of mean returns. For the Sharpe-ratio difference, the stationary bootstrap gives a 95 % interval of $[-0.0053, +0.1256]$ and a two-sided $p$-value of 0.113. The point estimate is economically positive, but its interval contains zero and the effect is not statistically established at the 5 % level.

The central comparison thus identifies an economically visible but statistically imprecise improvement in the point estimate. By itself, it is not sufficient to establish that the timing of $\rho_\tau$ is informative. The permutation tests presented next examine this question under a different null hypothesis: randomly reordered signal paths.

### 6.2.4 Validity of Signal Timing: Placebo Tests

The preceding comparison documents an observed gain but does not determine whether it genuinely arises from the timing of $\rho_\tau$. We therefore apply the three Level 1 placebos defined in Section 5.2.5. In each case, the modulation values are retained while their temporal order is rearranged. The universes, estimation matrices, holding returns, costs, and accounting framework remain unchanged.

The simple permutation completely destroys the temporal ordering of the signal while preserving its marginal distribution. The six- and twelve-month block permutations retain progressively more of its local persistence. For each permuted signal path, the dynamic portfolio is fully resolved and reconstructed over all 372 months. The Level 1 statistic is its net Sharpe ratio, consistent with the definition in Section 5.3.2.

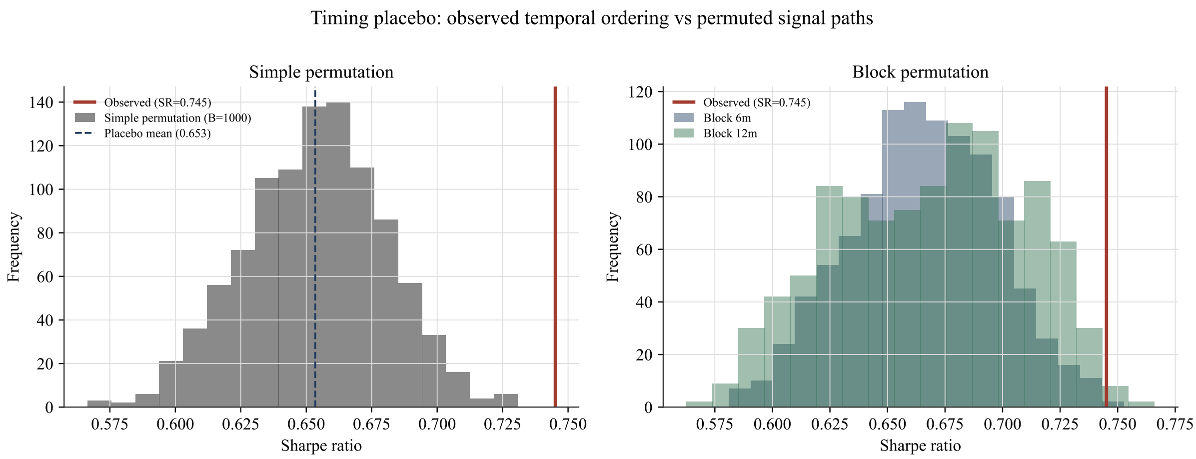


**Figure 6.16:** Distribution of the dynamic portfolio's net Sharpe ratio under $B_{\text{perm}} = 1000$ permuted signal orderings. The left panel shows the simple monthly permutation; the right panel shows the six- and twelve-month block permutations. The red line indicates the Sharpe ratio of the observed temporal ordering, equal to 0.745.

Figure 6.16 visually locates the observed temporal ordering within each null distribution. Table 6.5 complements this diagnostic by reporting the observed percentiles and corresponding $p$-values.

**Table 6.5:** Comparison of the observed net Sharpe ratio with the distributions obtained under simple and block permutation. The one-sided $p$-values are computed using the finite-sample correction defined in Section 5.3.2.

| Placebo | Preserves | Real percentile | $p$-value |
|---|---|---|---|
| Simple permutation | marginals | 100.0 | 0.001 |
| Block 6 months | marginals, short persistence | 99.8 | 0.003 |
| Block 12 months | marginals, long persistence | 99.2 | 0.009 |

The net Sharpe ratio of the observed temporal ordering equals 0.7451. Under simple permutation, none of the 1000 draws reaches or exceeds this value. The observed signal is therefore at percentile 100.0 of the placebo distribution, yielding

$$p_{\text{simple}} = \frac{0+1}{1000+1} = 0.001.$$

When local persistence is partially preserved, the conclusion gradually weakens but remains unchanged at the 5 % level. The six-month block permutation produces two exceedances ($p = 0.003$), whereas the twelve-month block permutation produces eight ($p = 0.009$). The means of the null distributions are 0.6535, 0.6655, and 0.6682, respectively, all below the observed ratio.

These results indicate that the performance of the dynamic portfolio is not reproduced by an arbitrary rearrangement of the modulation values. The persistence of the result under block permutations also suggests that it is not explained solely by the marginal distribution or by the share of local persistence retained by these rearrangements. This interpretation nevertheless remains specific to the three null hypotheses considered and does not constitute causal evidence.

Nor does it contradict the uncertainty identified in the central comparison. The bootstrap in Section 6.2.3 evaluates whether $\Delta\text{SR}$ differs from zero under resampling of realized returns. The placebos address a different question: whether the observed temporal ordering of the signal produces higher performance than that obtained after rearranging it. Under this second null hypothesis, all three tests reject at the 5 % level.

### 6.2.5 Conditional Attribution

The preceding results establish that the observed temporal ordering produces higher performance than the permuted signal paths. We now examine the periods with which this advantage is associated, following the descriptive protocol defined in Section 5.3.2. This attribution is conducted net of transaction costs.

The 372 months are first divided according to whether the value of $\sqrt{\rho_\tau}$, observed at the formation date, lies above or below its full-sample median. The high- and low-dispersion regimes therefore each contain 186 observations. This classification is ex post: it is used solely to describe the results and does not constitute a point-in-time investment rule.

We supplement this partition with five prespecified ±12-month windows around the stress episodes used to validate the signal. Each window contains 25 monthly returns. The Sharpe-ratio differences computed within them are descriptive diagnostics based on small subsamples, rather than additional hypothesis tests.

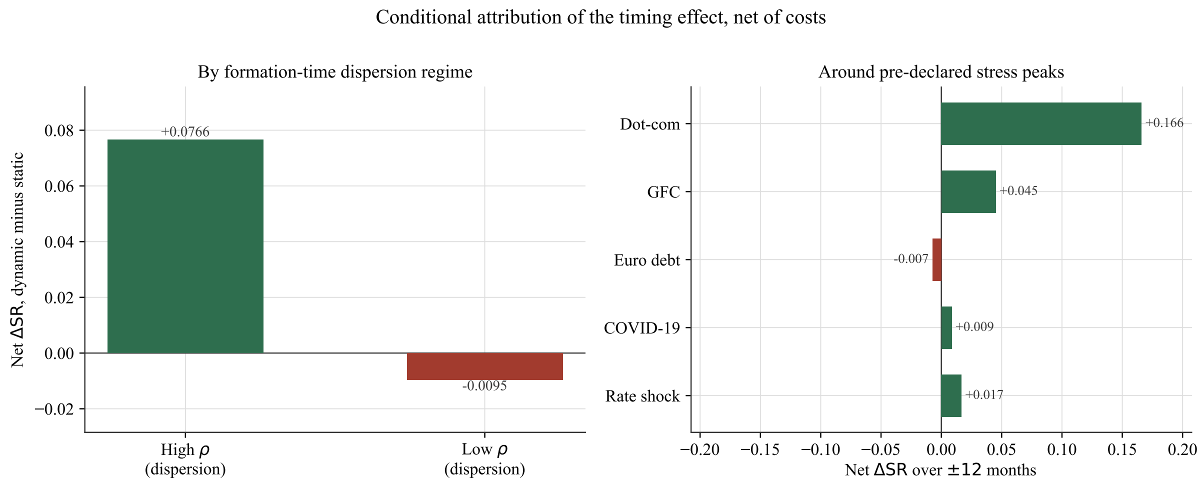


**Figure 6.17:** Net Sharpe-ratio difference between $P^{(\rho)}$ and $P^{(\text{stat})}$. The left panel distinguishes high- and low-dispersion regimes defined using the full-sample median of $\sqrt{\rho_T}$. The right panel presents the differences computed over the ±12-month windows surrounding the five prespecified stress episodes.

Figure 6.17 compares risk-adjusted performance within each subsample. Table 6.6 complements this interpretation with conditional Sharpe ratios and the additive attribution of the mean return difference.

**Table 6.6:** Net performance of the static and dynamic portfolios by dispersion regime and around stress episodes. For the regimes, the mean contribution represents the share of the full-sample mean difference, expressed in annualized return percentage points. Sharpe-ratio differences are not additive.

| Regime | $n$ | Sharpe $P^{(\text{stat})}$ | Sharpe $P^{(\rho)}$ | $\Delta$SR | Mean contribution (pp) |
|---|---|---|---|---|---|
| High $\rho$ | 186 | 0.471 | 0.547 | +0.0766 | +0.428 |
| Low $\rho$ | 186 | 1.069 | 1.059 | -0.0095 | +0.023 |
| All | 372 | 0.689 | 0.745 | +0.0559 | +0.451 |
| Panel B: pre-declared ±12-month stress windows | | | | | |
| Window | $n$ | Sharpe $P^{(\text{stat})}$ | Sharpe $P^{(\rho)}$ | $\Delta$SR | $\Delta\mu$ (pp, ann.) |
| Dot-com | 25 | -0.286 | -0.120 | +0.1660 | +5.373 |
| GFC | 25 | -0.505 | -0.460 | +0.0451 | +0.724 |
| Euro debt | 25 | 0.827 | 0.820 | -0.0074 | -0.100 |
| COVID-19 | 25 | 1.136 | 1.145 | +0.0086 | +0.090 |
| Rate shock | 25 | 0.201 | 0.218 | +0.0166 | +0.286 |

In the high-dispersion regime, the Sharpe ratio rises from 0.471 for $P^{(\mathrm{stat})}$ to 0.547 for $P^{(\rho)}$, giving $\Delta\mathrm{SR}_{\mathcal{H}} = +0.0766$. In the low-dispersion regime, it declines from 1.069 to 1.059, giving $\Delta\mathrm{SR}_{\mathcal{L}} = -0.0095$. The improvement in risk-adjusted performance is therefore concentrated in high-dispersion states; it reverses slightly in low-dispersion states, where the absolute Sharpe-ratio level nevertheless remains higher.

The attribution of mean returns leads to a similar conclusion. The high-dispersion regime contributes +0.428 percentage points of annualized return, compared with +0.023 percentage points for the low-dispersion regime. Their sum, +0.451 percentage points, recovers the annualized mean difference observed over the full sample. Approximately 95 % of this mean-return difference is therefore associated with high-dispersion months. This decomposition concerns mean returns rather than $\Delta$SR, which is not additive.

The episode-level analysis reveals more pronounced heterogeneity. The Sharpe-ratio gain is particularly high around the technology bubble (+0.1660) and then around the global financial crisis (+0.0451). It remains positive but smaller around the monetary tightening episode (+0.0166) and the pandemic shock (+0.0086). Around the European sovereign debt crisis, it is slightly negative (−0.0074).

The observed advantage is therefore neither uniform over time nor common to all stress episodes. It is primarily associated with periods of high dispersion and, among the prespecified windows, with the bursting of the technology bubble and the global financial crisis. These results locate the temporal origin of the gain without assigning it a causal interpretation. The next section positions the static and dynamic portfolios within the full DRO continuum.

### 6.2.6 Positioning within the DRO Continuum

The central comparison is positioned between the two poles of the linear-loss allocation family defined in Section 5.2.1. Portfolio $P^{(\mathrm{emp})}$ uses the numerical floor $\varepsilon_{\min} = 10^{-4}$ and constitutes an operational approximation of the empirical optimizer, rather than a solution obtained with an exactly zero radius. At the other extreme, $P^{(\mathrm{EW})}$ represents the diversified equal-weight limit. Portfolios $P^{(\mathrm{stat})}$ and $P^{(\rho)}$ lie between these two poles.

All four paths use the same point-in-time universes, holding returns, delisting conventions, weight-drift rules, and transaction costs. Covariance-based portfolios are not included because they optimize a different objective and do not belong to this continuum.

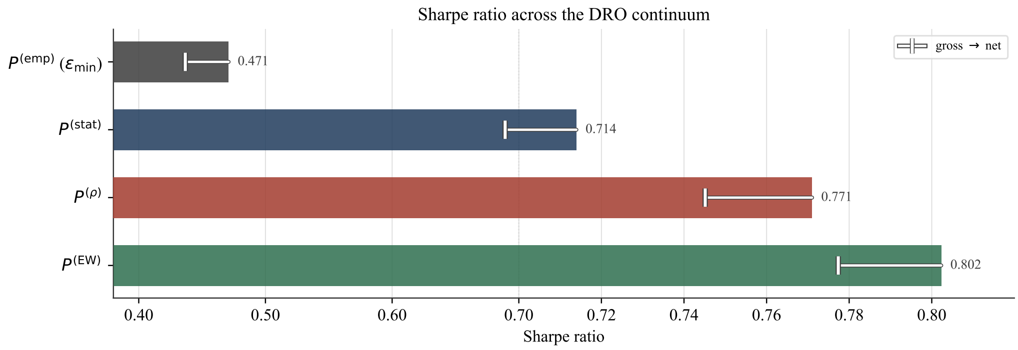


**Figure 6.18:** Gross Sharpe ratios of the four portfolios along the DRO continuum. For each strategy, the light segment represents the erosion in the Sharpe ratio between gross and net-of-cost results. The slope of the horizontal scale changes at 0.70 so that both the empirical pole and the smaller differences among the three diversified portfolios remain visible.

Figure 6.18 visually locates the four strategies according to their risk-adjusted performance. Table 6.7 complements this representation with the main net metrics and the gross Sharpe ratios retained as an audit of costs.

**Table 6.7:** Performance, risk, and turnover of the four portfolios over 1995–2025. The primary metrics are net of transaction costs; the gross Sharpe ratio is retained as a frictionless reference.

| | $P^{(\text{emp})}$ | $P^{(\text{stat})}$ | $P^{(\rho)}$ | $P^{(\text{EW})}$ |
|---|---|---|---|---|
| Annual return, net (%) | 21.26 | 11.20 | 11.66 | 11.55 |
| Volatility, net (%) | 48.68 | 16.26 | 15.64 | 14.86 |
| Sharpe, gross | 0.471 | 0.714 | 0.771 | 0.802 |
| Sharpe, net | 0.437 | 0.689 | 0.745 | 0.777 |
| Maximum drawdown, net | 0.961 | 0.617 | 0.528 | 0.509 |
| $\text{CVaR}_{95}$, net | 0.286 | 0.106 | 0.098 | 0.094 |
| Recurring two-way turnover | 0.4798 | 0.1116 | 0.1152 | 0.1035 |

The empirical pole has the highest annualized mean return (21.26 %), but at the cost of considerably greater risk exposure. Its volatility reaches 48.68 %, its maximum drawdown 96.1 %, and its monthly CVaR 28.6 %. Its net Sharpe ratio is therefore 0.437, well below those of the more robust portfolios. Its recurring turnover, equal to 0.4798, is also more than four times that of the strategies in the diversified region.

Regularization substantially transforms this profile. The static portfolio obtains a net Sharpe ratio of 0.689, compared with 0.745 for its dynamic counterpart. Equal weighting reaches 0.777, with volatility of 14.86 % and CVaR of 9.4 %. It therefore remains the highest-ranked portfolio by Sharpe ratio, approximately 0.032 above $P^{(\rho)}$.

Gross results produce the same ranking: 0.471 for $P^{(\mathrm{emp})}$, 0.714 for $P^{(\mathrm{stat})}$, 0.771 for $P^{(\rho)}$, and 0.802 for $P^{(\mathrm{EW})}$. Costs therefore reduce risk-adjusted performance without changing the ordering of the strategies.

This positioning clarifies the scope of the central comparison. Modulation by $\rho_\tau$ improves the static portfolio constructed around the same anchor, but does not dominate equal weighting in absolute terms. The contribution thus concerns the value of ambiguity-radius timing within the DRO family, rather than the construction of a portfolio that outperforms all passive benchmarks.

The favorable position of $P^{(\mathrm{EW})}$ naturally raises the question of which radius would be selected by directly maximizing past performance. The next subsection examines this cross-validation calibration.

### 6.2.7 Temporal Cross-Validation Calibration

We now examine the behavior of the radius when it is selected exclusively on the basis of past performance, following the procedure defined in Section 5.2.2. Sixteen radii between 0.05 and 10 are evaluated using strictly walk-forward candidate paths. At each date, the rule retains the radius that maximizes the gross Sharpe ratio computed from candidate returns already realized.

The first 36 holding months, from January 1995 to December 1997, constitute the initialization period. During this period, the radius is fixed at the numerical median of the grid, $\varepsilon_{\mathrm{init}} = 0.718$, and should not be interpreted as a cross-validation selection. The first genuine selection is made for the January 1998 holding month using the preceding 36 observations. The evaluation window then expands over time.

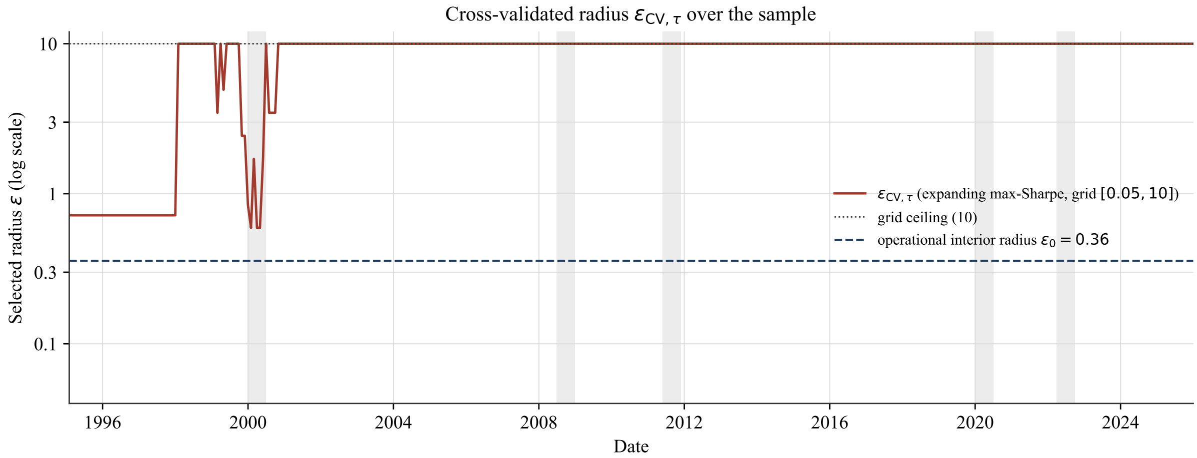


**Figure 6.19:** Path of the radius selected by expanding maximization of the past Sharpe ratio over the grid $[0.05, 10]$. The first 36 months correspond to initialization at radius $\varepsilon_{\text{init}} = 0.718$. From January 1998 onward, each selection uses only candidate returns realized before the current month. The upper horizontal line represents the grid ceiling and the lower line the operational radius $\varepsilon_0 = 0.3577$.

Figure 6.19 shows the complete selection path. Table 6.8 summarizes its behavior after the initialization period and compares the resulting portfolio with equal weighting.

**Table 6.8:** Behavior of the selected radius after the initialization period, weight concentration at the grid ceiling, and net performance of the resulting portfolio. The Sharpe ratios of the CV and equal-weighted portfolios are computed over their common period, 1998–2025.

| Diagnostic | Value |
|---|---|
| Initialization length (months) | 36 |
| Initialization radius (grid median) | 0.718 |
| First genuine CV holding month | 1998–01 |
| Median selected $\varepsilon_{\text{CV}}$ | 10.000 |
| Fraction at grid ceiling $\varepsilon = 10$ (%) | 96.1 |
| Fraction with $\varepsilon \geq 7$ (%) | 96.1 |
| Mean $\|w(10)\|_2$ | 0.1000 |
| Equal-weight $\|w^{(\text{EW})}\|_2$ | 0.1000 |
| Mean $\|w(10) - w^{(\text{EW})}\|_2$ | 0.0012 |
| Operational interior radius $\varepsilon_0$ | 0.3577 |
| Expanding-CV Sharpe, net (1998–2025) | 0.627 |
| Equal-weight Sharpe, net (1998–2025) | 0.642 |

After initialization, the median selected radius equals 10, the ceiling of the grid. This bound is retained in 96.1 % of months, and a radius greater than or equal to 7 is likewise retained in 96.1 % of cases. The procedure therefore almost never identifies an interior optimum: the past-performance criterion nearly always favors the most robust region allowed by the grid.

At $\varepsilon = 10$, mean concentration satisfies

$$\overline{\|w_\tau(10)\|_2} = 0.1000,$$

which, at the reported precision, equals the concentration of the equally weighted portfolio. Equality of the norm does not, however, imply exact identity of the weight vectors. Their mean distance from equal weighting remains 0.0012, revealing small compositional differences despite almost identical dispersion.

Over the period during which selection is effective, portfolio $P^{(\mathrm{CV})}$ obtains a net Sharpe ratio of 0.627, compared with 0.642 for $P^{(\mathrm{EW})}$. On this sample, expanding selection therefore fails to improve upon the diversified anchor it approaches.

This result characterizes a calibration that is empirically degenerate relative to its intended objective: rather than identifying an interior trade-off between exploiting estimated returns and robustness, the rule almost always selects the available upper bound. It does not establish that $\varepsilon = 10$ is a global optimum, because no larger radius is evaluated; it shows only that the optimum remains constrained by the grid ceiling.

For this reason, $P^{(\mathrm{CV})}$ is retained as a diagnostic of the behavior of performance-based calibration, but is not used to identify the central timing effect. The latter remains based on the prespecified comparison between $P^{(\mathrm{stat})}$ and $P^{(\rho)}$. We next examine whether this gain also appears in a different capitalization universe.

### 6.2.8 The Value of Timing across Capitalization Universes

We examine whether the gain associated with the timing of $\rho_\tau$ generalizes beyond the main universe. Two point-in-time universes are compared: the 100 large-capitalization stocks selected relative to the 90th NYSE percentile across the three eligible exchanges, and the 100 NYSE stocks between the 20th and 50th capitalization percentiles. Their construction is described in Section 3.4.

The dispersion signal is recomputed separately within each universe. Memberships, estimation matrices, and $\rho_\tau$ paths are therefore never transferred from one universe to the other. The two comparisons nevertheless use the same estimation window, monthly holding convention, and accounting framework.

We report results over the full sample and after excluding January 1999 through December 2001. The latter period assesses the sensitivity of the gain to the technology-bubble episode; it is not an alternative window for strategy selection.

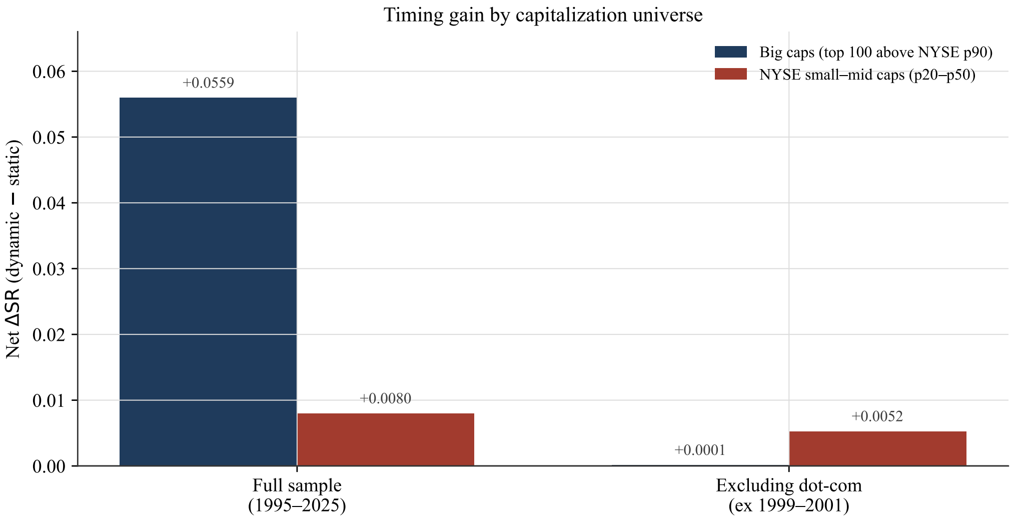


**Figure 6.20:** Net Sharpe-ratio difference between $P^{(\rho)}$ and $P^{(\text{stat})}$ in the large-capitalization universe and the NYSE small–mid-cap universe. Results are shown over the full sample and after excluding January 1999 through December 2001.

Figure 6.20 compares the four observed Sharpe-ratio differences. Table 6.9 complements this representation with marginal $p$-values from the simple monthly permutations, gross results retained as an audit, and recurring turnover. The $p$-values in this table are specific to each cell and do not yet adjust for multiplicity across the four comparisons.

**Table 6.9:** Sharpe-ratio differences between the dynamic and static portfolios in the two universes. Net results constitute the primary inference; gross results are retained as an audit. Marginal $p$-values are based on $B_{\text{perm}} = 1000$ joint monthly permutations and are not adjusted for the four reported cells.

| Metric | Big caps (top 100 above NYSE $P90$) | NYSE small–mid caps ($P20$–$P50$) |
|---|---|---|
| *Net timing inference* | | |
| Full sample ΔSR | +0.0559 | +0.0080 |
| Full sample $p$-value | 0.001 | 0.013 |
| Ex dot-com ΔSR | +0.0001 | +0.0052 |
| Ex dot-com $p$-value | 0.040 | 0.034 |
| *Gross audit* | | |
| Full sample ΔSR | +0.0571 | +0.0078 |
| Ex dot-com ΔSR | +0.0016 | +0.0055 |
| *Recurring turnover* | | |
| Static | 0.1116 | 0.4642 |
| Dynamic | 0.1152 | 0.4653 |

Over the full sample, the net gain reaches +0.0559 among large capitalizations and +0.0080 among NYSE small–mid caps. The point estimate is therefore markedly larger in the main universe.

Among large capitalizations, ΔSR falls to +0.0001 when 1999–2001 is removed. In the NYSE small–mid-cap universe, it declines from +0.0080 to +0.0052. A substantial part of the large-capitalization result is therefore associated with the technology bubble, whereas the contrast in the NYSE small–mid-cap universe is more stable in magnitude.

The marginal tests confirm this heterogeneity. For large capitalizations, the monthly permutation yields $p = 0.001$ over the full sample and $p = 0.040$ after excluding the dot-com period. In the NYSE small–mid-cap universe, the $p$-values are 0.013 and 0.034, respectively. All four cells thus reject their marginal null hypothesis at the 5 % level before adjustment for multiplicity.

Gross results are close to net results over the full sample: +0.0571 for large capitalizations and +0.0078 for the NYSE small–mid-cap universe. Excluding the dot-com period, they equal +0.0016 and +0.0055, respectively. Costs therefore determine neither the sign nor the ordering of the full-sample results, although the nearly zero large-capitalization contrast excluding the dot-com period remains more sensitive to their inclusion.

The absolute level of turnover nevertheless differs substantially. Recurring turnover rises from 0.1116 to 0.1152 between the static and dynamic strategies among large capitalizations, compared with 0.4642 and 0.4653 in the NYSE small–mid-cap universe. Within each universe, radius modulation therefore adds little trading relative to the static portfolio. The NYSE small–mid-cap universe nevertheless remains more sensitive to cost and liquidity assumptions.

These results combine a larger full-sample effect in the main universe with greater descriptive stability in the NYSE small–mid-cap universe. At this stage, however, the four marginal tests remain unadjusted; the next section controls their multiplicity using the max-$T_4$ and max-$T_{12}$ families.

**Family-wise multiplicity control.**

The preceding four $p$-values evaluate each cell separately. We now apply the max-$T$

procedure defined in Section 5.3.2 to control the probability of highlighting a favorable cell among several comparisons.

In each of the $B_{\text{perm}} = 1000$ replications, the same monthly permutation is applied simultaneously to the modulation paths in both universes. The dynamic portfolios are then fully reconstructed, and the four Sharpe-ratio differences are computed jointly. The maximum of these four statistics forms the null distribution for family $\mathcal{F}_4$. The marginal $p$-values and the max-$T_4$ $p$-values therefore come from the same panel of draws, rather than from two separate permutation experiments.

Cell $S_{\text{exDC}}$, corresponding to the NYSE small–mid-cap universe excluding the dot-com period, is retained as an illustrative reporting target. It permits a direct comparison of the corrections obtained under families $\mathcal{F}_4$ and $\mathcal{F}_{12}$. The main specification nevertheless remains $L_{\text{all}}$, whereas exclusion of the dot-com period is an exploratory sensitivity analysis.

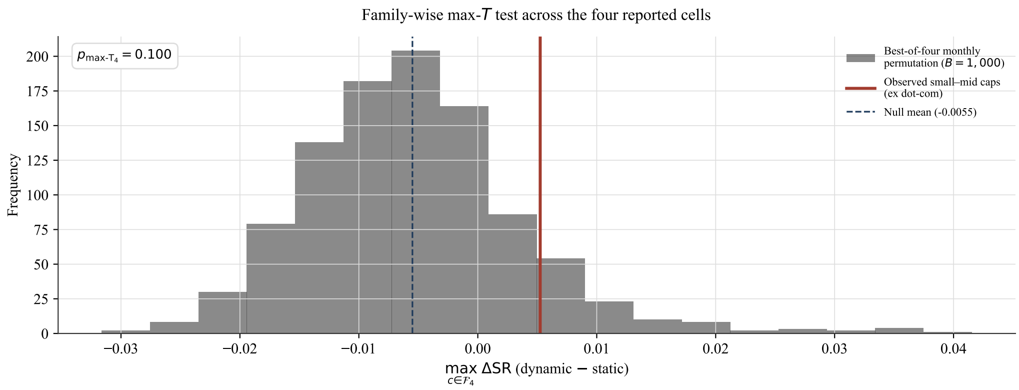


**Figure 6.21:** Distribution of the maximum of the four net Sharpe-ratio differences obtained in each monthly permutation. The red line represents the observed difference in the NYSE small–mid-cap universe excluding the dot-com period, $\Delta SR = +0.0052$, and the dashed line the mean of the null distribution. The corresponding family-wise $p$-value is $p_{\max\text{-}T_4} = 0.100$.

Figure 6.21 compares the target cell with the best difference produced by chance within the family of four cells. Table 6.10 reports the marginal and adjusted results for each of them.

**Table 6.10:** Net Sharpe-ratio differences, marginal $p$-values, and $p$-values adjusted using max-$T_4$. The four statistics are reconstructed jointly in each permutation.

| Cell | Net $\Delta$SR | Marginal $p$ | max-$T_4$ $p$ |
|---|---|---|---|
| Big caps, full sample | +0.0559 | 0.001 | 0.001 |
| Big caps, ex dot-com | +0.0001 | 0.040 | 0.224 |
| NYSE small–mid caps, full sample | +0.0080 | 0.013 | 0.065 |
| NYSE small–mid caps, ex dot-com | +0.0052 | 0.034 | 0.100 |

For large capitalizations, the full-sample cell retains an adjusted $p$-value of 0.001. Excluding the dot-com period, the observed gain is no longer significant, with $p_{\max\text{-}T_4} = 0.224$. In the NYSE small–mid-cap universe, the adjusted $p$-values reach 0.065 over the full sample and 0.100 excluding the dot-com period.

Within this restricted family, only the main $L_{\text{all}}$ specification remains significant at the 5 % level. Cell $S_{\text{all}}$ provides weaker evidence at the 10 % level, whereas $L_{\text{exDC}}$ and $S_{\text{exDC}}$ do not survive family-wise control. Inference is therefore concentrated on the main comparison rather than on uniform validation across universes and sample definitions.

**Extended family of twelve cells.**

As a more conservative exploratory diagnostic, family $\mathcal{F}_{12}$ extends $\mathcal{F}_4$ by adding six universe–subperiod combinations and two dispersion regimes for large capitalizations. The same monthly permutations are used, but the maximum is now computed over twelve statistics in each replication.

**Table 6.11:** Net Sharpe-ratio differences, marginal $p$-values, and $p$-values adjusted using the maximum across twelve cells. The extended family adds six universe–subperiod cells and two cells corresponding to the high- and low-dispersion regimes among large capitalizations.

| Cell | Net ΔSR | Marginal $p$ | max-$T_{12}$ $p$ |
|---|---|---|---|
| Big caps, full sample | +0.0559 | 0.001 | 0.034 |
| Big caps, ex dot-com | +0.0001 | 0.040 | 0.724 |
| NYSE small–mid caps, full sample | +0.0080 | 0.013 | 0.451 |
| NYSE small–mid caps, ex dot-com | +0.0052 | 0.034 | 0.547 |
| Big caps, 1995–2004 | +0.1317 | 0.001 | 0.001 |
| Big caps, 2005–2014 | -0.0019 | 0.191 | 0.769 |
| Big caps, 2015–2025 | +0.0064 | 0.106 | 0.503 |
| NYSE small–mid caps, 1995–2004 | +0.0101 | 0.034 | 0.379 |
| NYSE small–mid caps, 2005–2014 | +0.0063 | 0.082 | 0.507 |
| NYSE small–mid caps, 2015–2025 | +0.0096 | 0.134 | 0.398 |
| Big caps, high dispersion state | +0.0766 | 0.005 | 0.012 |
| Big caps, low dispersion state | -0.0095 | 0.212 | 0.953 |

For the main $L_{\text{all}}$ specification, the correction rises from $p_{\max\text{-}T_4} = 0.001$ to $p_{\max\text{-}T_{12}} = 0.034$. The null hypothesis therefore remains rejected at the 5 % level even when the family is extended to twelve cells. For the illustrative $S_{\text{exDC}}$ target, the correction rises from 0.100 to 0.547.

Three cells remain significant after controlling across all twelve comparisons: the main $L_{\text{all}}$

specification, with $p_{\max\text{-}T_{12}} = 0.034$; large capitalizations over 1995–2004, with $p_{\max\text{-}T_{12}} = 0.001$; and the high-dispersion regime, with $p_{\max\text{-}T_{12}} = 0.012$. The evidence therefore remains primarily localized in the first subperiod and in high-dispersion states.

Expanding the family weakens the inference without overturning the main specification on a net basis. On the gross basis retained as an audit, however, the extended correction reaches $p_{\max\text{-}T_{12}} = 0.0619$; rejection at the 5 % level is therefore sensitive to the treatment of costs in the permutation distribution. The results do not constitute uniform validation across the universes, regimes, and subperiods examined. They provide targeted evidence in favor of signal timing, whose generalization remains sensitive to the scope of multiplicity control.

### 6.2.9 Robustness

The preceding results rely on a prespecified reference specification. Consistent with the protocol in Section 5.2.6, we now examine whether the measured gain depends on the precise level of the radius, a particular construction choice, or a single historical period, and whether its effect can be reproduced by volatility-based timing rules.

All variants retain the same point-in-time universes, holding returns, delisting conventions, weight-drift rules, and transaction costs. Net returns remain the primary basis. Depending on the diagnostic, the contrast compares $P^{(\rho)}$ with either the static portfolio or the volatility benchmarks; gross results are retained as an audit.

#### 6.2.9.1 The Useful Range of the Radius

We first move the baseline radius $\bar{\varepsilon}$ over a logarithmic grid between 0.1 and 2. At each value, the static portfolio retains this radius, whereas the dynamic portfolio applies the same point-in-time modulation as in the main specification. Both paths are fully solved and accounted for over all 372 months.

This sweep is a sensitivity diagnostic: it is not used to select a new radius ex post. The operational point $\varepsilon_0 = 0.3577$ is explicitly included in the grid and must exactly reproduce the central comparison.

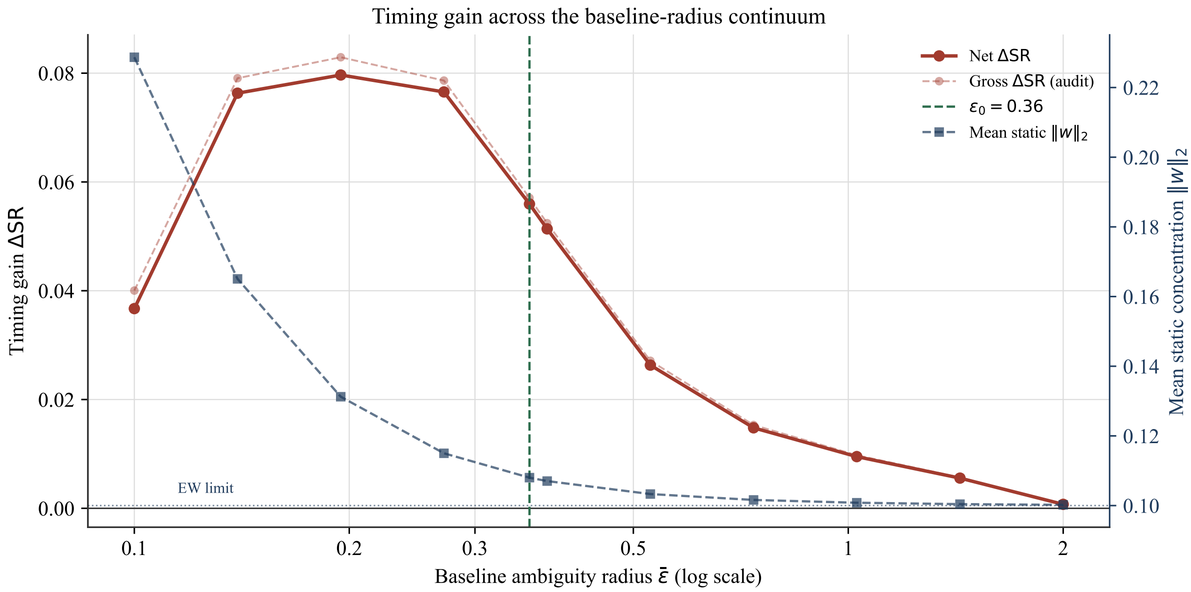


**Figure 6.22:** Sharpe-ratio difference between the dynamic and static portfolios as a function of the baseline radius $\bar{\varepsilon}$. The solid red curve presents the net result and the lighter red curve its gross counterpart. The right axis reports the static portfolio's mean concentration. The vertical line marks the operational radius $\varepsilon_0 = 0.3577$, while the dotted horizontal line represents equal-weight concentration $1/\sqrt{100} = 0.10$.

The net gain is hump-shaped along the continuum. It equals $+0.0367$ at $\bar{\varepsilon} = 0.1$ and then rises to a maximum of $+0.0797$ around $\bar{\varepsilon} = 0.195$. At the operational point, it exactly recovers the central result:

$$\Delta\mathrm{SR}^{\mathrm{net}} = +0.0559.$$

It then declines to $+0.0095$ around $\bar{\varepsilon} = 1.03$, and to $+0.0007$ when the radius reaches 2.

The static portfolio's mean concentration declines from 0.2287 at $\bar{\varepsilon} = 0.1$ to 0.1080 at the operational radius, and then to 0.1002 at $\bar{\varepsilon} = 2$. The declining portion of the gain therefore coincides with the gradual movement of both strategies toward the equal-weight limit. As the weights converge toward this limit, radius modulation has less scope to alter the allocation, and the timing gain tends toward zero without reaching it exactly over the grid examined.

Projection onto the safety cap is never active up to $\bar{\varepsilon} \simeq 1.03$. It affects approximately 5 % of months at $\bar{\varepsilon} \simeq 1.43$, and then 46 % of months when the baseline radius equals 2. The upper end of the curve therefore reflects both proximity to equal weighting and partial clipping of the dynamic path.

Gross and net results remain very close throughout the grid. The observed profile therefore does not arise from a particular variation in transaction costs. It reveals an intermediate range in which modulation has its greatest economic value in point estimates. At larger radii, this value weakens as robustness moves both strategies toward the same diversified limit.

#### 6.2.9.2 Stability across Construction Choices

We next vary one lever at a time around the main specification. Four prespecified dimensions are examined: the signal window $W_\rho \in \{36, 48, 60\}$ months, the modulation form, the rebalancing frequency, and the parameter $\beta \in \{0.05, 0.10, 0.20\}$. The constraints, accounting framework, and costs remain unchanged. The anchor changes only in the sensitivity analysis with respect to $\beta$; for the other levers, it remains fixed at its reference value. The identity, exponential, and rank forms are all constructed point-in-time. Likewise, the quarterly and annual variants retain positions between rebalancing dates: weights drift with realized returns and delisting proceeds remain in cash until the next scheduled transaction.

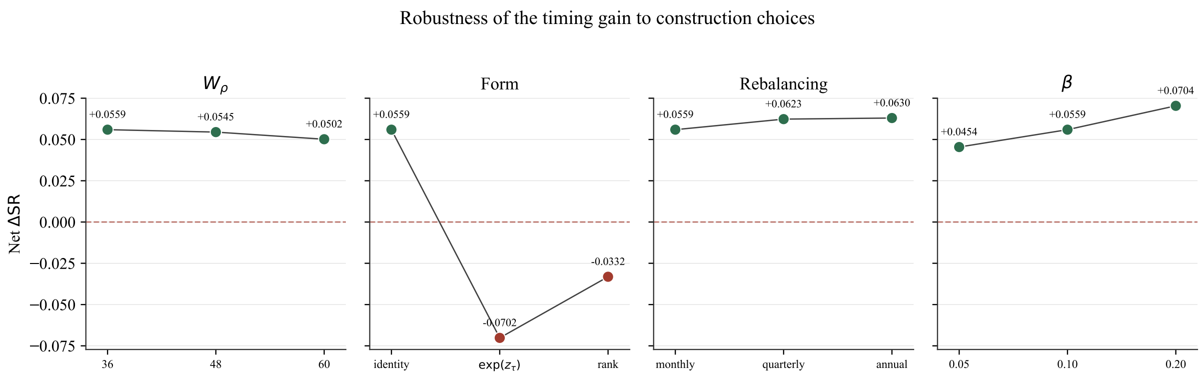


**Figure 6.23:** Net Sharpe-ratio difference between the dynamic and static portfolios when varying, in turn, the signal window, modulation form, rebalancing frequency, and parameter $\beta$. The horizontal dashed line represents zero gain. The main specification corresponds to $W_\rho = 36$, the identity form, monthly rebalancing, and $\beta = 0.10$.

The contrast remains positive across variants of the window, rebalancing frequency, and parameter $\beta$, but is not robust to the alternative functional forms. The signal window produces limited differences: $\Delta\mathrm{SR}^{\mathrm{net}}$ declines from +0.0559 with $W_\rho = 36$ to +0.0545 with 48 months and then to +0.0502 with 60 months.

The parameter $\beta$ has a larger effect on amplitude without changing the sign of the contrast. The gain reaches +0.0454, +0.0559, and +0.0704 for $\beta = 0.05$, 0.10, and 0.20, respectively.

These results remain consistent with the preceding diagnostic: relatively lower radius levels leave more scope for modulation.

Rebalancing frequency does not alter the sign of the contrast. The net gain equals +0.0559 at the monthly frequency, +0.0623 at the quarterly frequency, and +0.0630 at the annual frequency. At the same time, dynamic portfolio turnover declines from 0.1152 to 0.0672, and then to 0.0317. The sign of the gain therefore does not depend on monthly position adjustments. These results do not, however, establish that annual rebalancing is optimal, because each frequency generates a different holding path.

The functional form is the most sensitive lever. The identity form produces a net gain of +0.0559, whereas the exponential and rank transformations both reverse the sign of the contrast. Under the exponential form, ΔSR declines from −0.0506 gross to −0.0702 net, turnover reaches 0.3039, and the safety cap is active in approximately 0.5 % of decisions. Under the rank transformation, the contrast declines from −0.0100 gross to −0.0332 net, with turnover of 0.2787. Because both differences are already negative before costs, their reversal is not explained solely by the increase in turnover. The economic value of timing is therefore not robust to the choice of functional transformation.

The identity form remains the main specification because it directly transmits $\sqrt{\rho_\tau}$, never activates the safety cap, and generates the lowest turnover among the three forms considered. Its selection is prespecified and based on the transparency of the mechanism, rather than an ex post search for the best performance.

Detailed metrics for each configuration are presented in Table E.1, in Appendix E.1. The paths of the three modulation forms over the full sample and around the stress episodes are documented in Figures E.1 and E.2, in Appendix E.2.

#### 6.2.9.3 Volatility-Based Timing Benchmarks

The placebo tests in Section 6.2.4 show that the observed temporal ordering of $\rho$ is difficult to reproduce using the permutation mechanisms considered. A distinct question is whether its portfolio effect differs from that of rules based exclusively on volatility. Consistent with the protocol in Section 5.2.6, we compare $P^{(\rho)}$ with portfolios $P^{(\mathrm{IV})}$ and $P^{(\mathrm{RV}_{36})}$ over the full sample and in both point-in-time universes. All three strategies use the same anchor and allocation framework; only the modulation path differs.

**Table 6.12:** Net comparison between modulation by $\rho$ and modulations based on monthly implied volatility and 36-month realized volatility. The paired difference is defined as $\mathrm{SR}(P^{(\rho)}) - \mathrm{SR}(P^{(j)})$. The intervals and $p$-values are obtained from 10 000 replications of a paired stationary bootstrap. The $p$-values are two-sided, exploratory, and unadjusted for multiplicity.

| Universe | Strategy | $\mathrm{SR}_{\mathrm{net}}$ | $\Delta\mathrm{SR}_{\mathrm{stat}}$ | $\mathrm{SR}_\rho - \mathrm{SR}_j$ | 95% CI | $p_{\mathrm{boot}}$ |
|---|---|---|---|---|---|---|
| Big caps | $P^{(\rho)}$ | 0.7451 | +0.0559 | – | – | – |
| Big caps | $P^{(\mathrm{IV})}$ | 0.7063 | +0.0171 | +0.0388 | [-0.0076, +0.0957] | 0.170 |
| Big caps | $P^{(\mathrm{RV}_{36})}$ | 0.7475 | +0.0583 | -0.0023 | [-0.0107, +0.0052] | 0.538 |
| NYSE small–mid caps | $P^{(\rho)}$ | 0.6420 | +0.0080 | – | – | – |
| NYSE small–mid caps | $P^{(\mathrm{IV})}$ | 0.6395 | +0.0055 | +0.0024 | [-0.0161, +0.0239] | 0.840 |
| NYSE small–mid caps | $P^{(\mathrm{RV}_{36})}$ | 0.6386 | +0.0047 | +0.0033 | [-0.0092, +0.0165] | 0.665 |

In the main universe, $P^{(\rho)}$ has a net Sharpe ratio of 0.7451, compared with 0.7063 for $P^{(\mathrm{IV})}$ and 0.7475 for $P^{(\mathrm{RV}_{36})}$. Relative to the static portfolio, the corresponding gains are +0.0559, +0.0171, and +0.0583, respectively. The realized-volatility modulation therefore fully reproduces the point-estimate gain associated with $\rho$ and slightly exceeds it in this sample. The difference

$$\mathrm{SR}\big(P^{(\rho)}\big) - \mathrm{SR}\big(P^{(\mathrm{RV}_{36})}\big) = -0.0023$$

nevertheless remains indistinguishable from zero ($\mathrm{CI}_{95\%} = [-0.0107, 0.0052]$, $p = 0.538$). The difference from $P^{(\mathrm{IV})}$, equal to +0.0388, is likewise not established at the 5 % level ($\mathrm{CI}_{95\%} = [-0.0076, 0.0957]$, $p = 0.170$).

In the NYSE small–mid-cap universe, the net Sharpe ratios of $P^{(\rho)}$, $P^{(\mathrm{IV})}$, and $P^{(\mathrm{RV}_{36})}$ are 0.6420, 0.6395, and 0.6386, respectively. The differences between $P^{(\rho)}$ and the two benchmarks are +0.0024 and +0.0033, with respective $p$-values of 0.840 and 0.665. All four paired comparisons therefore have a 95 % interval containing zero. These results establish neither dominance by $P^{(\rho)}$ nor equivalence between the strategies. They show that the timing of $\rho$ is informative relative to the placebos considered, but that its incremental economic value relative to volatility rules is not established.

#### 6.2.9.4 Temporal Stability

The final diagnostic examines whether the gain is concentrated in a particular period. The sample is divided into three non-overlapping intervals: 1995–2004, 2005–2014, and 2015–2025. They contain 120, 120, and 132 monthly returns, respectively. The portfolios are not re-estimated for each subperiod: only the evaluation windows are extracted from the complete walk-forward paths.

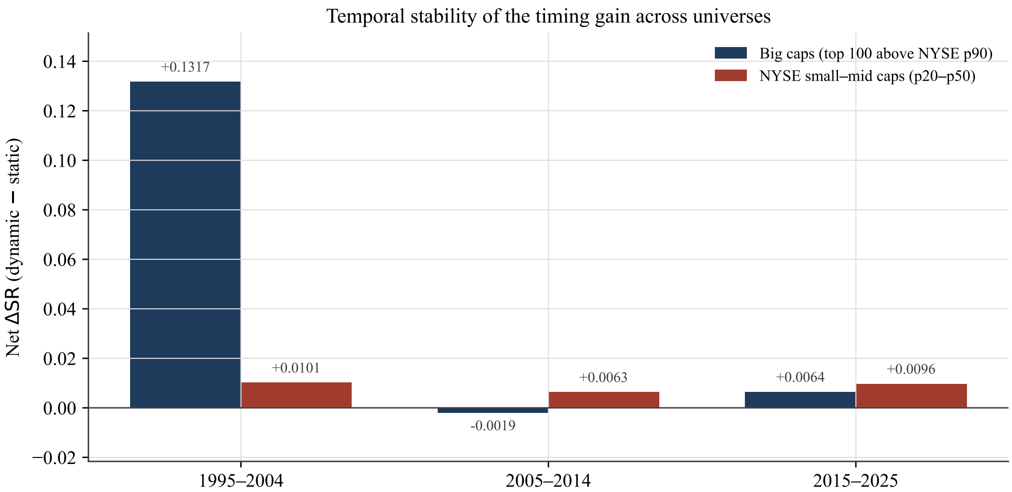


**Figure 6.24:** Net Sharpe-ratio difference between the dynamic and static portfolios over three non-overlapping subperiods, for large capitalizations and for the NYSE small–mid-cap universe. These cells are descriptive and also enter family $\mathcal{F}_{12}$.

Among large capitalizations, the gain is heavily concentrated in the first subperiod. It reaches $+0.1317$ between 1995 and 2004, becomes slightly negative between 2005 and 2014 ($-0.0019$), and then turns positive again, but remains small, between 2015 and 2025 ($+0.0064$). This profile confirms the importance of the dot-com period in the full-sample result and shows that the contrast is not temporally homogeneous.

In the NYSE small–mid-cap universe, the gain reaches $+0.0101$ over 1995–2004, $+0.0063$ over 2005–2014, and $+0.0096$ over 2015–2025. It therefore remains positive across all three subperiods, with a relatively stable magnitude that is nevertheless well below the gain observed among large capitalizations during the first decade.

These six estimates are not standalone significance tests. They are already included in the exploratory $\mathcal{F}_{12}$ family, whose results are presented in Table 6.11. Their purpose is to locate the temporal stability of the contrast rather than to multiply inferential conclusions.

Overall, the gain remains positive throughout the radius grid and across the window, rebalancing-frequency, and $\beta$-parameter variants. It does, however, change sign under the exponential and rank transformations. Across the six universe–subperiod cells, five estimates are positive; the large-capitalization cell over 2005–2014 is slightly negative.

The volatility benchmarks further qualify this robustness. In the main universe, $P^{(\mathrm{RV}_{36})}$ fully reproduces the point-estimate gain of $P^{(\rho)}$ and slightly exceeds it. In the NYSE small–mid-cap universe, $P^{(\rho)}$ has the highest point estimate of the Sharpe ratio, but the differences are very small. None of the four paired differences relative to the volatility benchmarks is established at the 5 % level.

The magnitude of the gain remains sensitive to the functional form, universe, and period examined. The evidence is strongest for large capitalizations over the full sample, during the first subperiod, and in high-dispersion states. All three placebos reject their respective Level 1 null hypotheses, but this evidence does not uniformly survive the extended family-wise corrections.

## 6.3 Synthesis of the Results

The results address two distinct questions. The first concerns the empirical validity of the inter-horizon misalignment signal; the second concerns the economic value of its timing when it modulates the ambiguity radius. This distinction is essential: the descriptive relevance of the signal does not mechanically imply a significant improvement in portfolio performance.

### 6.3.1 The Signal

The empirical diagnostics show that $\widehat{\rho}_{\tau}^{\mathrm{hyb}}$ responds around several externally dated stress episodes, without its maximum necessarily coinciding with the conventional peak of each crisis (Section 6.1.2). In parallel, the controlled placebo confirms that the signal distinguishes unperturbed windows from their paired counterparts after inter-frequency misalignment is injected (Section 6.1.3). These diagnostics support the internal validity of the measured statistical mechanism.

The resulting dynamics remain stable across the main numerical variants of the reference specification (Section 6.1.4). They become less concordant, however, when the geometry or economic definition of the signal is substantially altered, notably under realized-volatility normalization. The regressions further confirm that the signal shares an important component with volatility measures, particularly at the 36-month horizon, while retaining a residual component in the linear specifications considered (Section 6.1.5). It should therefore be interpreted as a measure closely related to volatility dynamics, but not fully reducible to them within the empirical framework examined.

### 6.3.2 The Portfolio

The transmission of the signal to the allocation problem is mechanically verified: an increase in the ambiguity radius moves the weights toward the equal-weight limit (Sections 6.2.1 and 6.2.2). The static portfolio is already located in a highly diversified region, with a mean concentration of 0.1080 at the operational radius. Modulation nevertheless retains sufficient scope to alter the weights and realized performance.

In the central comparison, the net Sharpe ratio rises from 0.6892 to 0.7451, giving

$\Delta\mathrm{SR}^{\mathrm{net}} = +0.0559$, compared with $+0.0571$ before costs. The annualized mean return increases from 11.20 % to 11.66 %, while volatility and maximum drawdown decline from 16.26 % to 15.64 % and from 0.617 to 0.528, respectively. Recurring turnover rises only from 0.1116 to 0.1152. This point-estimate improvement nevertheless remains statistically imprecise: the HAC test applied to the mean return difference yields $p = 0.287$, while the paired bootstrap for $\Delta\mathrm{SR}$ yields $p = 0.113$, with a 95 % interval containing zero (Section 6.2.3).

The permutation tests address a different null hypothesis concerning the interchangeability of the observed temporal ordering. The simple monthly permutation and the six- and twelve-month block permutations produce $p$-values of 0.001, 0.003, and 0.009, respectively (Section 6.2.4). After family-wise adjustment, only the main specification remains significant in $\mathcal{F}_4$, with $p_{\max\text{-}T_4} = 0.001$. In $\mathcal{F}_{12}$, it retains a $p$-value of 0.034. The large-capitalization cell over 1995–2004 and the high-dispersion-regime cell also remain significant, with adjusted $p$-values of 0.001 and 0.012. The evidence is therefore localized rather than uniformly generalizable across universes, regimes, and subperiods (Section 6.2.8).

The volatility benchmarks further qualify this interpretation. In the main universe, $P^{(\mathrm{RV}_{36})}$ obtains a Sharpe ratio of 0.7475, slightly above the 0.7451 obtained by $P^{(\rho)}$. In the NYSE small–mid-cap universe, $P^{(\rho)}$ has the highest point estimate, but the differences remain small. None of the four paired comparisons with the volatility rules is established at the 5 % level. The incremental economic value of $\rho$ relative to these benchmarks has therefore not been established (Section 6.2.9.3).

Finally, the sign of the gain is robust to variants of the signal window, rebalancing frequency, and parameter $\beta$, but reverses under the exponential and rank transformations. The dynamic portfolio therefore improves upon its static reference under the prespecified identity form without dominating equal weighting, whose net Sharpe ratio reaches 0.777. Cross-validation also selects the ceiling of its grid in 96.1 % of the months following initialization, leading to nearly equal-weighted allocations (Sections 6.2.6, 6.2.7, and 6.2.9).

Overall, the results primarily support a methodological contribution. The signal detects the misalignment injected in the controlled experiment, and its observed temporal ordering is associated with an economically visible point-estimate improvement in the main specification. This improvement is not, however, supported by the conventional inference on returns or the Sharpe ratio, is not uniformly generalizable after family-wise control, and is not statistically distinguishable from volatility-based rules. The contribution therefore lies in identifying and testing the timing of the ambiguity radius, whose economic value remains conditional on the selected transformation, the anchor, and the empirical environment. These findings structure the discussion that follows.

# 7 Discussion

The preceding results permit a joint interpretation of three dimensions of the proposed approach: the nature of the information contained in the signal, the mechanism through which it affects robust allocation, and the economic scope of the resulting gains. To this end, the discussion distinguishes what the empirical analysis establishes, the interpretations it renders plausible, and the conclusions it does not support.

The central contribution lies first in separating the average level of the ambiguity radius from its timing. By decomposing $\varepsilon_\tau$ into an anchor $\bar{\varepsilon}$ and an endogenous modulation with an approximately unit mean (Equation 18), the protocol makes it possible to examine a question generally conflated with calibration: at a comparable average intensity of robustness, does the timing of the radius have economic value of its own?

The results provide a nuanced answer. In the central comparison, the point estimate of the gain is positive, but the paired tests of the mean-return and Sharpe-ratio differences do not establish a general improvement at the 5 % level (Section 6.2.3). By contrast, all three placebos reject the hypothesis that the observed temporal ordering is interchangeable with a randomized signal ordering, including when the permutations preserve part of its local dependence (Section 6.2.4).

Multiplicity control nevertheless circumscribes this evidence without eliminating it in the main specification. Within $\mathcal{F}_4$, only the full-sample large-capitalization cell remains significant after adjustment ($p_{\max\text{-}T_4} = 0.001$). Within $\mathcal{F}_{12}$, the same cell retains a $p$-value of 0.034. The evidence also remains significant among large capitalizations over 1995–2004 ($p_{\max\text{-}T_{12}} = 0.001$) and in high-dispersion states ($p_{\max\text{-}T_{12}} = 0.012$), but not uniformly across the remaining cells (Section 6.2.8). Evidence on the value of timing should therefore be interpreted as targeted and conditional on the period, market segment, and scope for adjustment left by the initial radius level, rather than as evidence of general dominance.

The point-estimate improvement is economically visible, but its scope remains constrained by the geometry of the allocation problem. The linear loss was selected precisely to make this channel identifiable: holding the data and constraints fixed, modulation acts directly on the coefficient $\varepsilon_\tau$ of the $\|w\|_2$ penalty. This design isolates the transmission from the radius to the weights without prejudging the magnitude that this effect might attain under richer objective functions.

In this formulation, the radius regularizes estimated mean returns and gradually moves the allocation toward equal weighting. The static anchor is already located in a highly diversified region (Sections 6.2.2 and 6.2.7). The radius sweep nevertheless reveals an intermediate range: the gain initially rises, reaches a maximum before the operational point, and then declines as the static and dynamic portfolios converge toward the same equal-weight limit (Section 6.2.9.1). The value of timing therefore depends on the remaining scope for movement within the weight space, rather than on a numerical failure of the solution procedure.

This dependence also appears across functional forms. The sign of the gain is robust to variants of the window, rebalancing frequency, and parameter $\beta$, but reverses under the exponential and rank rules. These transformations simultaneously alter the temporal distribution of the effective radius and turnover; they are therefore sensitivity tests of the complete rule, not comparisons of curvature at a strictly identical ex post mean radius. The evidence thus concerns the prespecified identity form and does not establish the general effectiveness of every monotonic transformation of the signal.

The second empirical contribution concerns the characterization of inter-horizon misalignment. The diagnostics in Section 6.1 show that the signal responds around several stress episodes, distinguishes the perturbations introduced in the controlled placebo, and remains stable under the main variants that preserve its economic definition.

The regressions nevertheless reveal a substantial component shared with volatility, particularly at the 36-month horizon (Section 6.1.5). The signal therefore cannot be presented as autonomous or structurally independent of volatility. The linear controls do not, however, fully reproduce its statistical variation.

The portfolio comparison reinforces this qualification. In the main universe, $P^{(\mathrm{RV}_{36})}$ obtains a slightly higher Sharpe ratio than $P^{(\rho)}$, whereas the latter has the highest point estimate in the secondary universe (Section 6.2.9.3). None of the four paired differences is established at the 5 % level. The residual statistical component of the signal therefore remains compatible with information specific to inter-horizon misalignment, without the analysis establishing incremental economic value, a causal relationship, or non-redundancy relative to volatility.

This interpretation remains consistent with the heterogeneous market hypothesis of Müller et al. [1], according to which participants operating at different horizons may encode different information. The link between the statistical signal and the actual heterogeneity of market participants nevertheless remains an interpretative conjecture, rather than a causal mechanism identified by the analysis.

Finally, the proposed approach is positioned as a complement to, rather than a substitute for, other calibration strategies. After initialization, expanding cross-validation selects the upper bound of the grid in 96.1 % of months and therefore leads to portfolios close to equal weighting without improving upon its performance (Section 6.2.7). Portfolio $P^{(\rho)}$ likewise does not dominate $P^{(\mathrm{EW})}$ in absolute terms. Its contribution concerns a narrower question: the value of ambiguity-radius timing within a given DRO family at a comparable average level of robustness.

Relative to regime-switching models [12], this modulation does not require transition dynamics to be specified, at the cost of a less detailed representation of market states. These comparisons do not establish any general dominance. Instead, they position inter-horizon misalignment as a modulation mechanism that may be combined with other calibration methods. The principal contribution of this study therefore lies less in constructing a universally dominant portfolio than in providing an architecture that makes it possible to separate, identify, and empirically test the average level of robustness and the value of its timing. The main avenues for extension are presented in Section 10.

# 8 Limitations

The conclusions of this thesis should be interpreted in light of several methodological, statistical, and empirical limitations.

The first limitation follows directly from the chosen identification design. Under the linear loss, the Wasserstein DRO program reduces to regularizing estimated mean returns through a penalty on weight concentration. This deliberate choice makes it possible to attribute changes in weights transparently to the timing of the radius, without simultaneously varying a covariance matrix, a risk-aversion parameter, or a preference regarding tail risk.

The corresponding cost of this transparency is to restrict the economic channel available to the signal. The radius does not directly robustify the covariance structure, extreme dependence, or tail risk; these dimensions are evaluated out of sample but do not enter the objective function. A mean–variance, quadratic, shortfall, or CVaR-type loss could therefore respond differently to the same modulation. The conclusions thus concern the identification of the mechanism under a linear loss and cannot be generalized to all DRO allocation problems without further testing.

The separation between level and timing also remains approximate along the realized sample path. The expanding normalization is entirely point-in-time and keeps the modulation centered around one, but it does not impose exact ex post equality between the mean dynamic radius and the static radius. The central comparison therefore identifies the complete dynamic rule relative to its static benchmark, rather than an abstract causal effect of timing under perfect equality of the realized mean radii.

The second limitation concerns the theoretical status of the radius. As noted in Remark 4.6.1, the parameter $\beta$ in the reduced-form rule cannot be interpreted as a coverage probability. The assumptions required by the concentration bounds are not imposed in the empirical setting. The radius is therefore an empirically motivated adaptive regularization device, rather than an ambiguity set endowed with an exact frequentist guarantee. Establishing such a guarantee for an endogenous, time-varying radius under temporally dependent observations lies beyond the scope of this study.

The third limitation is inferential. The 1995–2025 evaluation period contains only a limited number of major stress episodes. The conditional attribution associates a substantial share of the contrast with certain high-dispersion periods, but it neither identifies a causal effect of crises nor permits these episodes to be treated as independent observations. The procedures address distinct null hypotheses: the HAC test and bootstrap characterize uncertainty surrounding the observed contrast, whereas the permutations assess the value of its temporal ordering. The $\mathcal{F}_4$ and $\mathcal{F}_{12}$ families rely on joint monthly permutations, which preserve dependence across cells but not the serial dependence of the modulation; block permutations remain confined to level 1. Their conclusions are therefore not interchangeable.

The scope of this inference also depends on the selected family of hypotheses. Within the restricted $\mathcal{F}_4$ family, only the main $L_{\text{all}}$ specification remains significant after multiplicity adjustment. The two cells in the NYSE small–mid-cap universe have positive point estimates, but their $p$-values under the max-$T_4$ correction do not permit rejection of the null hypothesis at the 5 % level. When the adjustment is extended to the twelve cells of $\mathcal{F}_{12}$, the $p$-value for $L_{\text{all}}$ reaches 0.034. The main specification, large capitalizations over 1995–2004, and high-dispersion states then remain significant, with respective adjusted $p$-values of 0.034, 0.001, and 0.012. The results therefore support targeted evidence, but not uniform validation across the universes, regimes, and subperiods examined.

A fourth limitation concerns identification of the signal's distinct content. The daily, weekly, and monthly measures are all constructed from the same daily process and rely on overlapping temporal aggregations. They therefore do not constitute independent sources of information. The signal measures disagreement among several temporal representations of the same process, rather than among three autonomous samples.

The regressions in Section 6.1.5 further show that the signal shares a substantial component with volatility measures, particularly at the formation horizon. The benchmarks in Section 6.2.9.3 directly compare its modulation with rules based on monthly implied volatility and 36-month realized volatility, without establishing any paired difference at the 5 % level. This failure to reject does not, however, demonstrate equivalence between the strategies. These comparisons remain exploratory, are not adjusted for multiplicity, and are limited to two representative rules rather than an exhaustive grid of horizons. The analysis therefore cannot precisely isolate the share of economic value attributable to inter-horizon misalignment rather than to its common component with volatility.

The fifth limitation concerns the representation of distributional geometry. In theory, the Sliced-Wasserstein distance defined over all directions characterizes the distributions being compared. In practice, it is estimated from a finite number of projections, after which the multi-frequency information is compressed into a scalar signal of barycentric dispersion. The convergence diagnostics and use of common random directions reduce numerical uncertainty without eliminating the information loss induced by this approximation and aggregation. The exact relationship between the resulting signal and complete multivariate misalignment is therefore not characterized theoretically in this study.

The robustness analyses should be interpreted from the same perspective. At the signal level, they show primarily that the conclusions remain similar under several numerical variants that preserve its economic definition. They do not protect against a common specification bias, notably in the choice of daily, weekly, and monthly frequencies, the formation window, or the compression of misalignment into a scalar indicator. Normalization by realized volatility illustrates that an economically different transformation can produce substantially different dynamics.

At the portfolio level, the sign of the gain is robust to variants of the window, rebalancing frequency, and parameter $\beta$, but reverses under the exponential and rank transformations. These rules alter both the temporal distribution of the realized radius and turnover; their comparison therefore cannot isolate a pure curvature effect at a strictly constant mean level. The economic conclusions thus concern the prespecified identity form and cannot be generalized to every monotonic transformation of the signal.

Finally, the external scope of the results remains circumscribed. The analysis examines two constant-cardinality universes of securities listed on U.S. markets, evaluated over a single historical period and under a long-only portfolio constraint. The transaction-cost model is proportional and relies on aggregate estimates [33]; it captures neither market impact nor security-specific liquidity constraints. This qualification is particularly important for the NYSE small–mid-cap universe, whose turnover is substantially higher. Applying the mechanism to other asset classes, geographies, universe dimensions, or rebalancing frequencies therefore remains an open question.

# 9 Conclusion

This thesis was motivated by a question that remains underexplored in the literature on distributionally robust optimization: can the Wasserstein ambiguity radius, generally treated as a static parameter to be calibrated, be modulated over time on the basis of observable information, and does its timing have economic value in its own right?

To address this question, we constructed a measure of misalignment among the empirical return distributions observed at daily, weekly, and monthly frequencies. This misalignment is quantified by the Sliced-Wasserstein dispersion of the frequency-specific measures around their multi-frequency barycenter, following a rescaling designed to render their geometries comparable. The resulting scalar signal, $\widehat{\rho}_\tau^{\mathrm{hyb}}$, is then used to modulate the ambiguity radius around a reference level within a Wasserstein DRO program solved at each formation date.

The signal results provide several pieces of evidence supporting its empirical validity. Its dynamics respond around various externally dated stress episodes, and its discriminatory power increases with the intensity of the misalignment introduced in the controlled placebo. It also remains close to the reference specification under several numerical variants that preserve its economic definition. The signal nevertheless shares a substantial component with volatility measures, particularly at the 36-month formation horizon. It retains residual variation in the linear specifications examined, but this variation does not establish structural independence or economic non-redundancy.

The portfolio comparison reinforces this qualification. In the main universe, the modulation based on 36-month realized volatility obtains a slightly higher Sharpe ratio than $P^{(\rho)}$. In the NYSE small–mid-cap universe, $P^{(\rho)}$ has the highest point estimate, but the differences are small. None of the four paired differences is established at the 5 % level. The signal's residual variation is therefore insufficient to demonstrate incremental economic value relative to the volatility-based rules considered.

The portfolio evaluation shows that, at a comparable average level of robustness, endogenous modulation of the radius is associated with an economically visible point-estimate improvement. In the main large-capitalization specification, the Sharpe-ratio gain reaches

+0.0559 net of costs, compared with +0.0571 before costs. The dynamic portfolio combines a higher mean return with lower volatility, tail risk, and maximum drawdown, while the increase in turnover remains limited. The HAC test of the mean-return difference and the paired bootstrap of the Sharpe-ratio difference nevertheless fail to establish this effect at the 5 % level. By contrast, the three permutation tests indicate that the performance associated with the observed temporal ordering is difficult to reproduce through the monthly and block reorderings considered.

The scope of this evidence depends on the inferential family. Within the restricted $\mathcal{F}_4$ family, only the main $L_{\text{all}}$ specification remains significant after multiplicity adjustment, with $p_{\text{max-}T_4} = 0.001$. When the adjustment is extended to the twelve cells of $\mathcal{F}_{12}$, its $p$-value reaches 0.034. Three cells then remain significant: the main specification, large capitalizations over 1995–2004, and high-dispersion states, with respective adjusted $p$-values of 0.034, 0.001, and 0.012. The result therefore constitutes targeted evidence in favor of the value of ambiguity-radius timing, rather than uniform validation across universes, regimes, and subperiods.

The analysis of portfolio geometry clarifies the nature of this contribution. The static reference level is already located in a highly diversified region, where increasing the radius rapidly moves the weights toward equal weighting. The radius sweep nevertheless reveals an intermediate range within which its modulation produces an economically discernible gain. Under the prespecified identity form, $P^{(\rho)}$ improves upon the static portfolio constructed at the same anchor without dominating equal weighting in absolute terms.

This conclusion is not invariant to the choice of functional transformation: the exponential and rank rules reverse the sign of the contrast. The identified economic value therefore concerns radius timing under the selected main rule, rather than the general effectiveness of every modulation or the construction of a strategy that universally outperforms passive benchmarks.

The contribution of this study is primarily methodological. It provides, first, an inter-horizon misalignment signal based on multi-frequency distributional geometry and, second, a framework for empirically distinguishing the average level of robustness from the value of its timing. The associated protocol — point-in-time construction, level–modulation separation, signal-ordering placebos, regime-based attribution, and multiplicity control — can be reused to examine other mechanisms of dynamic robustness.

The choice of a linear loss is both a strength for identification and a limitation in scope. It transparently isolates the transmission of ambiguity-radius timing to the weights, but does not determine whether an objective function that directly incorporates covariance or tail risk would amplify, attenuate, or otherwise alter this mechanism. The substantial relationship between the signal and volatility, the limited number of stress episodes, and the external scope of the sample also delimit the conclusions. These qualifications do not undermine the internal coherence of the design; they circumscribe the conditions under which its results may be generalized.

The extensions presented in Section 10 naturally continue this research program by considering richer objective functions, learned modulation rules, and other empirical settings. The central contribution of this thesis is to formulate the timing of the radius as a component distinct from its average level and to make it explicitly measurable, implementable, and empirically testable.

# 10 Extensions

The proposed architecture — an observable distributional signal that modulates a robustness parameter — is deliberately modular. Each of its three components, the objective function, the signal, and the modulation rule, can be enriched independently of the other two, thereby opening a relatively broad range of extensions. We outline the main ones below, in decreasing order of proximity to the current framework.

## 10.1 Changing the Objective Function

The most direct extension would replace the linear loss with an objective function whose robustification involves more than mean returns. The robust mean–variance framework of Blanchet et al. [11] is a natural candidate: its robustness penalty acts on the second-order structure, such that modulation of the radius by $\widehat{\rho}_\tau^{\text{hyb}}$ would affect regularization of the covariance itself, precisely the component left unchanged by the current linear framework. A reasonable conjecture — albeit one that would require testing — is that the value of distributional timing would be greater in this setting, because the signal captures shape distortions that the linear loss cannot exploit. Objectives directed toward tail risk, such as robust CVaR, would offer a second field of application, with the additional advantage of retaining tractable reformulations [4]. In both cases, the level–modulation separation and placebo protocol developed here would remain directly applicable, enabling controlled comparisons across objective functions.

## 10.2 Statistical Learning for the Modulation Signal

The modulation rule adopted in this thesis is deliberately parsimonious: a fixed transformation of the signal with no learned parameter. A natural extension would be to learn the function linking the market's distributional state to the ambiguity radius. Several degrees of ambition are possible. At the first level, a monotonic transformation of the signal — for example, a parametric link function estimated through sequential validation — could optimize the sensitivity of the modulation while limiting, though not eliminating, the risk of overfitting. At the second level, a supervised model could combine $\widehat{\rho}_\tau^{\text{hyb}}$ with other state variables to predict a robustness target directly, such as a target radius defined ex post solely within the training sample and then predicted through a strictly walk-forward procedure. At the third level, end-to-end approaches in which the modulation rule is learned by directly optimizing out-of-sample portfolio performance are conceivable, but would raise substantial inferential and validation difficulties, as the boundary between genuine timing and temporal overfitting would become difficult to establish. The placebo protocol developed here would provide precisely the safeguard required for this evaluation: any learned rule should demonstrate that its value disappears when the temporal ordering of the signal is destroyed.

## 10.3 Distributional Signal Engineering

The signal $\widehat{\rho}_\tau^{\mathrm{hyb}}$ summarizes inter-horizon disagreement in a scalar, which facilitates interpretation but necessarily compresses the available information. A first priority is to isolate its distinct content relative to volatility more effectively. One extension could construct, on a strictly point-in-time basis, a modulation based on the component of the signal unexplained by realized and implied volatility measures, and then compare it with a prespecified family of volatility rules. Such an analysis would permit a more direct assessment of the incremental economic value of the distributional information.

Other extensions concern the representation of the signal. Its contributions by frequency, projection direction, or asset group could be retained separately to form a vector of distributional features suitable for the learning methods discussed above. Implied distributions extracted from option prices could also introduce a forward-looking horizon absent from historical frequencies and make it possible to study disagreement between historical and implied measures. Finally, the comparison geometry could be refined through generalized or projected variants of sliced distances [37], whose finite-sample statistical properties are documented by Nadjahi et al. [17]. The common challenge across these extensions is to preserve the signal's stability and interpretability as its dimension increases.

## 10.4 Complementary Directions

Three more exploratory directions complete this research program. The first is theoretical: establishing coverage guarantees, even asymptotic ones, for an endogenous, time-varying radius under weakly dependent data would bridge the gap between the concentration-based motivation and its empirical use in this study. The second is structural: combining endogenous modulation with a regime-switching specification [12], with the signal serving as an observation variable in the transition model, would make it possible to test whether the two information sources are substitutes or complements. The third is empirical: applying the mechanism to other asset classes — interest rates, credit, and currencies, where the multi-horizon structure is plausibly different — would provide a demanding out-of-sample test of its scope. If inter-horizon misalignment captures a general property of market distributions rather than a feature specific to the listed-equity universes examined here, its value should also manifest itself in these settings; otherwise, the geography of its successes and failures would itself be informative.

# Appendix

# A Transport Plan

Using two empirical measures of different sizes, this appendix illustrates how the Wasserstein distance compares two distributions without imposing a one-to-one correspondence between their observations, an essential property in our multi-frequency setting.

## A.1 Empirical Measures with Heterogeneous Supports

In the empirical application, each observation is a return vector in $\mathbb{R}^N$, where $N$ denotes the number of assets. To make the example easy to visualize, we work in two dimensions:

$$x = (r^{(1)}, r^{(2)}) \in \mathbb{R}^2.$$

The dimension of the space represents the number of assets, whereas the number of atoms in a measure corresponds to the number of observations available at a given frequency.

Consider two stylized empirical measures. The first represents a monthly frequency with two observations, whereas the second represents a weekly frequency with three observations. Both measures lie in the same space $\mathbb{R}^2$, but they do not have the same number of atoms.

The supports of the two measures are given by the matrices

$$X_m = \begin{pmatrix} 0 & 0 \\ 3 & 3 \end{pmatrix} \in \mathbb{R}^{2\times 2}, \qquad X_w = \begin{pmatrix} 0 & 0 \\ \frac{3}{2} & \frac{3}{2} \\ 3 & 3 \end{pmatrix} \in \mathbb{R}^{3\times 2}.$$

Each row corresponds to an atom of the empirical measure. The first column can be read as the return of the first asset and the second as the return of the second asset.

| Measure | Atoms | | | Empirical weights |
|---|---|---|---|---|
| $\widehat{\mu}_m$ | $x_1^{(m)} = (0, 0)$ | $x_2^{(m)} = (3, 3)$ | $\varnothing$ | $\frac{1}{2}, \frac{1}{2}$ |
| $\widehat{\mu}_w$ | $x_1^{(w)} = (0, 0)$ | $x_2^{(w)} = \left(\frac{3}{2}, \frac{3}{2}\right)$ | $x_3^{(w)} = (3, 3)$ | $\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$ |

The symbol $\varnothing$ indicates that the monthly measure has no third atom. This difference in cardinality is precisely what the transport plan can accommodate.

The two empirical measures can therefore be written compactly as

$$\widehat{\mu}_m = \frac{1}{2}\delta_{x_1^{(m)}} + \frac{1}{2}\delta_{x_2^{(m)}},$$

$$\widehat{\mu}_w = \frac{1}{3}\delta_{x_1^{(w)}} + \frac{1}{3}\delta_{x_2^{(w)}} + \frac{1}{3}\delta_{x_3^{(w)}}.$$

Both measures lie in the same space $\mathbb{R}^2$ but do not have the same number of atoms. The total mass of each measure equals one.

For the remainder of the example, we adopt a precise notational convention: the monthly measure $\widehat{\mu}_m$ is taken as the source measure and the weekly measure $\widehat{\mu}_w$ as the target measure. The transport plan is therefore read as transporting mass from $\widehat{\mu}_m$ to $\widehat{\mu}_w$. This convention determines only the matrix organization of the plan: rows correspond to monthly atoms and columns to weekly atoms. It does not imply a temporal transformation from monthly to weekly frequency; it serves only to write the coupling between the two empirical measures.

## A.2 Transport Plan and Mass Constraints

Under this convention, a transport plan between $\widehat{\mu}_m$ and $\widehat{\mu}_w$ is represented by a matrix

$$\pi \in \mathbb{R}_+^{2\times 3},$$

whose rows correspond to the two atoms of the monthly measure and whose columns correspond to the three atoms of the weekly measure. The coefficient $\pi_{ij}$ denotes the amount of mass coupled between monthly atom $x_i^{(m)}$ and weekly atom $x_j^{(w)}$.

If the convention were reversed, the same coupling would be represented by a $3 \times 2$ matrix with transposed indices. The value of the Wasserstein distance does not depend on this convention.

The plan must satisfy two sets of constraints:

$$\begin{aligned}
&\sum_{j=1}^{3} \pi_{ij} = \frac{1}{2}, \qquad i = 1,2, \qquad \text{row constraints: monthly mass preserved,} \\
&\sum_{i=1}^{2} \pi_{ij} = \frac{1}{3}, \qquad j = 1,2,3, \quad \text{column constraints: weekly mass received.}
\end{aligned} \tag{26}$$

The row constraints require all the mass of each monthly observation to be distributed. The column constraints require each weekly observation to receive exactly its empirical mass. The set of admissible plans is denoted by $\Pi(\widehat{\mu}_m, \widehat{\mu}_w)$.

## A.3 Transport Cost

For an admissible plan $\pi$, the total transport cost aggregates the elementary costs of each displacement:

$$\mathcal{C}(\pi) = \sum_{i=1}^{2} \sum_{j=1}^{3} \pi_{ij} \left\| x_i^{(m)} - x_j^{(w)} \right\|_2^2. \tag{27}$$

Each term is the product of the mass transported and the elementary cost of the displacement $x_i^{(m)} \to x_j^{(w)}$. To make this calculation explicit, we introduce the matrix of elementary costs $D \in \mathbb{R}^{2\times 3}$, whose entry $(i,j)$ is obtained by computing the squared distance between monthly atom $x_i^{(m)}$ and weekly atom $x_j^{(w)}$.

$$\left( \left\| x_i^{(m)} - x_j^{(w)} \right\|_2^2 \right)_{ij} = \begin{pmatrix} \|(0,0)-(0,0)\|_2^2 & \left\|(0,0)-\left(\frac{3}{2},\frac{3}{2}\right)\right\|_2^2 & \|(0,0)-(3,3)\|_2^2 \\ \|(3,3)-(0,0)\|_2^2 & \left\|(3,3)-\left(\frac{3}{2},\frac{3}{2}\right)\right\|_2^2 & \|(3,3)-(3,3)\|_2^2 \end{pmatrix}.$$

This gives

$$D_{ij} = \left\| x_i^{(m)} - x_j^{(w)} \right\|_2^2, \qquad D = \begin{pmatrix} 0 & \frac{9}{2} & 18 \\ 18 & \frac{9}{2} & 0 \end{pmatrix}.$$

The total cost can then be rewritten as a Hadamard product followed by a summation:

$$\mathcal{C}(\pi) = \mathbf{1}^\top (\pi \odot D)\, \mathbf{1} = \sum_{i,j} \pi_{ij} D_{ij},$$

where $\odot$ denotes the elementwise product. The squared Wasserstein distance is the minimum cost over the set of admissible plans:

$$W_2^2(\widehat{\mu}_m, \widehat{\mu}_w) = \min_{\pi \in \Pi(\widehat{\mu}_m, \widehat{\mu}_w)} \mathcal{C}(\pi).$$

## A.4 Computing the Cost of an Optimal Plan

In this symmetric example, the following admissible plan is optimal:

$$\pi = \begin{pmatrix} 1/3 & 1/6 & 0 \\ 0 & 1/6 & 1/3 \end{pmatrix}. \tag{28}$$

This plan can be read as a table of mass flows:

| | $x_1^{(w)}$ | $x_2^{(w)}$ | $x_3^{(w)}$ | initial mass |
|---|---|---|---|---|
| $x_1^{(m)}$ | 1/3 | 1/6 | 0 | 1/2 |
| $x_2^{(m)}$ | 0 | 1/6 | 1/3 | 1/2 |
| mass received | 1/3 | 1/3 | 1/3 | 1 |

The rows sum to the monthly masses 1/2 and the columns to the weekly masses 1/3. The central observation $x_2^{(w)} = (3/2, 3/2)$ simultaneously receives mass from both monthly observations — the transport plan therefore imposes no one-to-one correspondence between the points of the two measures.

The cost calculation using the Hadamard product gives

$$\pi \odot D = \begin{pmatrix} \frac{1}{3} & \frac{1}{6} & 0 \\ 0 & \frac{1}{6} & \frac{1}{3} \end{pmatrix} \odot \begin{pmatrix} 0 & \frac{9}{2} & 18 \\ 18 & \frac{9}{2} & 0 \end{pmatrix} = \begin{pmatrix} 0 & \frac{3}{4} & 0 \\ 0 & \frac{3}{4} & 0 \end{pmatrix}.$$

The zeros in the matrix $\pi \odot D$ have two distinct sources: some flows are zero, whereas some transports have zero cost because the corresponding points coincide. In particular, $\pi_{13} = \pi_{21} = 0$, whereas $D_{11} = D_{23} = 0$. To make the structure of this calculation transparent, it can be decomposed term by term:

$$\mathcal{C}(\pi) = \underbrace{\frac{1}{3}\cdot 0}_{x_1^{(m)}\to x_1^{(w)}} + \underbrace{\frac{1}{6}\cdot\frac{9}{2}}_{x_1^{(m)}\to x_2^{(w)}} + \underbrace{0\cdot 18}_{\pi_{13}=0} + \underbrace{0\cdot 18}_{\pi_{21}=0} + \underbrace{\frac{1}{6}\cdot\frac{9}{2}}_{x_2^{(m)}\to x_2^{(w)}} + \underbrace{\frac{1}{3}\cdot 0}_{x_2^{(m)}\to x_3^{(w)}} = \frac{3}{4}+\frac{3}{4} = \frac{3}{2}.$$

Thus, the minimum quadratic cost is

$$W_2^2(\widehat{\mu}_m, \widehat{\mu}_w) = \frac{3}{2}. \tag{29}$$

The Wasserstein distance itself is therefore

$$W_2(\widehat{\mu}_m, \widehat{\mu}_w) = \sqrt{\frac{3}{2}}.$$

## A.5 Interpretation for the Empirical Application

This example highlights the central property of optimal transport in our setting. The transport plan naturally accommodates the difference in size between the supports of the empirical measures. No manual aggregation is required: the three-atom weekly measure and the two-atom monthly measure are compared directly, and the optimal plan distributes mass in a geometrically coherent manner.

In the actual application, the empirical measures associated with daily, weekly, and monthly frequencies contain approximately $T_d \approx 756$, $T_w \approx 156$, and $T_m \approx 36$ observations, respectively, over a three-year rolling window. Weekly and monthly returns are constructed in advance by geometrically compounding daily returns, as described in Section 3.5.

Once the three measures have been constructed, the $M = 50$-atom barycenter is linked to each of them through a transport plan of dimension $M \times T_k$. These plans do not construct the aggregated frequencies: they compare the measures already formed, accommodate their differences in cardinality without a one-to-one correspondence, and serve to update the atoms of the free-support barycenter.

# B Conic Reformulation of the Robust Problem

This appendix details the conic reformulation used to solve numerically the DRO problem presented in the main text. The objective is not to introduce a new economic criterion, but to show how the Wasserstein robustness penalty can be represented in a form compatible with a second-order cone solver.

## B.1 From the Norm to the Conic Constraint

In the main text, Wasserstein duality applied to the linear-loss case leads to a robust problem of the form

$$\min_{w \in \mathcal{W}} \left\{ -w^\top \widehat{\boldsymbol{\mu}}_{r,\tau} + \varepsilon_\tau \|w\|_2 \right\}.$$

This appendix takes this formulation as its starting point. The objective is to explain how the norm term

$$\|w\|_2$$

can be represented in a form compatible with a second-order cone solver.

To this end, we introduce a scalar auxiliary variable $t \geq 0$ and impose

$$\|w\|_2 \leq t. \tag{30}$$

This constraint means that $t$ acts as an upper bound on the Euclidean norm of the portfolio. The robustness term

$$\varepsilon_\tau \|w\|_2$$

can then be replaced in the objective by the linear term

$$\varepsilon_\tau t.$$

The problem therefore becomes equivalent to minimizing

$$-w^\top \widehat{\boldsymbol{\mu}}_{r,\tau} + \varepsilon_\tau t$$

subject to the additional constraint $\|w\|_2 \leq t$.

When $\varepsilon_\tau > 0$, the objective directly penalizes the variable $t$. The solver therefore has an incentive to choose the smallest value of $t$ compatible with the conic constraint. At the optimum, this gives

$$t^\star = \|w^\star\|_2.$$

The variable $t$ therefore does not alter the original economic problem: it serves only to represent the Wasserstein robustness penalty in a form suitable for numerical computation. The following subsection explains why the constraint $\|w\|_2 \leq t$ is precisely a second-order cone constraint.

## B.2 Geometric Intuition of the Second-Order Cone

A second-order cone of dimension $n+1$ is defined by

$$\mathcal{Q}^{n+1} = \left\{(t, x) \in \mathbb{R} \times \mathbb{R}^n : \|x\|_2 \leq t\right\}.$$

The notation $n+1$ reflects the fact that the cone is defined over a pair comprising a scalar $t \in \mathbb{R}$ and a vector $x \in \mathbb{R}^n$. Thus, $(t, x)$ belongs to the space $\mathbb{R}^{n+1}$. The constraint requires the scalar component $t$, which can be interpreted geometrically as a height, to be at least as large as the norm of the vector $x$. In particular, it implies $t \geq 0$.

Figure B.1 illustrates this geometry for $\mathcal{Q}^2$ and $\mathcal{Q}^3$. In two dimensions,

$$\mathcal{Q}^2 = \{(t, x) \in \mathbb{R} \times \mathbb{R} : |x| \leq t\},$$

which corresponds to an upward-opening V-shaped region. The lines $t = |x|$ form the boundary of the cone. A point $(x, t)$ therefore belongs to the cone when its height $t$ is at least equal to its horizontal distance from the origin. For example, the point $(1.5, 2.5)$ belongs to the cone, whereas the point $(2.5, 1.5)$ does not.

In three dimensions,

$$\mathcal{Q}^3 = \left\{(t, x_1, x_2) \in \mathbb{R} \times \mathbb{R}^2 : \|(x_1, x_2)\|_2 \leq t\right\}.$$

For a given height $t$, the admissible points form a disk of radius $t$. Thus, at height $t = 1$, the admissible cross-section is a disk of radius 1; at height $t = 2$, it is a disk of radius 2, and so forth. As $t$ increases, the set of admissible vectors expands.

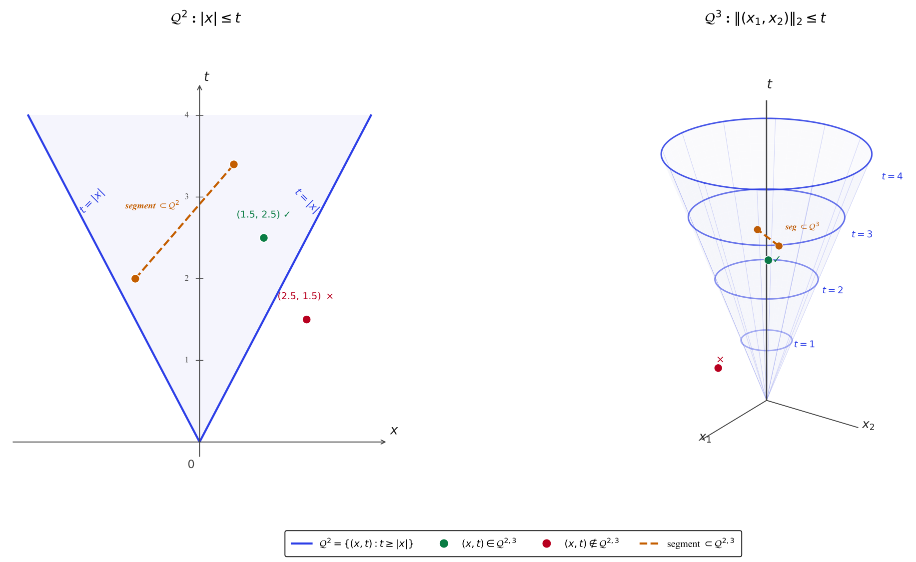


**Figure B.1:** Geometry of the second-order cone. The left panel represents the cone $\mathcal{Q}^2 = \{(t,x) : |x| \le t\}$, namely the set of points whose height $t$ dominates the norm $|x|$. The green point satisfies this constraint, whereas the red point violates it. The orange segment connects two points that belong to the cone and remains entirely within it, illustrating the convexity of $\mathcal{Q}^2$. The right panel represents the cone $\mathcal{Q}^3 = \{(t, x_1, x_2) : \|(x_1, x_2)\|_2 \le t\}$. Its horizontal cross-sections at height $t$ are disks of radius $t$, as illustrated for $t = 1, 2, 3, 4$.

This geometric representation clarifies the reformulation of the robust problem. In our case, the role of the vector $x$ is played by the portfolio $w$, and the role of the height by the auxiliary variable $t$. The constraint

$$\|w\|_2 \le t$$

therefore means that the pair $(t, w)$ must belong to the cone

$$\mathcal{Q}^{N+1} = \left\{(t, z) \in \mathbb{R} \times \mathbb{R}^N : \|z\|_2 \le t\right\}.$$

## B.3 SOCP Interpretation of the Allocation Problem

After introducing the auxiliary variable $t$, the robustness term $\varepsilon_\tau \|w\|_2$ is represented by the pair

$$\varepsilon_\tau t, \qquad \|w\|_2 \le t.$$

The conic nature of the program therefore follows from the constraint

$$\|w\|_2 \le t.$$

Because $w \in \mathbb{R}^N$, the pair $(t, w)$ belongs to $\mathbb{R}^{N+1}$. This constraint can then be written as membership in the second-order cone:

$$(t, w) \in \mathcal{Q}^{N+1} = \left\{(t, z) \in \mathbb{R} \times \mathbb{R}^N : \|z\|_2 \leq t\right\}. \tag{31}$$

The remaining constraints of the problem,

$$\mathbf{1}^\top w = 1, \qquad w_i \geq 0, \quad i = 1, \ldots, N, \qquad t \geq 0,$$

are linear. The objective,

$$-w^\top \widehat{\boldsymbol{\mu}}_{r,\tau} + \varepsilon_\tau t,$$

is also linear in $(w, t)$. The problem therefore combines a linear objective, linear constraints, and a second-order cone constraint. It consequently belongs to the class of SOCPs.

The value of this reformulation is computational. It transforms the nonlinear penalty $\varepsilon_\tau \|w\|_2$ into a standard conic constraint that can be handled directly by a solver such as MOSEK. The conic formulation does not alter the economic problem: it merely provides a representation suited to repeated numerical solution in the rolling-window setting.

$$\text{Primal DRO} \quad \longrightarrow \quad \text{dual formulation} \quad \longrightarrow \quad \text{penalty } \varepsilon_\tau \|w\|_2 \quad \longrightarrow \quad \text{SOCP.}$$

## B.4 Operational Interpretation

The SOCP reformulation does not alter the economic objective of the problem. It only provides a more suitable numerical representation. The ambiguity radius $\varepsilon_\tau$ retains the same interpretation: it controls the degree of prudence imposed on the portfolio. When $\varepsilon_\tau$ increases, the weight assigned to the variable $t$ also increases, thereby strengthening the penalty on the portfolio norm.

In the fully invested, long-only case, this norm can also be interpreted as a measure of portfolio concentration. Indeed, under the constraints $1^\top w = 1$ and $w_i \geq 0$, a high value of $\|w\|_2$ corresponds to a more concentrated allocation, whereas a lower value corresponds to a more diversified allocation.

In this thesis, problem (7) is solved at each rebalancing date using a conic solver. This step is compatible with a rolling-window setting because the problem solved at each date remains convex and finite-dimensional. The conic formulation thereby links the theoretical definition of Wasserstein DRO to its repeated numerical implementation in the empirical analysis.

## B.5 Summary of the Reformulation Chain

The preceding reformulation can be summarized as a four-step pipeline. The starting point is the primal DRO program, formulated as a min–max problem over a set of plausible distributions. This problem is conceptually clear but difficult to solve directly because the inner problem is defined over a probability measure $\mathbb{Q}$. Wasserstein duality then replaces this optimization over distributions with a penalty on perturbations around the empirical scenarios. Under the linear loss $\ell(w, r) = -w^\top r$, this representation yields a closed form containing the robustness penalty $\varepsilon_\tau \|w\|_2$. Finally, this norm is represented by an auxiliary variable $t$ and a second-order cone constraint, yielding an SOCP that can be solved numerically.

The reformulation chain can be summarized in four steps.

1. **Primal DRO program.**

$$\min_{w \in \mathbb{R}^N} \quad \sup_{\mathbb{Q} \in \mathcal{P}_2(\mathbb{R}^N)} \mathbb{E}_{\mathbb{Q}}[\ell(w, r)]$$

$$\text{s.t.} \quad \begin{cases} w \in \mathcal{W}, \\ W_2\left(\mathbb{Q}, \widehat{\mu}_\tau^{\text{nom}}\right) \leq \varepsilon_\tau. \end{cases}$$

2. **Wasserstein dual form.**

$$\min_{w \in \mathbb{R}^N} \quad \inf_{\eta \in \mathbb{R}} \left\{ \eta \varepsilon_\tau^2 + \frac{1}{T_\tau} \sum_{i=1}^{T_\tau} \sup_{r \in \mathbb{R}^N} \left[ \ell(w, r) - \eta \|r - r_{i,\tau}\|_2^2 \right] \right\}$$

$$\text{s.t.} \quad \begin{cases} w \in \mathcal{W}, \\ \eta \geq 0. \end{cases}$$

3. **Linear-loss case.** For $\ell(w, r) = -w^\top r$, this gives

$$\min_{w \in \mathbb{R}^N} \quad -w^\top \widehat{\boldsymbol{\mu}}_{r,\tau} + \varepsilon_\tau \|w\|_2$$
$$\text{s.t.} \quad \left\{ \; w \in \mathcal{W}. \right.$$

4. **Conic reformulation.** Introducing an auxiliary variable $t$, the problem becomes

$$\min_{w \in \mathbb{R}^N,\, t \in \mathbb{R}} \quad -w^\top \widehat{\boldsymbol{\mu}}_{r,\tau} + \varepsilon_\tau t$$
$$\text{s.t.} \quad \begin{cases} \|w\|_2 \le t, \\ \mathbf{1}^\top w = 1, \\ w_i \ge 0, \quad i = 1, \dots, N, \\ t \ge 0. \end{cases}$$

# C Free-Support Wasserstein Barycenter

This appendix complements the presentation of the barycenter introduced in Subsection 4.3.4. Appendix A presented the elementary mechanism of a transport plan between two empirical measures of different sizes. We now show how this mechanism is extended simultaneously to several frequency-specific measures in order to update the atoms of the free-support barycenter.

The purpose is therefore not to repeat the general definition of optimal transport, but to clarify its role in the barycentric algorithm used to construct the multi-frequency consensus distribution.

## C.1 Barycentric Problem

At each rebalancing date $\tau$, the inputs to the problem are the three normalized empirical measures

$$(\widehat{\mu}_{k,\tau})_{k\in\mathcal{K}}\,, \qquad \mathcal{K} = \{d, w, m\},$$

defined in Section 4.2.1.

The free-support barycenter is computed using uniform frequency weights:

$$\widehat{\mu}_{\tau}^{*,W_2} \in \arg\min_{\nu\in\mathcal{P}_2(\mathbb{R}^N)} \sum_{k\in\mathcal{K}} \alpha_k W_2^2(\nu, \widehat{\mu}_{k,\tau})\,, \qquad \alpha_k = \frac{1}{3}. \tag{32}$$

These barycentric weights remain fixed across the sensitivity variants. The alternative weighting schemes apply to the final aggregation of distances around the consensus, rather than to the barycentric support itself.

## C.2 Free-Support Representation

The barycenter is represented by a discrete measure with $M$ atoms:

$$\nu_\tau = \sum_{j=1}^{M} b_j \delta_{y_{j,\tau}}\,, \qquad y_{j,\tau} \in \mathbb{R}^N, \quad b_j = \frac{1}{M}.$$

The positions $y_{1,\tau}, \ldots, y_{M,\tau}$ are the optimization variables. They are not constrained to coincide with historical observations: the term *free support* means precisely that these positions are optimized directly in $\mathbb{R}^N$. In the main specification, the number of atoms is fixed at

$$M = 50.$$

This choice provides an intermediate resolution for the barycentric consensus. It is sufficiently rich not to constrain the support to the number of monthly observations, while retaining a reasonable computational cost. Sensitivity to the choice of $M$ is assessed by comparing

$$M \in \{36, 50, 72\}.$$

## C.3 Iterative Algorithm

The free-support barycenter is computed through an alternating two-step procedure. Let $Y_\tau^{(q)} = (y_{1,\tau}^{(q)}, \ldots, y_{M,\tau}^{(q)})$ denote the current support at iteration $q$, and let $\nu_\tau^{(q)} = \sum_j b_j \delta_{y_{j,\tau}^{(q)}}$ denote the corresponding barycentric measure.

**Initialization.**

The initial support is obtained through weighted k-means++ initialization [25]. Each frequency receives a total mass of 1/3, regardless of the number of observations it contains. This step is purely numerical and precedes the iterative computation of the barycenter.

**Step 1 — Computation of the optimal transport plans.** For each frequency $k \in \mathcal{K}$, we compute the optimal transport plan between the current barycenter $\nu_\tau^{(q)}$ and the empirical measure $\widehat{\mu}_{k,\tau}$:

$$\pi^{(k,q)} \in \arg \min_{\pi \in \Pi(b, a^{(k)})} \sum_{j=1}^{M} \sum_{i=1}^{T_{k,\tau}} \pi_{ji} \left\| y_{j,\tau}^{(q)} - \widetilde{r}_{k,i,\tau} \right\|_2^2, \tag{33}$$

where $a_i^{(k)} = 1/T_{k,\tau}$ denotes the mass of the $i$th observation at frequency $k$. This yields three transport plans, $\pi^{(d,q)}, \pi^{(w,q)}, \pi^{(m,q)}$, each indicating how the mass of every barycentric atom is associated with the observations at the corresponding frequency.

Equivalently, defining the cost matrix

$$C_{ji}^{(k,q)} = \left\| y_{j,\tau}^{(q)} - \widetilde{r}_{k,i,\tau} \right\|_2^2,$$

the transport cost can be written as

$$\langle \pi^{(k,q)}, C^{(k,q)} \rangle = \sum_{j=1}^{M} \sum_{i=1}^{T_{k,\tau}} \pi_{ji}^{(k,q)} C_{ji}^{(k,q)} = \mathbf{1}^\top \left( \pi^{(k,q)} \odot C^{(k,q)} \right) \mathbf{1},$$

where $\odot$ denotes the Hadamard product.

**Step 2 — Updating the barycentric positions.** The barycenter atoms are moved toward a weighted average of the observations to which they are connected by the optimal transport plans. With fixed barycentric masses $b_j$, the update is

$$y_{j,\tau}^{(q+1)} = \frac{1}{b_j} \sum_{k \in \mathcal{K}} \alpha_k \sum_{i=1}^{T_{k,\tau}} \pi_{ji}^{(k,q)} \widetilde{r}_{k,i,\tau}. \tag{34}$$

Each atom $y_{j,\tau}^{(q+1)}$ is therefore a Euclidean barycenter of the observations to which it contributes, weighted jointly by the flows in the transport plans and the barycentric weights $\alpha_k$. The new positions $Y_\tau^{(q+1)}$ become the support for the next iteration.

The procedure can be summarized by the following diagram:

$$Y_\tau^{(q)}$$

$$\downarrow$$

$$\left(\pi^{(k,q)}\right)_{k \in \mathcal{K}} \quad \text{optimal transport plans}$$

$$\downarrow$$

$$Y_\tau^{(q+1)} \quad \text{updated atoms}$$

$$\downarrow$$

$$\text{convergence or } q = q_{\max}$$

The algorithm is repeated until the positions converge or the maximum number of iterations

$q_{\max}$ is reached. The convergence criterion is

$$\max_{j=1,\ldots,M} \left\| y_{j,\tau}^{(q+1)} - y_{j,\tau}^{(q)} \right\|_2 < \varepsilon_{\text{tol}},$$

where $\varepsilon_{\text{tol}}$ is a fixed numerical tolerance.

## C.4 Illustration of a Barycentric Update

Consider three empirical measures in $\mathbb{R}^2$, respectively representing the monthly, weekly, and daily frequencies:

$$\widehat{\mu}_m = \frac{1}{2}\left(\delta_{(0,0)} + \delta_{(3,3)}\right),$$

$$\widehat{\mu}_w = \frac{1}{3}\left(\delta_{(0,0)} + \delta_{(3/2,3/2)} + \delta_{(3,3)}\right),$$

and

$$\widehat{\mu}_d = \frac{1}{4}\left(\delta_{(0,0)} + \delta_{(1,1)} + \delta_{(2,2)} + \delta_{(3,3)}\right).$$

We seek a barycenter with $M = 2$ atoms and masses $b_1 = b_2 = 1/2$, assigning the weights $\alpha_m = \alpha_w = \alpha_d = 1/3$ to the three frequencies. The initial support is

$$Y^{(0)} = \begin{pmatrix} 0 & 0 \\ 3 & 3 \end{pmatrix}.$$

For this initialization, the optimal transport plans are

$$\pi^{(m,0)} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix}, \qquad \pi^{(w,0)} = \begin{pmatrix} 1/3 & 1/6 & 0 \\ 0 & 1/6 & 1/3 \end{pmatrix},$$

and

$$\pi^{(d,0)} = \begin{pmatrix} 1/4 & 1/4 & 0 & 0 \\ 0 & 0 & 1/4 & 1/4 \end{pmatrix}.$$

Letting $X^{(k)}$ denote the matrix whose rows are the atoms at frequency $k$, their aggregate contribution is

$$\sum_{k\in\{m,w,d\}} \alpha_k\, \pi^{(k,0)} X^{(k)} = \begin{pmatrix} 1/6 & 1/6 \\ 4/3 & 4/3 \end{pmatrix}.$$

Updating the barycentric support then gives

$$\begin{aligned} Y^{(1)} &= \operatorname{diag}(b)^{-1} \sum_{k\in\{m,w,d\}} \alpha_k\, \pi^{(k,0)} X^{(k)} \\ &= \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1/6 & 1/6 \\ 4/3 & 4/3 \end{pmatrix} = \begin{pmatrix} 1/3 & 1/3 \\ 8/3 & 8/3 \end{pmatrix}. \end{aligned}$$

The two atoms thus leave their initial monthly positions and move toward the center shared by the three distributions. This step illustrates the role of the barycenter: constructing a consensus support without imposing a one-to-one correspondence between measures with different numbers of observations.

In the complete algorithm, the transport plans are recomputed from $Y^{(1)}$, after which the support is updated again:

$$Y^{(q)} \longrightarrow \left(\pi^{(k,q)}\right)_{k\in\mathcal{K}} \longrightarrow Y^{(q+1)}.$$

This alternation continues until the convergence criterion is satisfied or the maximum number of iterations is reached.

## C.5 Numerical Limitations and Dependence on Initialization

The free-support problem is not convex with respect to the positions $y_{1,\tau}, \ldots, y_{M,\tau}$. When it converges, the alternating procedure reaches a stationary solution or local approximation whose quality may depend on the initialization of the barycentric support; global optimality is not guaranteed.

The Sliced-Wasserstein distances used in $\widehat{\rho}_\tau^{\text{hyb}}$ vary continuously with the positions of the atoms. This property suggests local stability of the signal under small perturbations of a given support. It guarantees neither that distinct initializations lead to the same local optimum nor that the corresponding signals are numerically equivalent. This sensitivity is therefore assessed empirically through the robustness diagnostics.

The weighted initialization is described in Section C.3. It reduces dependence on an arbitrary initial configuration without guaranteeing global optimality of the barycenter. Sensitivity to support resolution is assessed by comparing $M \in \{36, 50, 72\}$.

The resulting support constitutes the multivariate consensus used to measure inter-frequency dispersion according to the construction presented in Section 4.4.

# D Monte Carlo Convergence of the Sliced Signal

Figure D.1 documents the numerical stability of the sliced signal as the number of random projections increases. The diagnostic uses $B_{\mathrm{MC}} = 30$ replications for each value of $L \in \{10, 20, 50, 100, 200, 500, 1000\}$.

Two 36-month windows are retained: a stress window ending in September 2009 and a calm window ending in June 2016. The barycenter is computed only once per window. Within each replication, the same random seeds and nested directions are used across the different values of $L$.

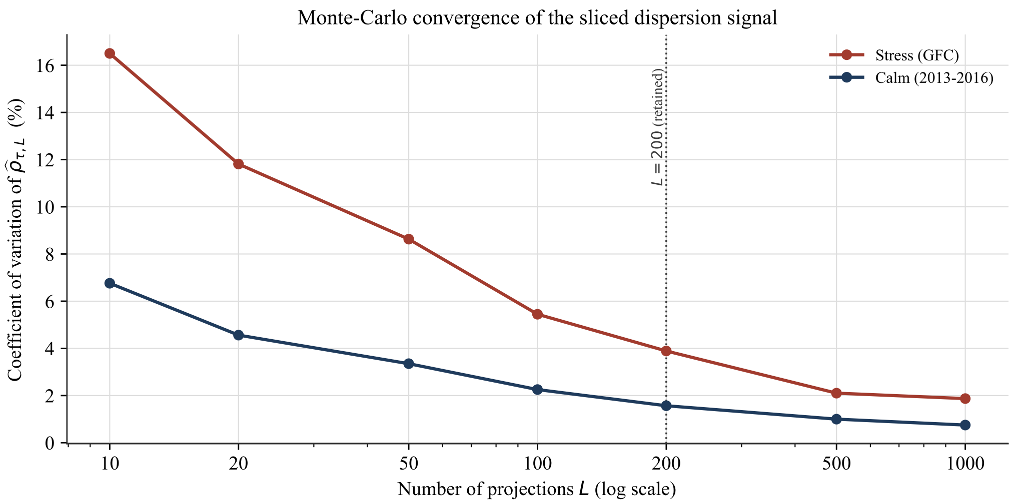


**Figure D.1:** Coefficient of variation of $\widehat{\rho}$ across $B_{\mathrm{MC}} = 30$ sets of projections as a function of the number of directions $L$. The stress window ends in September 2009 and the calm window in June 2016. The vertical line indicates the selected specification, $L = 200$, for which the coefficient of variation equals 3.89 % and 1.57 %, respectively.

The decline in the coefficient of variation confirms the progressive stabilization of the approximation as $L$ increases. At $L = 200$, relative dispersion is limited to 3.89 % in the stress window and 1.57 % in the calm window. This choice therefore represents an explicit trade-off between numerical stability and the cost of walk-forward recomputation.

# E Portfolio Robustness

## E.1 Detailed Metrics

Table E.1 complements the analysis presented in Section 6.2.9.2. Each block varies a single lever around the main specification — window $W_\rho = 36$, identity form, monthly rebalancing, and $\beta = 0.10$ — while all other components of the portfolio framework remain unchanged. Sharpe-ratio differences are calculated relative to the static portfolio corresponding to each configuration.

**Table E.1:** Dynamic-portfolio metrics and Sharpe-ratio difference relative to the corresponding static portfolio over 1995–2025. The table reports the net Sharpe ratio, annualized net return and volatility, maximum drawdown, recurring two-way turnover, gross and net Sharpe-ratio differences, and the proportion of decisions for which the radius reaches the safety cap. Each block varies one lever at a time.

| Lever | Value | $SR_{dyn,net}$ | $\mu_{net}$ (%) | $\sigma_{net}$ (%) | MaxDD (%) | Turnover | $\Delta SR_{gross}$ | $\Delta SR_{net}$ | Clipped (%) |
|---|---|---|---|---|---|---|---|---|---|
| $W_\rho$ | 36 | 0.745 | 11.66 | 15.64 | 52.79 | 0.1152 | +0.0571 | +0.0559 | 0.0 |
| | 48 | 0.744 | 11.66 | 15.68 | 53.37 | 0.1133 | +0.0552 | +0.0545 | 0.0 |
| | 60 | 0.739 | 11.62 | 15.71 | 53.98 | 0.1122 | +0.0506 | +0.0502 | 0.0 |
| Form | identity | 0.745 | 11.66 | 15.64 | 52.79 | 0.1152 | +0.0571 | +0.0559 | 0.0 |
| | $\exp(z_\tau)$ | 0.619 | 13.90 | 22.45 | 63.43 | 0.3039 | -0.0506 | -0.0702 | 0.5 |
| | rank | 0.656 | 12.60 | 19.21 | 56.88 | 0.2787 | -0.0100 | -0.0332 | 0.0 |
| Rebalancing | monthly | 0.745 | 11.66 | 15.64 | 52.79 | 0.1152 | +0.0571 | +0.0559 | 0.0 |
| | quarterly | 0.743 | 11.71 | 15.76 | 53.77 | 0.0672 | +0.0630 | +0.0623 | 0.0 |
| | annual | 0.759 | 12.00 | 15.82 | 53.57 | 0.0317 | +0.0632 | +0.0630 | 0.0 |
| $\beta$ | 0.05 | 0.756 | 11.69 | 15.47 | 50.84 | 0.1083 | +0.0464 | +0.0454 | 0.0 |
| | 0.10 | 0.745 | 11.66 | 15.64 | 52.79 | 0.1152 | +0.0571 | +0.0559 | 0.0 |
| | 0.20 | 0.724 | 11.60 | 16.01 | 56.47 | 0.1287 | +0.0721 | +0.0704 | 0.0 |

The window, rebalancing-frequency, and $\beta$ variants retain a positive net contrast ranging from +0.0454 to +0.0704. Lengthening the $W_\rho$ window progressively reduces $\Delta SR_{net}$, from +0.0559 to +0.0502, whereas quarterly and annual rebalancing and higher values of $\beta$ produce larger point estimates of the contrast. These variations constitute sensitivity analyses rather than ex post selection criteria for the main specification.

The functional form is the principal source of instability. The exponential and rank transformations reverse the contrast, with net Sharpe-ratio differences of −0.0702 and −0.0332, respectively. They are also associated with the highest turnover and more pronounced gross-to-net erosion. The identity form remains the main specification because of its direct interpretation, absence of clipping, and lower turnover among the forms considered, not because it maximizes realized performance. The detailed comparisons are interpreted in Section 6.2.9.2.

## E.2 Signal Modulation by Functional Form

Figures E.1 and E.2 document the paths produced by the three functional forms examined in Section 6.2.9.2. They complement the sensitivity analysis with descriptive diagnostics and do not constitute additional performance tests.

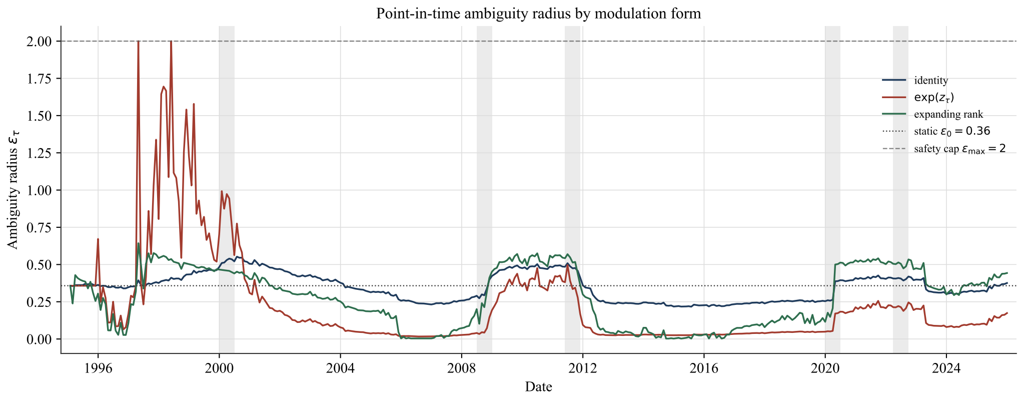


**Figure E.1:** Point-in-time ambiguity-radius paths over 1995–2025 under the identity, exponential, and expanding-rank forms. The dotted line represents the operational static radius $\varepsilon_0 = 0.3577$, and the dashed line the safety cap $\varepsilon_{\max} = 2$. The shaded bands surround the five prespecified stress episodes. The exponential form amplifies extreme values more strongly and reaches the cap for 0.5 % of decisions; the identity and rank forms are never clipped.

Figure E.2 then centers the analysis on the five stress episodes. To make the magnitudes comparable, each point-in-time modulation $m_\tau^{(F)}$ is standardized over the sample displayed:

$$z_\tau^{(F)} = \frac{m_\tau^{(F)} - \overline{m}^{(F)}}{s(m^{(F)})}.$$

This standardization is performed solely for graphical representation. The portfolios are always constructed from the unstandardized point-in-time modulations.

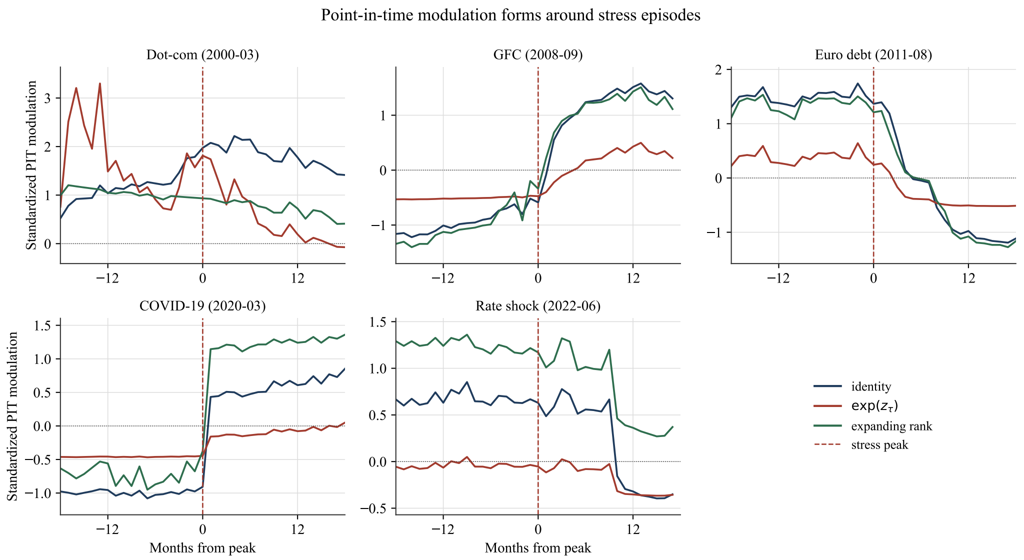


**Figure E.2:** Point-in-time modulations standardized over windows of ±18 months around the five prespecified episodes. The vertical line marks the peak date. The exponential form accentuates extreme fluctuations, whereas the identity form transmits the normalized variations of the signal more directly and the expanding-rank form depends on its position in the available historical distribution. The standardization is exclusively visual and enters neither the construction of the radius nor the backtest.

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